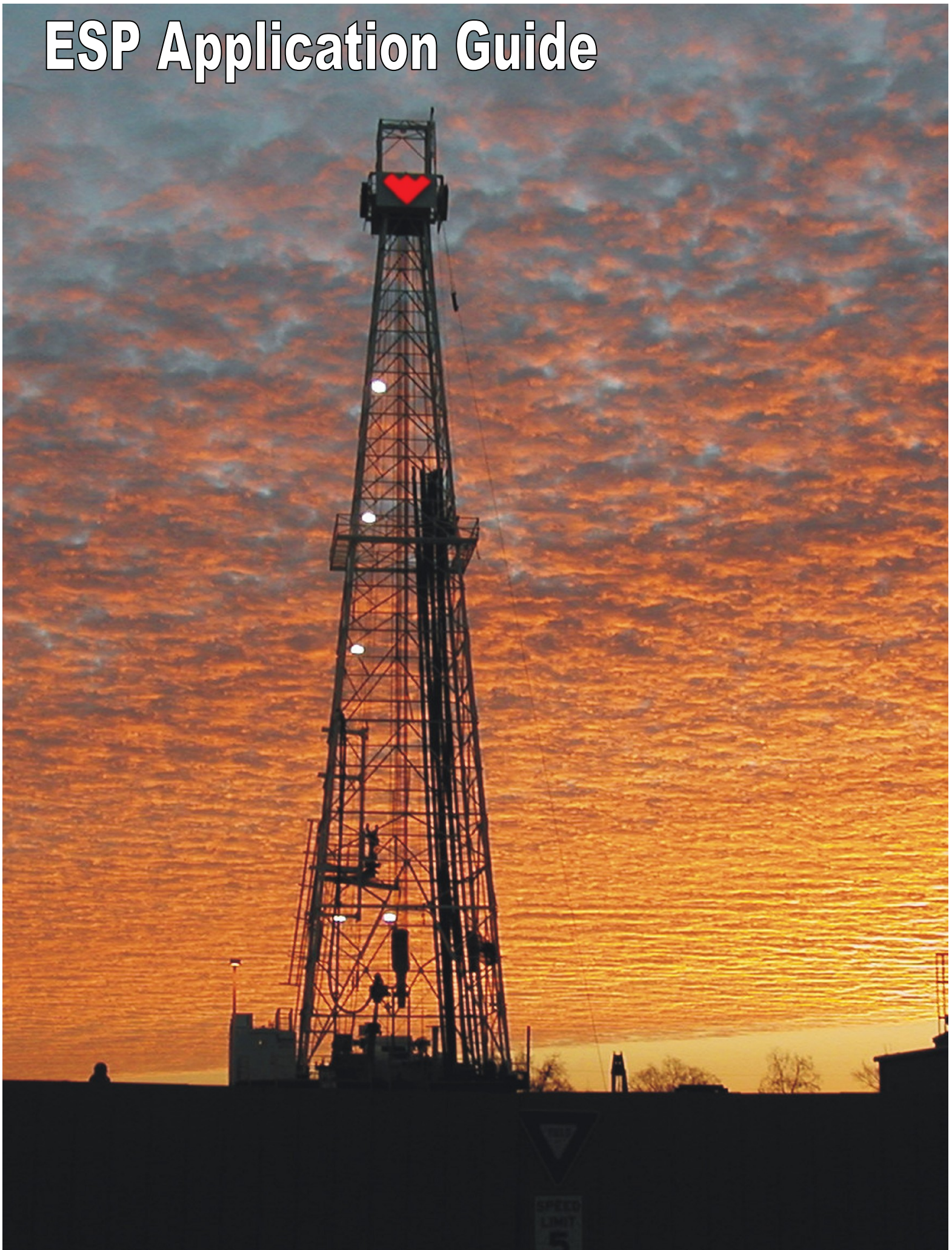


ESP Application Guide





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1 Introduction

Featuring operating depths up to 17,000 TVD and operating volumes to 40,000 BFPD, Weatherford's Electric Submersible Pumping (ESP) systems are often considered the high volume and depth champion among lift systems. This system requires very little surface space; works well in highly deviated wells and is ideally suited for offshore applications and vertical wells.

The durability and service life of an ESP system relies heavily on the quality of each system component. From robust, dependable downhole motors to an extensive range of intakes and multistage centrifugal pumps and surface components, ESP systems can be customized and assembled for a variety of applications for long-term efficiency and extended service life.

Figure 1 shows a typical ESP System installation, which incorporates an electric motor and centrifugal pump unit run on a production string and connected to the surface switchboard and transformer via an electric power cable.

The downhole components are suspended from the production tubing above the wells' perforations. In most cases the motor is located on the bottom of the string. Above the motor are the seal section, the intake or gas separator, and the pump. The power cable is banded to the tubing and plugs into the top of the motor.

As the fluid enters the well it must pass by the motor and into the pump. This fluid flow past the motor cools the motor. The fluid then enters the intake of the pump. The submersible pump consists of multiple stages, each one composed of two key components: an impeller, that drives the fluid, and one diffuser, that directs the fluid to the next stage. Each stage adds head to the produced fluid. The total head available at the pump discharge is designed to equal or exceed the Total Dynamic Head (TDH) required to lift the designed flow rate to surface and move it to the surface production facility.

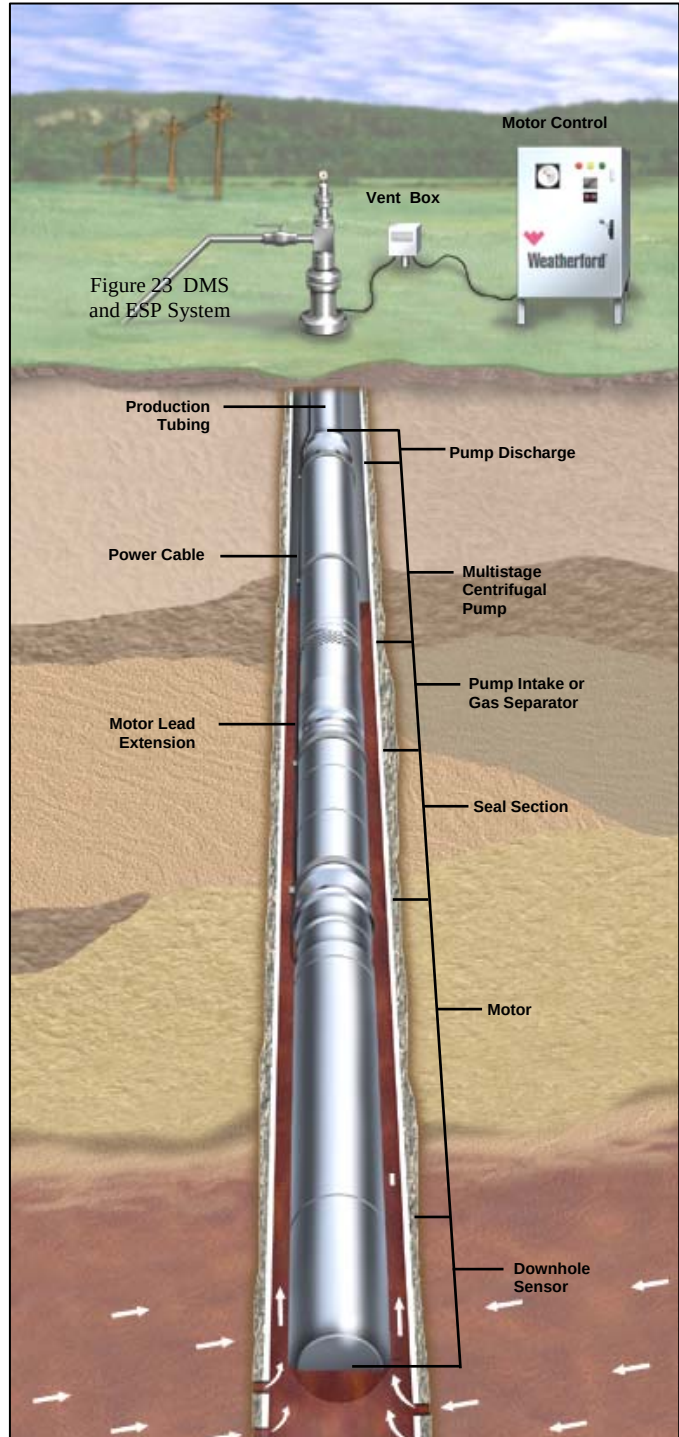


Figure. 1.- Typical ESP System Configuration



2 ESP System Applications

ESP Systems can be applied in a wide range of applications. Common applications include:

- High volume lift requirements (>200 BPD)
- A variety of well types including highly deviated or non-vertical well bores
- Water floods and high water-cut wells
- Wells with H₂S and CO₂ including CO₂ floods and WAG operations.
- Well testing operations
- Abrasive, gassy and viscous fluids
- Coal Bed Methane (CBM) gas well deliquification applications
- The following table summarizes typical application ranges for the ESP System, as well as maximum limits under special conditions:

APPLICATION CONSIDERATIONS	TYPICAL RANGE	MAXIMUM
Operating Depth	1,000–10,000 feet (300-3,000 meters) TVD	17,000 feet (4,500 meter) TVD
Operating Volume	120–20,000 BFPD (20-,3200 m ³ /day)	40,000 BFPD (6,350 m ³ /day)
Motor Operating Temperature	100–302°F (38-150°C)	356°F (180°C)
Wellbore Deviation		0° – 90° Pump Displacement <3°/100ft Build Angle
Corrosion Handling	Good	
Gas Handling	Fair	
Solids Handling	Fair	
Fluid Gravity	> 10° API	
Servicing	Workover or Pulling Unit	
Prime Mover Type	Electric Motor	
Offshore Application	Excellent	



2.1 ESP System Advantages & Benefits

- Extended service life in deep wells, deviated wells, and vertical wells with doglegs.
- High operating efficiency and lower overall operating costs in wells with production volumes greater than 500 BFPD.
- Minimal maintenance requirements result in greater production with less downtime.
- Minimal surface requirements enable lower installation costs and are well suited for environmentally sensitive and space sensitive (off shore) applications.
- Wells with casing sizes 4-1/2 inches and larger can readily be fitted with an ESP system.
- High resistance to corrosive downhole environments.
- Optionally can be installed with real time instrumentation (DMS) system that report intake pressure, discharge pressure, motor temperature, fluid temperature, system vibration and current leakage to surface.
- Well testing applications when the well PI is unknown.

2.2 ESP System Limitations

- System is limited to areas where electric power or generators are available.
- Limited adaptability to major changes in reservoir due to pump range of operation; can be improved when a Variable Frequency Controller (VFD) is used.
- Higher energy requirement when high viscosity fluids are pumped.
- High intervention costs.



3 ESP System Components

3.1 Submersible Electric Motors

The ESP system's prime mover is the submersible motor (see Figure 2). The motor is a two-pole, three-phase, squirrel-cage induction type. Motors run at a nominal speed of 3500 RPM in 60-Hz operation (2917 RPM on 50-Hz power). Motors are filled with high dielectric oil that provides bearing lubrication, and thermal conductivity.

Heat generated by motor operation is transferred to the well fluid as it flows past the motor housing. A minimum fluid velocity of 1 ft/sec is typically recommended to provide adequate cooling. Because the motor relies on the flow of well fluid for cooling, a standard ESP must never be set at or below perforations or producing zone unless the motor is shrouded.

Motors are manufactured in five different diameters (series) as 3.75, 4.56, 5.40, 5.62 and 7.38 inches. Thus, motors can be used in casing sizes as small as 4.50 inches. 60 Hz horsepower capabilities range from a low of 7.5 HP in 3.75-inch series to a high of 1,200 HP in the 5.62 inch series. Motor construction may be a single section or multiple sections bolted together to reach a specific horsepower. Motors are selected on the basis of the maximum OD that can be run in a given casing size and the HP required to operate the pump.

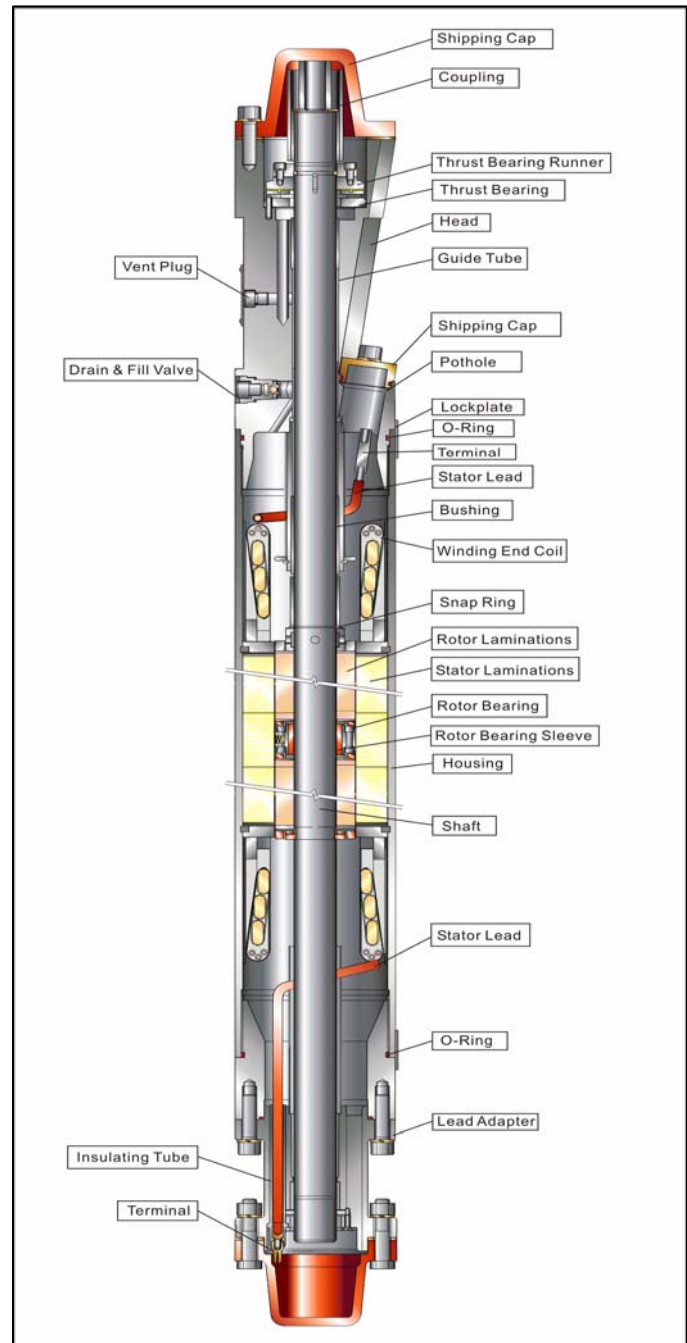


Figure 2 Upper tandem motor



3.2 Multistage Centrifugal Pumps

The ESP pump is a multistage centrifugal pump type (see Figure 3). A stage consists of an impeller and a diffuser (Figure 4). The impeller is keyed to the shaft and rotates at the RPM of the motor. Centrifugal force causes the fluid to move from the center (or eye) of the impeller outward. These forces impart kinetic or velocity energy to the fluid.

The diffuser is stationary and its function is to direct the fluids to flow efficiently from one impeller to another and to convert a portion of the velocity (kinetic) energy into pressure (potential) energy.

The stages (an impeller-diffuser combination) are placed onto a keyed shaft and then loaded into a steel housing. When the threaded head and base are screwed into the housing they compress against the outside edge of the diffuser. It is this compression that holds the diffusers stationary. If this compression is lost then the diffusers would be free to rotate. This rotation would cause the pump to lose almost all of its ability to produce any head (or lift).

The impellers incorporate a fully enclosed curved vane design, whose maximum efficiency is a function of impeller design and type.

The fluid enters the impeller at the eye (Figure 4). The vanes in the impeller create channels through which the fluid is directed. The size of the impeller (or the volume between the upper and lower shroud) determines the volume per unit time (or fluid rate) that can be produced.

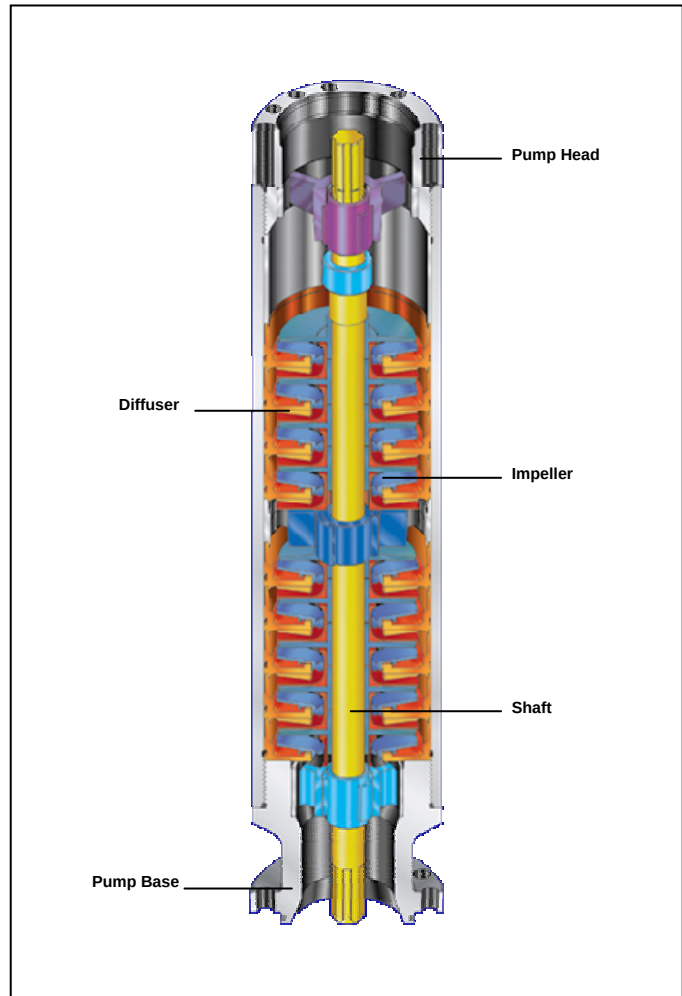


Figure 3.- Centrifugal Pump

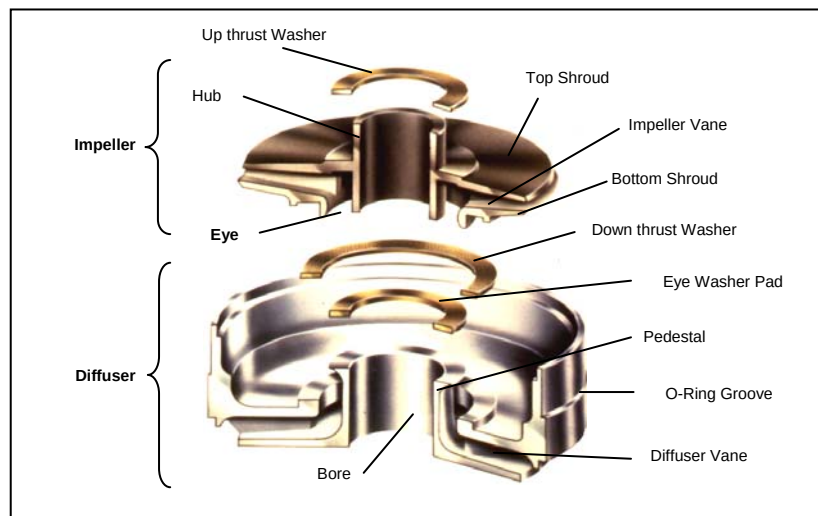


Figure 4.- Impeller and Diffuser description



There are two types of impellers used in oil well submersible pumps. These are the mixed flow (Figure 5) and the radial flow (Figure 6). The radial stages generally range from 150 BFPD to 1600 BFPD in 4.00 OD pumps and 1300 to 4600 BFPD in 5.38 OD pumps. The radial stage is a flat stage and is the most efficient design for these lower flow rates. The mixed flow stage is used for higher flow rate applications.

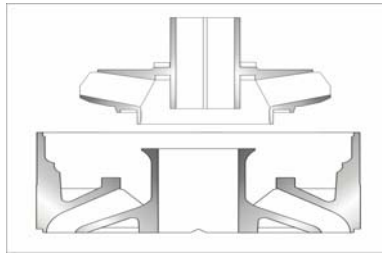


Figure 5 Radial flow stage

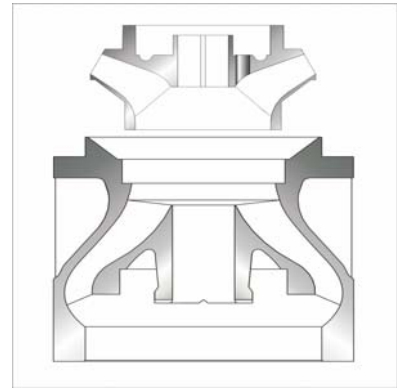


Figure 6 Mixed flow stage

Through the use of the corrosion-resistant materials, cast Ni-resist (high nickel-iron) impellers and diffusers with K-monel shafting, pump wear and corrosion can be minimized. However, unless otherwise specified, the housings, heads and bases of the pumps, protectors, and motors will be carbon steel. In corrosive applications the equipment may be coated with a corrosion resistant coating or premium Stainless Steel housing / heads and bases should be specified. In addition Monel fasteners, vent plug / drain and fill valves will need to be specified. These multistage pumps may be assembled as a floater or fixed-impeller compression pump design, depending on how the axial thrust of the pump is handled and well conditions.

3.2.1 Floater Pump Design

The impellers are free to move axially along the shaft (Figure 7). Thrust washers installed on the impeller support the axial thrust of each impeller. A thrust bearing in the seal section supports the weight and thrust of the shaft.

3.2.2 Compression Pump Design

The impeller is locked to prevent axial movement along the shaft (Figure 8). The axial thrust of each impeller is transferred through the shaft to the thrust bearing located in the seal section below the pump. A thrust bearing in the seal section supports the weight of the shaft, the thrust generated by the stages and the thrust generated by the pump discharge pressure acting on the end of the shaft.

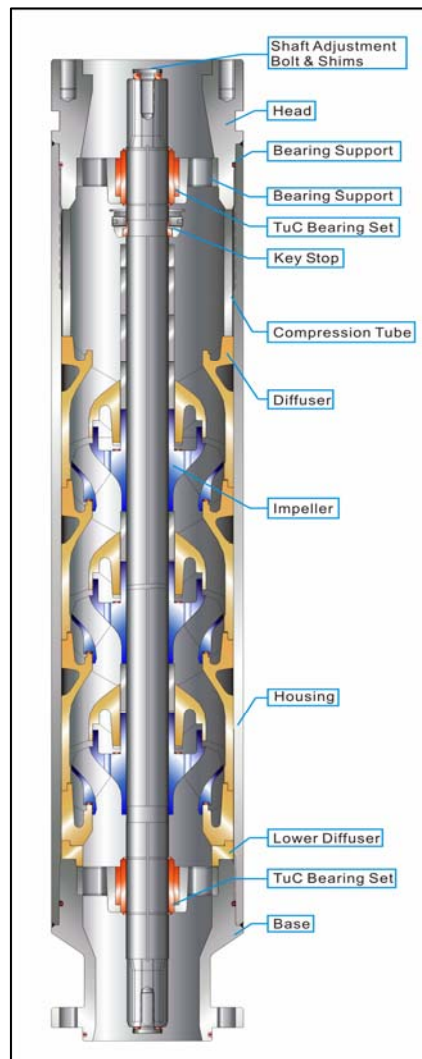


Figure 7 Floater Pump

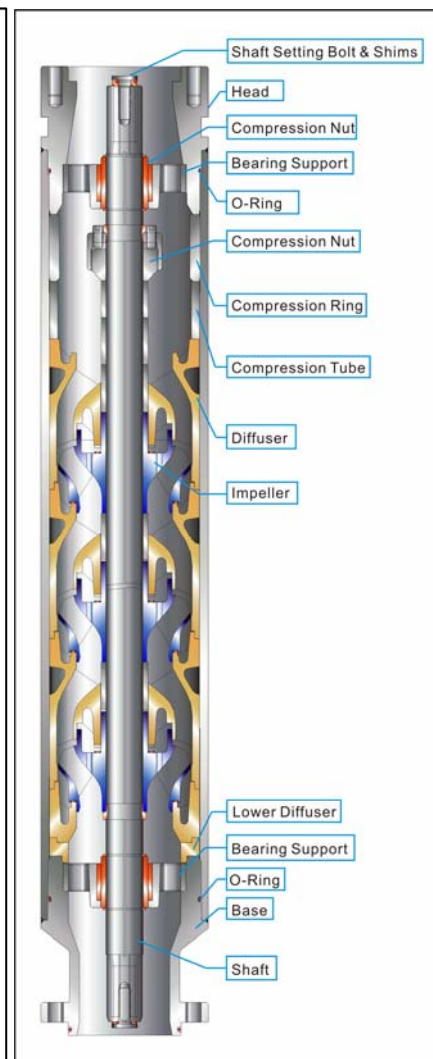


Figure 8 Compression Pump



3.3 Seal Section

The seal section's primary purpose is to isolate the motor oil from the well fluid, equalize the motor internal pressure with the annulus pressure and to house the thrust bearing (Figure 9) that carries pump thrust. There are two types of seal section design – the bag (Figure 10) and the labyrinth chamber (Figure 11) path. The bag type seal design relies on an elastic, fluid-barrier bag to allow for the thermal expansion of motor fluid in operation, while still isolating the well fluid from the motor oil. The labyrinth path design uses the specific gravity of the well fluid and motor oil to prevent the well fluid from entering the motor. This is accomplished by allowing the well fluid and motor oil to communicate through tube paths connecting segregated chambers. Various chamber combinations, elastomers, housing materials, fasteners, shaft materials and thrust bearings are available which allow the motor seal to be optimized for the well conditions.

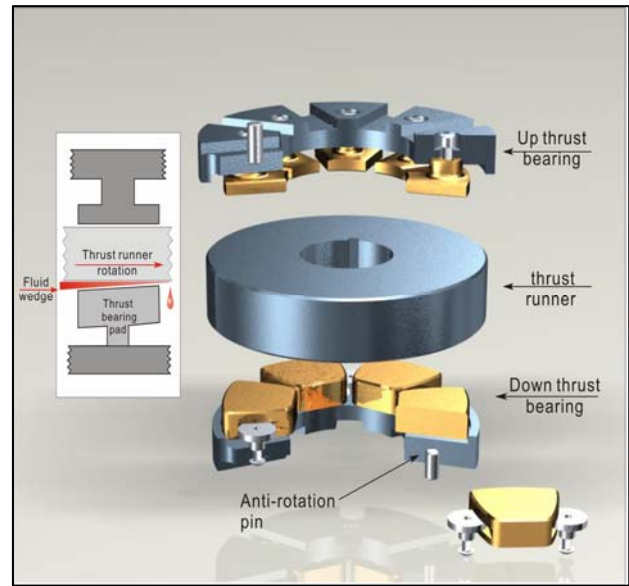


Figure 9 Typical thrust bearing

The seal section performs four basic functions:

- a) Transfers power from the motor to the intake / pump.
- b) houses a thrust bearing to absorb pump shaft axial thrust;
- c) isolates motor oil from well fluid while allowing wellbore-motor pressure equalization;
- d) acts as a reservoir for thermal expansion and contraction of motor oil due to operating heat rise and thermal contraction of the motor oil after shutdown.

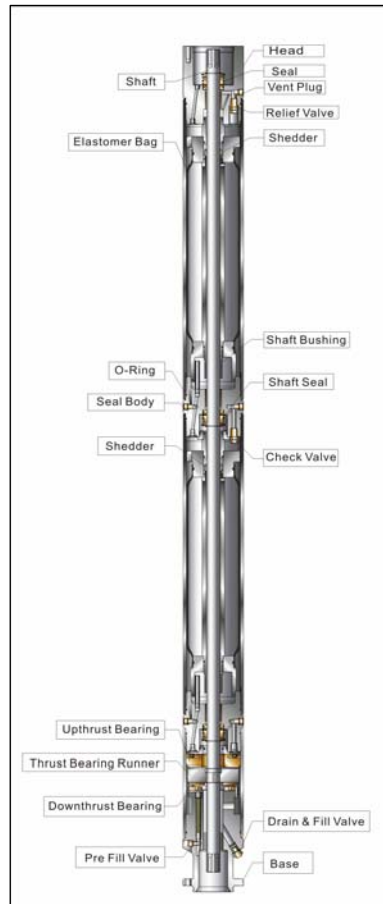


Figure 10 Labyrinth chamber motor seal

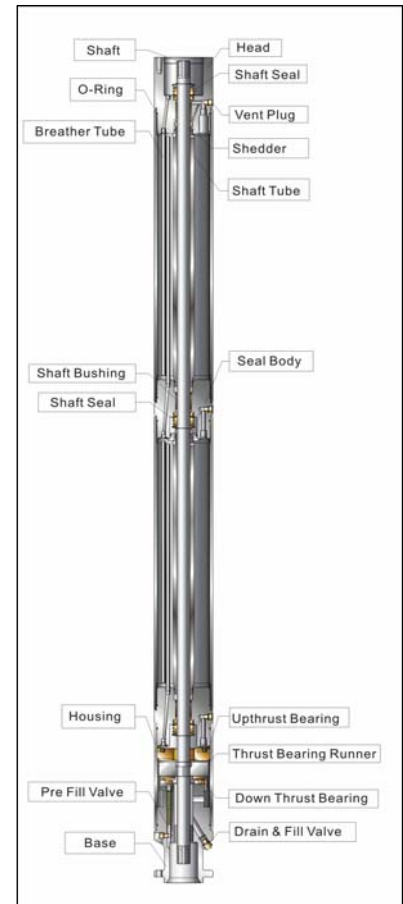


Figure 11 Bag chamber motor seal



3.4 Pump Intake / Gas Separator

There are two types of pump intakes: Standard and Dynamic Gas Separators.

Standard intakes (Figure 12) are used in wells that produce with a very low free gas or vapor to liquid ratio (VLR). In general amount of free gas by volume at pump intake conditions should be no more than 10% for a radial flow stage and 20% for a mixed flow stage. The standard intake has several fairly large ports, allowing fluids to flow into the lower section of the pump and enter the bottom stage in the pump. Most models are equipped with a screen to keep large debris out of the pump. The intake is bolted to the bottom of the pump. There are Tungsten Carbide bushings at the top and bottom of the intake to provide enhanced resistance to abrasive wear.

The vortex gas separator (Figure 13) will separate free gas with an efficiency of up to 90% under some conditions. Vortex gas separator should be used where the free gas available at the intake exceeds 10% with a radial flow stage and 20% with a mixed flow stage. Use of a vortex separator must be carefully considered. Even though the vortex gas separator is very efficient, there can still be cases where the pump will gas lock. Tandem gas separators are available for extreme applications; however there will still be applications where the VLR will be high enough that there will be gas interference or gas locking of the pump.

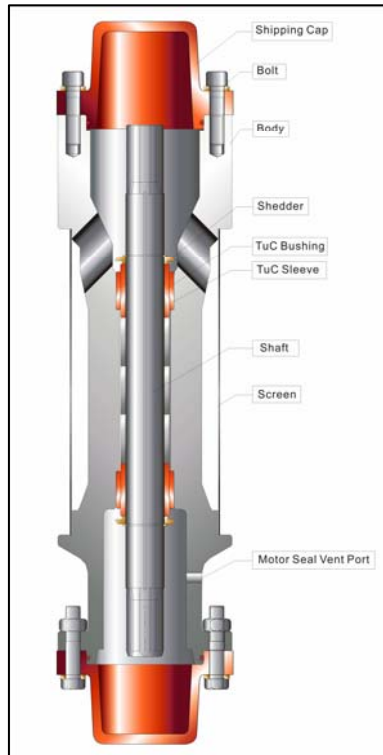


Figure 12 Standard intake

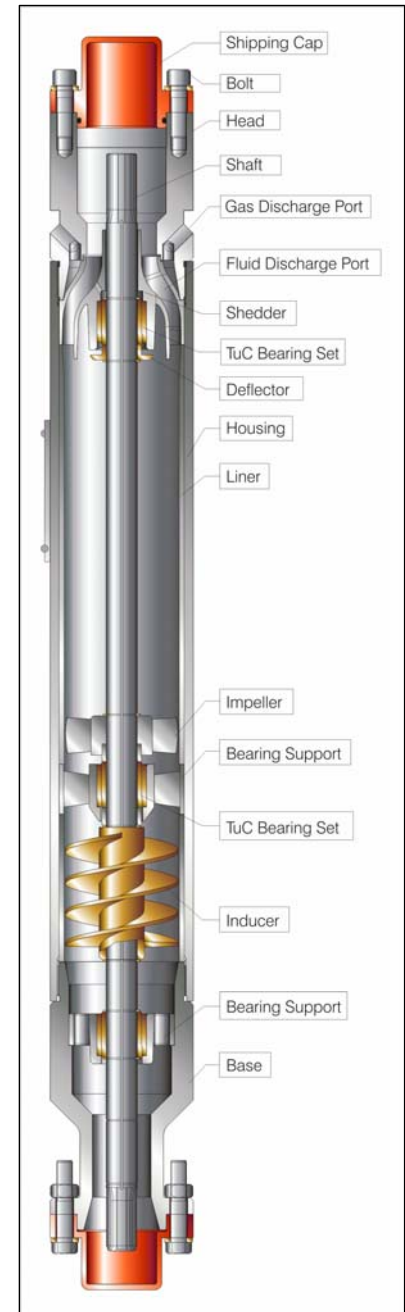


Figure 13 Vortex gas separator



3.5 Power Cable

Electric power is supplied to the downhole motor by a special submersible three-phase cable (Figure 14). There are two cable configurations, flat (parallel) and round. Round construction is typically used unless casing clearance requires the lower profile of flat construction. The standard range of conductor sizes is 1/0 to 6 AWG (American Wire Gauge). The conductor will be stranded or solid copper with a tin coating that reduces the potential of corrosion damage. A number of insulation types and layouts are available and selection is based on the well bore operating environment.

Mechanical protection is provided by armor made from galvanized steel. Stainless steel and Monel are available for corrosive environments.

Cable is constructed with three individual conductors, one for each power phase. Each conductor is enclosed by insulation and sheathing material. The thickness and composition of the insulation and sheathing determines the conductor's resistance to current leakage, its maximum temperature capability, and its resistance to permeation by well fluid and gas. Electric power cable is rated to operate at temperatures as high as 450°F (232°F) at 5,000 psi and 5 kV.

Chemical injection lines can also be incorporated into the Power Cable during manufacture.



Fig 14.- Power Cable

3.6 Motor Lead Extension

The motor lead extension (MLE) is the lowest section of the power cable string. The motor lead extension has a lower profile than standard flat power cable so that it can run the length of the pump, seal and intake sections in limited clearance situations. The length of the MLE is determined by the system length (discharge head + pumps + intake + motor seal + 2 feet). A minimum of 7 additional feet is required to allow for splicing of the MLE to the power cable. If high bottom hole temperatures, or extreme gas interference / intermittent operation is anticipated then consideration should be given to increasing the MLE length an additional 20 to 30 feet, thus moving the splice well above the DHE.

The motor lead extension (Figure 16) is manufactured with a pothead (termination and terminals) (Figure 17), designed to allow mate with the ESP motor while sealing the connection and motor from well fluid entry.



Figure 16 MLE

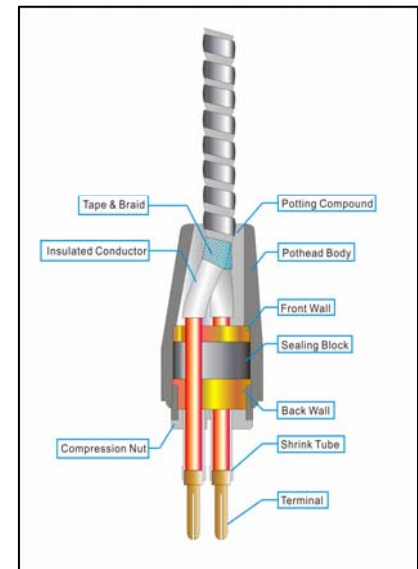


Figure 17 Pothead



3.7 Switchboard

The switchboard (Figure 18) is basically a motor control device. Voltage rating ranges from 600 to 5,000 volts. Typically the enclosures are NEMA 3R, which is suitable for virtually all outdoor applications.

Several models of motor controller are available for the switchboard. All motor controllers monitor motor current and the incoming power supply. Monitoring these parameters allow for protection of the ESP system from damage caused by conditions such as pump-off, gas lock, tubing leaks, power supply problems and shut-off operations. The higher end motor controllers allow for more elaborate protection from a much greater list of potential problems. Most motor controllers also incorporate data logging functions.

A valuable switchboard feature is the recording ammeter. Its function is to record, on a circular chart, the input amperage to the downhole motor. The ammeter chart record shows, whether the downhole unit is performing as designed or whether abnormal operating conditions exist. Abnormal conditions can occur when a well's inflow performance is not matched correctly with pump capability or when electric power is of poor quality. Abnormal conditions that are indicated on the ammeter chart record are primary line voltage fluctuations, low current, high current, and erratic current.



Figure 18 5 kV switchboard

3.8 Variable Frequency Drives

The variable-frequency drive (VFD) is a highly sophisticated switchboard-motor controller. The VFD performs three distinct functions. It varies the capacity of the ESP by varying the motor speed, protects downhole components from power transients, and provides "soft-start" capability. Each of these functions is discussed in more detail below.

A VFD changes the capacity of the ESP by varying the motor speed. By changing the power frequency supplied to the motor and thus motor RPM, the capacity of the pump is also changed in a linear relationship. Thus, well production can be optimized by balancing flow performance with pump performance. This applies to both long-range reservoir changes as well as short-term transients such as those associated with high-GOR wells. This may eliminate the need to change the capacity of a pump to match changing well conditions or it may mean improved run life by preventing cycling of the system. This capability is also useful in determining the productivity of new wells by allowing evaluation and measurement of pressure and production values over a range of drawdown rates. The change in frequency can be made manually or automatically. A VFD can automatically adjust the operating frequency to maintain a target pressure, flow rate, current or other set points when operating in a "closed loop" mode.

The VFD also protects the downhole motor from poor quality electricity power. VFDs are relatively insensitive to incoming power balance and regulation while providing closely regulated and balanced output. The VFD will not pass transients through to the downhole motor but it can be shut down or damaged by such transients. Given the choice, most operators prefer to repair surface installation equipment rather than pull and run downhole equipment. Within limits, the VFD upgrades poor-quality electric power by "rebuilding". The VFD takes a given frequency and voltage AC input, converts the AC to DC, and then converts the DC to an AC waveform at the desired frequency and voltage.

The soft-start capability of a VFD provides two major benefits. First, it reduces the startup drain on the power system. Second, the strain on the pump shaft (and its associated components) is significantly reduced when compared with that of a standard start. This capability is valuable in gassy or sandy wells. In some cases, slowly ramping the pump up to operating speed may reduce inflow of abrasives into the well bore thus reducing pump damage.



3.9 Other Elements and Accessories

The following is a partial listing of other elements and accessories usually installed with ESP Systems:

3.9.1 Transformers

The ESP system involves three different transformer configurations: single-phase transformers, three-phase dual wound (Figure 19) or three-phase autotransformers. Transformers generally are required because primary line voltage does not meet the downhole motor voltage requirement. Oil-immersed self-cooled (OISC) transformers are typically used. Dry type transformers are available for offshore applications where the operator excludes oil-filled transformers.

3.9.2 Wellhead

Two typical types of wellhead used by the industry are illustrated below. Based on local regulatory agencies, well characteristics, environmental factors and client standards flanged (Figure 20 – high pressure) and (Figure 21 – low pressure) wellheads are available. The wellhead provides a pressure tight pack-off around the tubing and power cable as well as suspending the tubing string.



Figure 19 3 Phase Transformer

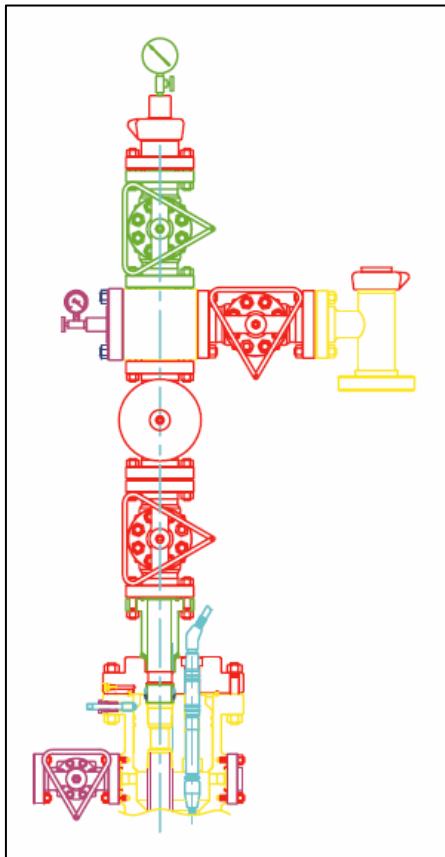


Figure 20 Flanged Wellhead with
Penetrator System

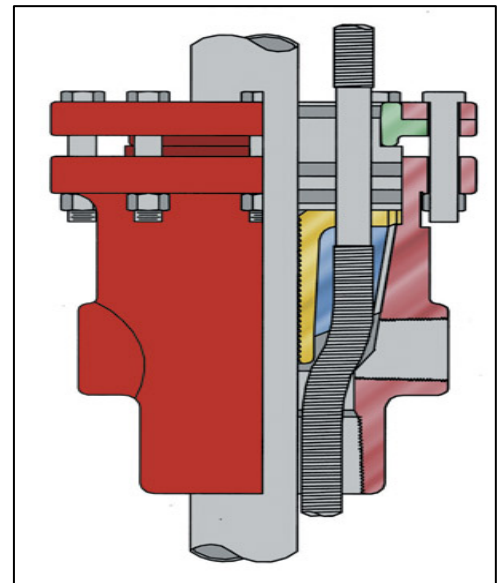


Figure 21 Low Pressure Wellhead



3.9.3 Junction Box

A junction box (Figure 22) connects the power cable from the switchboard/VFD to the well's power cable. The junction box is necessary to vent to the atmosphere of any gas that may migrate up the power cable from the well. This prevents accumulation of gas in the switchboard/VFD that could result in an explosive and unsafe operating condition. A junction box is required on all ESP installations that do not have a wellhead penetrator system. A junction box is recommended on all installations even when a wellhead penetrator system is in place

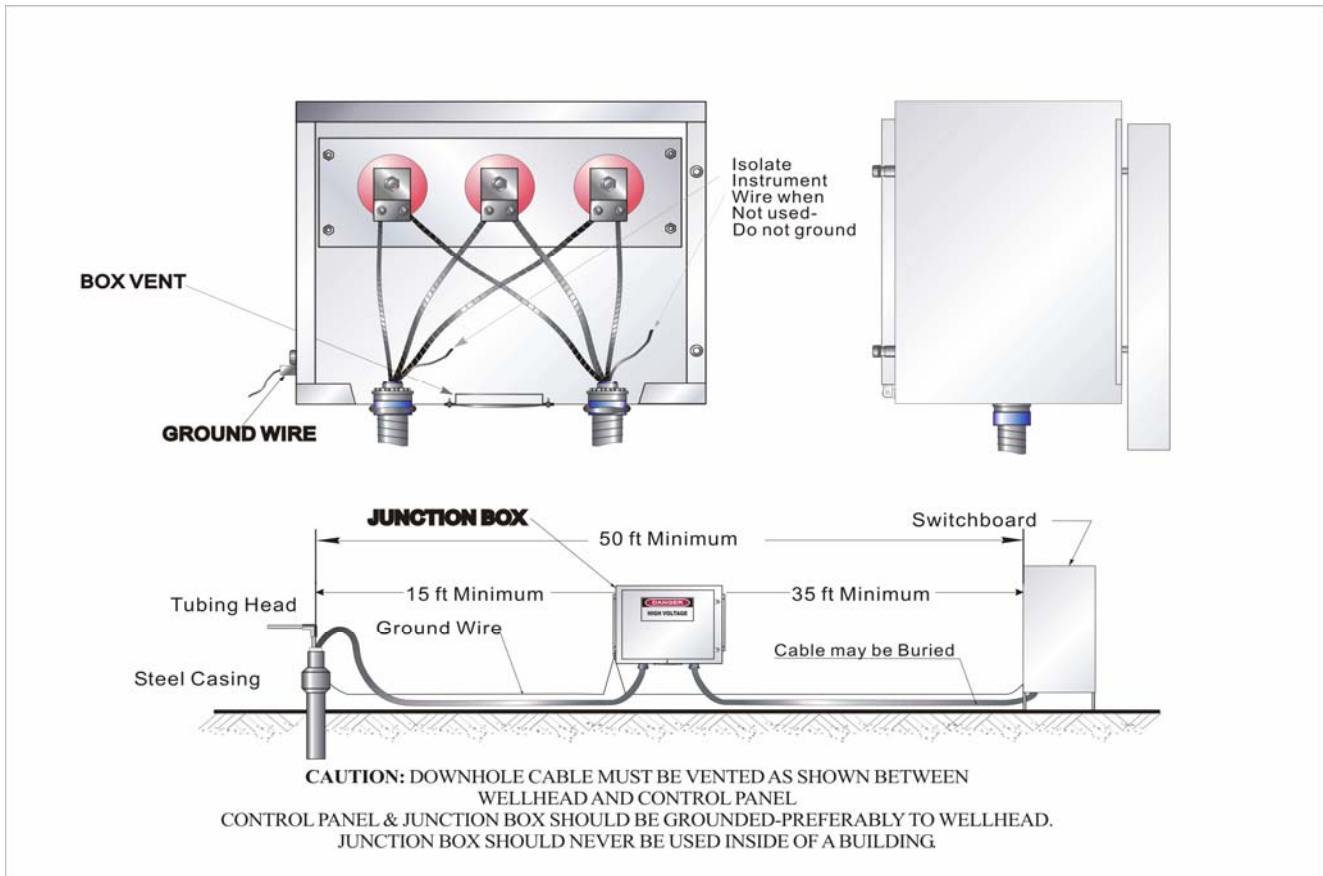


Figure 22 Recommended Junction
Box Layout



3.9.4 Downhole Monitoring System



Figure 23 DMS
and ESP System

The downhole monitoring system (DMS) (Figure 23) provides the operator with precise downhole pressure and temperature data. This instrument has two components: The downhole instrument and a surface readout unit. The downhole instrument (Figure 24) connects electrically and mechanically to the base of the motor. Data is transmitted to the surface readout (Figure 25) through the motor windings and the power cable on a DC carrier signal. The downhole instrument receives operating power from the motor's neutral point.

The primary function of the DMS is to assist in determining the producing potential of a well. This is accomplished by determining both static and dynamic reservoir pressures. By correlating the change in pressure with a given producing rate, a well's inflow performance can be accurately quantified. This in turn will allow equipment selection that optimizes well production for future installations.

NOTE: The DMS / ESP system illustrated in Figure 24 includes the optional pump discharge monitoring feature.

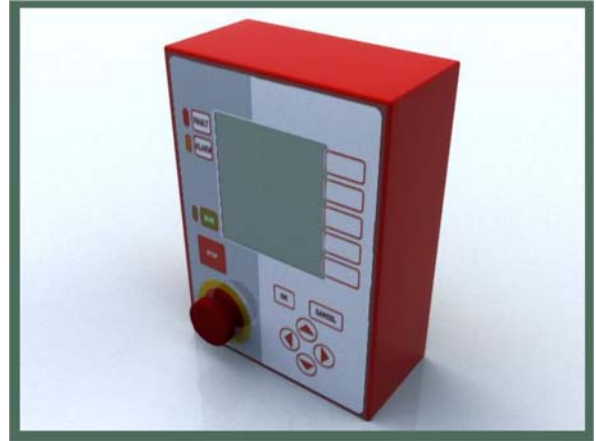


Figure 25 ALS
Controller



Figure 24 DMS
Instrument



4 Pump Performance Curves

The Pump Performance Curve is useful for understanding the operating range of an ESP. The curves in Figure 26 describe the performance of a particular impeller (or stage) type. All the manufacturers represent their pump performance with this type of curve. The left vertical axis is scaled in feet (and meters) of head (or lift). The bottom horizontal axis is scaled in BPD (and cubic meters per day). The curve labeled "Head Capacity" defines the lift (or head) the impeller can produce at all of the available flow rates. For example, at 2200 BPD the 1 stage 400-2200 in Figure 10 will produce 24.5 feet of lift (or head).

It should be noted that centrifugal pumps are measured by the head they produce, not the pressure. The 25.0 feet of lift in the example above represents 10.82 psi for a specific gravity of 1.00 fluid. However, the impeller will produce the same 25.0 feet of lift with a specific gravity 0.85 fluid with an associated pressure of 9.2 psi. This occurs because the centrifugal forces acting on the fluid are the same regardless of the fluid's density.

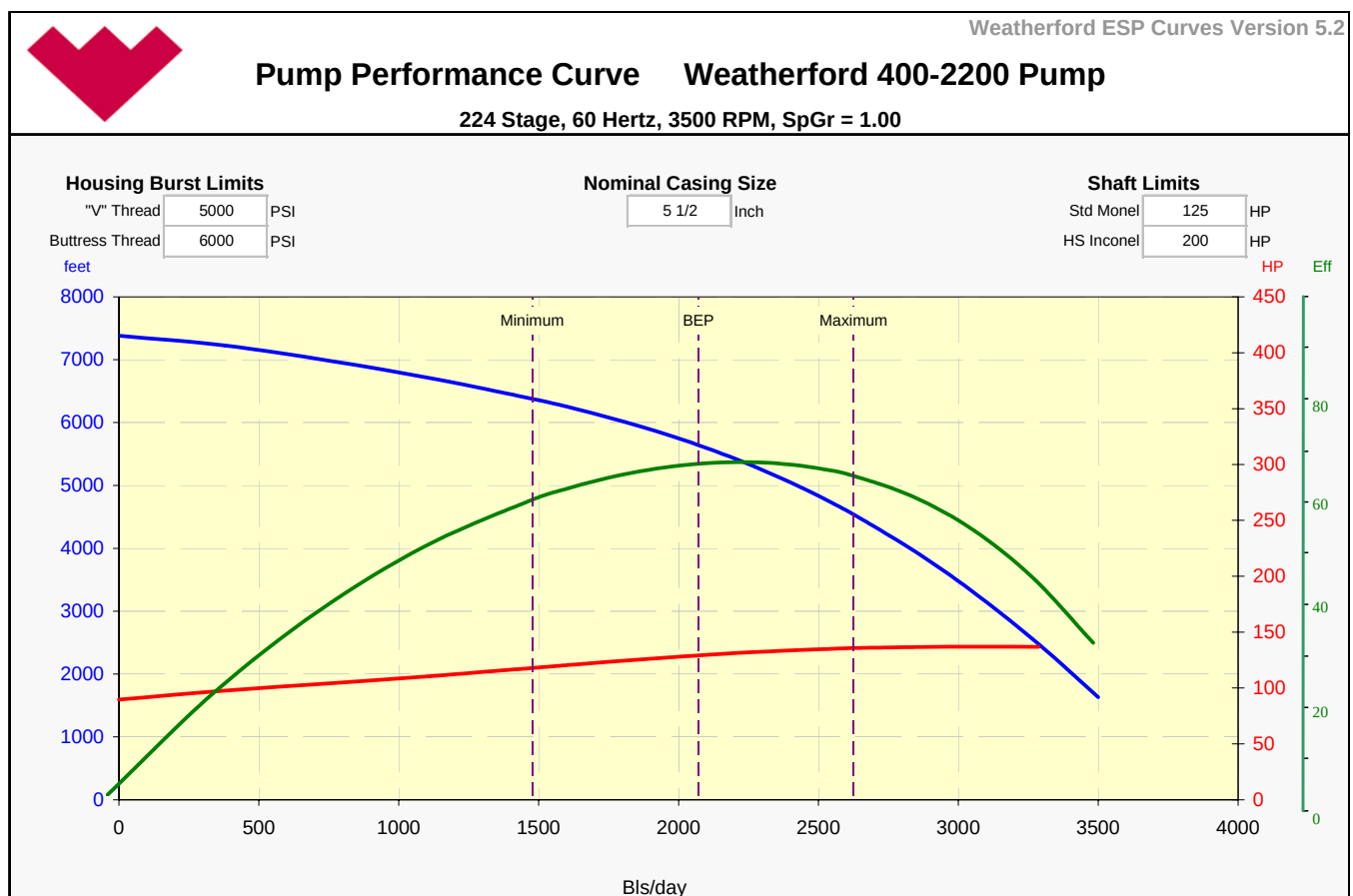


Figure 26 Typical Pump Performance Curve



Density does affect the horsepower required to lift the fluid. The curve in Figure 26 labeled “Horsepower Motor Load” defines the horsepower requirements for this stage at different flow rates. The first vertical axis on the right is scaled in horsepower motor load. This horsepower is based on pumping specific gravity 1.00 water. As an example, at 2200 BPD the 1 stage pump in figure 26 will require 0.58 HP if the fluid is specific gravity 1.00. For a specific gravity 0.85 fluid the pump will only require 0.49 HP.

The output horsepower (or hydraulic horsepower) the pump develops can be calculated from the head capacity curve at any flow rate. The input horsepower (or brake horsepower, BHP) can be determined from the horsepower motor load curve at any flow rate by dividing the output horsepower by the input horsepower at every BPD across the curve. “Pump Only Efficiency” curve can be developed as follows:

$$HHP = \frac{2200 \text{ BPD} \times 25 \text{ ft} \times 1.00}{136,000} = 0.4044 \text{ hp}$$

$$BHP = 0.58 \text{ hp}$$

$$\text{Efficiency} = \frac{0.404 \text{ hp}}{0.58 \text{ hp}} \times 100$$

$$\text{Efficiency} = 69.6\%$$

The far right vertical axis of Figure 26 is scaled in percent efficiency. Sometimes the curves will not match exactly with the calculation due to errors in reading and reproducing the curves. Because of this, API¹ and the industry have established that mathematical coefficients should be used to determine an impeller's head, horsepower and efficiency.

The curve will usually be for a single-stage pump but sometimes the curve will be on a 100-stage basis. In the example above, if head was at 2200 BPD of a 100-stage curve we would read 2500 feet. The curves are also RPM dependent and the RPM for the curve will be listed. Changing the RPM of the impeller will affect the head and horsepower curves according to the pump and affinity laws (see section 7.1).

Every centrifugal stage is designed to produce at a certain flow rate. There is a best efficient point (BEP) for each stage design. Every impeller type has a recommended range. In Figure 26 the recommended operating range (ROR) is the darker zone (labeled “*Recommended Operating Range*”). For the example stage 400-2200 this range is 1550 BPD to 2650 BPD. Operation of the pump outside of the ROR must be carefully reviewed on a case by case basis. The primary item that must be evaluated is pump thrust versus thrust load capacity of the impeller thrust washer loading, however abrasives, gas and temperature may also need to be considered in extreme applications. Typically the stage will operate in down thrust, but within the load capacity of the thrust washers. As the flow rate decreases / head increases the amount of down thrust increases and as the rate increases / head decreases the stage will move from down thrust to up thrust. Loading of the pump thrust washers beyond 100% capacity will impact the operation life of the pump.

¹ API RP11S2



5 ESP System Design

5.1 Data Required

Designing an efficient ESP is not a complicated task, but reliable and accurate information must be available for the calculation process in order to select the appropriate equipment...

The data requirements for selection of an ESP are categorized as mechanical data, production data, fluid data, and power supply.

5.1.1 Mechanical Data

- Casing size and weight
- Tubing size, weight and thread
- Well depth (both measured and true vertical)
- Perforation depth (both measured and true vertical)
- Unusual conditions such as tight spots, doglegs, liners and deviation from true vertical at desired setting depth.
- Well bore survey if the well is deviated or directional.

The casing size and weight determines the maximum diameter of the motor, pump, and seal section that will fit in the well. In general, the most efficient installation is obtained when the largest possible diameter pump in the target flow range is selected.

The depth of the well and the perforations determine the maximum setting depth of the ESP. If the motor is to be set below the perforations, a motor shroud must be used to provide a flow of well fluid past the motor for cooling.

5.1.2 Production Data

- Current and desired production rate
- Oil production rate
- Water production rate
- GOR, free gas, solution gas, and gas bubble point
- Static BHP and fluid level
- Producing BHP and stabilized fluid level
- Bottom hole temperature
- System backpressure from flow lines, separator, and wellhead choke

The inflow performance of a well establishes the maximum economical and efficient rate at which it can be produced. Liquid-level data may be used as a substitute for producing pressures and rates in water wells or in low-oil-cut wells with no gas. In these cases, a straight line PI may be used as reasonable approximation of well capacity.

Most oil wells do not exhibit a straight-line PI due to interference caused by gas. The Vogel technique yields a downward-sloping curve that corrects for gas interference. The IPR curve applies when wellbore pressure in the producing zone drops below the bubble point, which results in two-phase flow as the gas breaks out of the fluid. Again, the data obtained for this approach in sizing an ESP must be both accurate and reliable to ensure proper equipment selection.



5.1.3 Fluid Data

- Oil API gravity, viscosity, pour point, paraffin content, sand, and emulsion tendency
- Water specific gravity, chemical content, corrosion potential, and scale-forming tendency
- Gas specific gravity, chemical content, and corrosion potential
- Reservoir FVF, bubble point pressure, and viscosity/temperature curve.

The specific gravity of the produced fluid has a direct impact on the horsepower required to operate a given size pump. Although relative few applications encounter fluid viscosities high enough to influence pump performance, it is important to be aware that capacity, head, and horsepower correction factors may be required. In wells with water cut of 65% or higher, the fluid will not require viscosity correction factors (except for emulsions). The PVT data are required when gas is present in order to have an accurate calculation of free gas volume at pump intake conditions.

5.1.4 Power Supply

- Primary grid voltage
- Primary grid frequency
- Capacity of the service
- Quality of service (spikes, sags, etc.)
- Power supply source (commercial grid, on site generator, shared generator, operator owned grid etc.)
- Any special requirements such as high ambient temperatures, hazardous locations etc.

The power system data is very important as it factors into transformer, switchboard, VFD sizing as well as other design considerations.

5.2 Determining Reservoir Inflow Capacity (Productivity Index)

The reservoir inflow capacity will be governed by the IPR (Inflow Performance Relationship) curve. This curve shows the flow rate associated with each bottom hole flowing pressure for a specific reservoir condition. Depending on how stable the reservoir static pressure (P_{ws}) is, this information could be valid for an extended period of time (if PI is high) or only for current well condition (if PI is low). The most common method used to calculate this IPR curve is the Straight Line method (if Flowing pressure, P_{wf} , is higher than the bubble point pressure, P_b) and the Vogel method (if P_{wf} is lower than P_b). Figure 27, shows a typical IPR curve.

If the bottom hole flowing pressure (P_{wf}) is higher than or equals to the bubble point pressure (P_b), no free gas is present at the reservoir so compressibility of the liquid is insignificant. Under this assumption, a straight line (or constant Productivity Index, PI) behavior could be considered for the relation between P_{wf} and Flow Rate (Q):

$$PI = \frac{Q}{P_{ws} - P_{wf}}$$

If P_{wf} is lower than P_b , free gas will be liberated from the solution. This means that PI will decrease while the pressure decreases. Under these conditions, the method of Vogel is one of the most appropriate procedures to establish the relationship between P_{wf} and Q . Following, the equation:

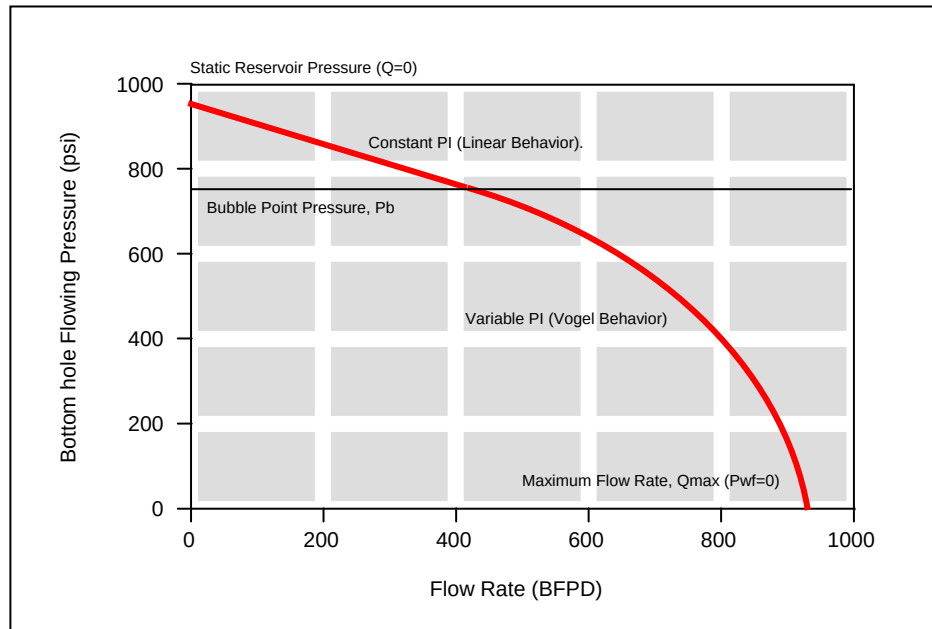
$$\frac{Q}{Q_{\max}} = 1 - 0.2 \cdot \left(\frac{P_{wf}}{P_{ws}} \right) - 0.8 \cdot \left(\frac{P_{wf}}{P_{ws}} \right)^2$$

Figure 26 is a graphic example of the Vogel method.

$Q_{\max}$ will be the maximum flow rate that reservoir can produce when P_{wf} is equal to zero (maximum drawdown).



5.3 Example Vogel Calculation



Desired Flow =
2000 BPD

Q_{max} = 2332
BPD

Static (P_r)= 2500
psi

Fig. 27.- Typical Reservoir Inflow Performance Relationship (IPR) Curve.

$$P_{wf} = 0.125 * P_r \left(-1 + \sqrt{81 - 80 \left(Q_o / Q_o(\max) \right)} \right)$$

$$P_{wf} = 0.125 * 2500 \left(-1 + \sqrt{81 - 80 \left(2000 / 2332 \right)} \right)$$

$$P_{wf} = 787.4541 \text{ psi } P_{wf}$$

5.4 Determining Fluid Properties at Pumping Condition

Pressure and temperature conditions vary depending on specific production conditions and the mechanical configuration of the well. Due to these changes, produced fluid properties also change affecting not only their physical characteristics but also their relative volumes. The relationship between Pressure, Volume and Temperature is known as PVT properties of fluids. The best way to attain these properties is with laboratory analysis. Another more common method is with PVT correlations such as Standing, Vasquez & Beggs, Lasater, etc.

Determining fluid properties as fluid specific gravity and viscosity at pump intake conditions is very important because they have a large influence on the pump performance curve. In previous sections, the effect of specific gravity on the head capacity and horsepower requirement was explained. Viscosity has a different effect on the pump, increasing the horsepower requirement (up to 2.5 times) and reducing displacement (up to 40%) and head capacities (up to 30%), as it increases. Therefore, knowledge of these two parameters is extremely important to select the correct system.

PVT properties will also help to determine the equivalent volumes of oil, gas, and water produced by the well at pump intake conditions. The next section will explain a detailed calculation procedure to determine such volumes.

5.5 Determining Total Fluid Volume at Pump Intake Conditions



5.5.1 Oil Volume at Pump Intake

Calculation of gas solubility, R_s :

- PIP = Pump Intake Pressure (psi)
- γ_g = Gas Specific Gravity (dimensionless)
- T = Temperature at Pump Intake (°F)
- API = Oil Density (°API)

$$R_s = \gamma_g \left[\frac{PIP + 14.7}{18 \times 10^{(0.0009(T+460) - 0.0125API)}} \right]^{1.204}$$

Calculation of oil volumetric factor, B_o :

where:

- γ_o = Oil Specific Gravity (dimensionless)

$$B_o = 0.972 + 0.000147 \left(R_s \left(\frac{\gamma_g}{\gamma_o} \right)^{0.5} + 1.25T \right)^{1.175}$$

$$\gamma_o = \frac{141.5}{131.5 + API}$$

Calculation of oil volume at intake, V_o :

$$V_o = Q_o \times B_o$$

5.5.2 Water Volume at Pump Intake

Volume of water at pump intake conditions (V_w) can be assumed as equal to the water flow rate at stock conditions (Q_w) because its relative insignificant compressibility. In addition thermal expansion of the fluid is normally ignored.

5.5.2.1 Free Gas Volume at Pump Intake

Calculation of gas compressibility factor, z :

$$z = A + B \cdot Pr + (1 - A) \cdot e^{-C} - H \cdot \left(\frac{Pr}{10} \right)^4$$



where:

$$\begin{aligned}Tr &= (T + 460) / (175 + 307\gamma_g) \\Pr &= Pa / (701 - 47\gamma_g) \\A &= -0.101 - 0.36Tr + 1.3868(Tr - 0.919)^{0.5} \\B &= 0.021 + 0.0425 / (Tr - 0.65) \\C &= Pr(D + E \cdot Pr + F \cdot Pr^4) \\D &= 0.6222 - 0.224Tr \\E &= 0.0657 / (Tr - 0.86) - 0.037 \\F &= 0.32e^{(-19.53(Tr-1))} \\H &= 0.122e^{(-11.3(Tr-1))}\end{aligned}$$

NOTE: The Z factor is calculated for each application, it is not a constant. The above example is applicable only to this example.

Calculation of gas volumetric factor, Bg:

$$Bg = \frac{0.0283 \cdot z \cdot (T + 460)}{PIP + 14.7}$$

Calculation of free gas volume at intake, Vg:

$$Vg = 0.17811 \cdot Qo \cdot (GOR - Rs) \cdot Bg$$

where:

- Qo: = Oil Flow Rate, stock conditions(BPD)
- GOR: = Produced Gas-Oil Relationship (scf/sbl)

Calculation of total volume at intake, Vt:

$$Vt = Vo + Vw + Vg$$

Calculation of free gas content, Fg:

$$Fg = \frac{Vg}{Vt}$$

If Fg is higher than 10% with radial-flow impeller pumps or 20% with mixed-flow impeller pumps, the use of a gas separator is recommended in order to minimize gas interference at the pump.

5.6 Determining Total Dynamic Head (TDH)

The Total Dynamic Head could be defined as the differential pressure (or energy) that the pump must supply to get the desired flow rate to the surface facility. This differential pressure is defined by the pump discharge pressure (function of surface pressure, flow losses through tubing string, and weight of liquid column inside the tubing) and the pump intake pressure (function of the reservoir inflow performance). The best way to estimate the discharge pressure is by using multi-phase flow correlations that consider elevation, acceleration and friction forces. The intake pressure could be calculated as a static column above the perforations, using as reference the bottom hole flowing pressure corresponding to a specific flow rate.



The following is a simplified calculation procedure that assumes a single-phase flow pattern into the tubing string. This single-phase fluid will be a liquid; properties are equal to the average properties of current produced fluids (water, oil, and gas).

1. Calculation of fluid specific gravity, γ_f :

$$\gamma_f = \frac{(\gamma_o \cdot V_o) + (\gamma_w \cdot V_w) + (\gamma_g \cdot V_g)}{V_t}$$

2. Calculation of net suction head, H_s (in feet):

$$H_s = \frac{PIP}{(0.433 \cdot \gamma_f)}$$

3. Calculation of equivalent vertical head, H_d (in feet):

$$H_d = H_{sd} - H_s$$

where: H_{sd} = Pump Seating Vertical Depth (feet)

4. Calculation of surface back-head, P_d (in feet):

$$P_d = \frac{P_{surface}}{(0.433 \cdot \gamma_f)}$$

5. The friction losses in the tubing string (F_t) could be estimated using the Hazen-Williams correlation, which is shown graphically on Table 13 of the "Engineering Tables" section of this manual (formula is also shown in this section).

6. Calculation of Total Dynamic Head, TDH (in feet):

$$TDH = H_d + P_d + F_t$$

5.7 Selection of Pump, Motor and Seal Section

Typically the pump with the largest OD that can be run in the casing is the optimum pump series for the well. The pump must have the target capacity (V_t) within its recommended operating range and preferably close to its Best Efficient Point (BEP).

- Remember to allow for the MLE and cable guards when calculating pump to casing clearance.
- Remember to take into consideration the VLR. In some cases a mixed flow stage in a smaller pump series is preferable to a radial slow stage in a larger pump series.

The individual pump curve should then be reviewed to determine the optimal producing range and the proximity of the design-producing rate to the pump's BEP (section 4 shows the pump performance curve basics). It is very important to choose a producing rate that is in the recommended capacity range of the specific pump.

Once the pump is chosen, the number of stages (N_{stages}) required can be calculated using the head per stage (H_{stage}) reading from the pump performance curve, as follows:

$$N_{stages} = \frac{TDH}{H_{stage}}$$



The horsepower required by the pump design can then be calculated. To accomplish this, the horsepower required per stage is read from the specific pump performance curve. The required motor horsepower (BHP_{motor}) is determined by multiplying the horsepower required per stage (BHP_{stage}) by the number of design stages (N_{stages}). The performance curve horsepower data apply only to specific gravity 1.0 fluids. For other fluids (other specific gravities), the water horsepower also must be multiplied by the specific gravity of the fluid pumped (γ_f). Thus, the following equation for the motor horsepower calculation:

$$BHP_{motor} = BHP_{stage} \cdot N_{stages} \cdot \gamma_f$$

Once the design motor horsepower is determined, specific motor selection is based on setting depth, casing size, and motor voltage. Although the cost of the motor is generally unrelated to voltage, overall ESP system cost may be reduced by using higher-voltage motors in deep applications. This lower cost will sometimes occur because a higher voltage / lower amperage motor may lower the cable conductor size required. A smaller conductor size, lower-cost cable may more than offset the increased cost of a higher-voltage switchboard. Setting depth is a major consideration in motor selection because of starting and voltage drop losses that are a function of the motor amperage and cable conductor size.

The seal section selection variables are: pump and motor series (sizes), motor horsepower, well temperature and fluid properties. Normally the seal section is the same series as the pump and motor. Large horsepower motors may require multiple sections to accommodate the motor fluid expansion and contraction. Well bore trajectory and produced fluid properties will influence the type of chambers selected. Temperature and produced fluid properties will influence the elastomers selection.

Finally, in order to ensure the appropriate selection of pump and motor, the following checks must be made:

5.8 Equipment Checks

5.8.1 Pump, Motor, Seal Section and Power Cable to Casing Clearance:

Check for outside diameter of these elements and confirm that they can be run into the specific casing size. Remember to allow for the MLE when checking clearance on the pump, intake and motor seal.

5.8.2 Pump Housing Limit:

Pumps are typically available with two different types of housing thread. Maximum pressure (worst case scenario) to be contained by the housing would be operating at zero flow (surface valves closed) and the annulus fluid level drawn down to the pump intake. This is also called "Shut Off Head" by some users. To calculate this value, read the pump head at 0 BPD from the pump performance curve (where the pump head curve crosses the left-vertical axis) and multiply it by the number of stages. Then, find the equivalent pressure to this maximum head and check the limit of each type of housing provided for that specific pump model. See Table 10 for details.

5.8.3 Pump, Intake, Seal Section and Motor Shaft Limits:

Check for horsepower limits of pump, motor and seal section shafts in order to determine if a standard or high strength material must be used. See Tables 4, 5 and 6 for details

5.8.4 Seal Thrust Bearing Capacity:

Check for maximum axial load that thrust bearing can support. For floater pumps multiply the differential pressure through the pump by the shaft cross-sectional area, axial load on the shaft is obtained. Hydraulic data specific to each stage is required to calculate pump thrust generated for compression pumps.

5.8.5 Motor Heat Rise:

In order to guarantee enough fluid cooling capacity, the recommended minimum fluid velocity passing the motor is 1 ft/sec. Knowing the flow rate, casing size and motor series the "Fluid Velocity Table" in the ESP Product Catalog is used to determine this value. Software modeling is required to fully evaluate motor heat rise, especially on well with high BHT, low flow rates or high oil cuts. See Table 12 for details.

5.8.6 Selection of Downhole Power Cable

Selection of proper type and size of downhole power cable will depend on a number of factors.

- bottom hole temperature



- motor operating current
- casing and tubing sizes
- pump setting depth
- well fluid and environment (presence of H₂S, CO₂, free gas, treating chemicals, etc.)
- power cost considerations.

The cable type, configuration, construction and conductor size are selected based on environmental conditions, ambient temperature, motor current / voltage and fluid composition.

Once the cable conductor size is selected, the "Cable Voltage Drop Graph" (Table 15) is used to determine the voltage drop. Using motor nameplate the voltage drop per 1000 feet can be read per each size of cable. Industry practice is to limit cable voltage drop to a maximum of 30 Volts/1000 feet. If voltage drop is higher than such limit, a larger size cable should be selected.

NOTE: Also, note that the "Cable Voltage Drop Graph" is based on a conductor operating temperature of 77°F. In order to correct such temperature to the ambient bottom hole condition, the value obtained from the Graph must be corrected based on the read value in Table 15 must be multiplied by the correction factor read on Table 16. Again, results should not exceed 30 Volts/1000 feet.

NOTE: Cable Ampacity is based on conductor operating temperature and not wellbore temperature. Charts are available in the ESP Product Catalog that define ampacity versus cable operating temperature for each cable type. Computer software is used to calculate actual conductor temperature based on the projected operating conditions.

Verify that voltage at the motor terminals during start-up conditions is adequate to start the unit. NOTE: current draw during startup is typically 5X motor name plate current for a period of less than 1 second. Use motor name plate current X 5 for this calculation. Calculate motor terminal voltage in the following manner. Using Table 15 obtain the voltage loss per 1000 feet of cable. Now multiply this value the cable length from the switchboard / VFD to the motor terminals. Now multiply this value X 5. The result is the voltage loss for the planned system at start. If the voltage loss is greater than 40% (voltage available is less than 60% of motor nameplate) of the motor nameplate current then the system design must be reviewed and modified. A larger conductor cable or higher a voltage motor may be a better choice for this application. Exercise caution when designing a unit for operation on a dedicated generator. A careful review of these applications is needed to insure the generator is capable of not only operating, but also starting the ESP.

5.9 Selection of Switchboard

All applications, except where Variable Speed Drives are used, will require a surface switchboard or control panel. Switchboard selection will be based on voltage and current requirements. The surface voltage will be the result of adding motor nameplate voltage plus cable voltage losses. Amperage will be equal to motor nameplate current.

Switchboards are available in 600, 1500, 3600, and 5000 volt rating. 600 volt panels are available with several different current ratings. The 1500, 3600 and 5000 volt panels are available only in a 200 amp rating.

5.10 Selection of Transformers

Distribution of service transformer are electrically rated by input/output voltage and KVA (KVA is the abbreviation for Kilo-Volts-Amperes or thousand of Volts-Amperes, a measure of apparent power). The minimum required total transformer KVA rating can be found using the following formula for three phase operation:

$$KVA = \frac{V_{surface} \cdot I_{mn} \cdot \sqrt{3}}{1000}$$

where:

- $V_{surface}$:- Required voltage at surface
- I_{mn} :- Motor nameplate full-load current



When using a single auto-transformer or three-phase transformer, the calculated KVA value must not exceed the transformer's rating. Three single-phase transformers have a total KVA rating of three times their individual rating.

Transformer sizing for normal installations is relatively straight forward. However care is needed to insure the correct transformer is supplied. To follow is a check list of those items.

- Confirm the client does not have any special / unique ambient temperature requirements. As an example transformers supplied to the Middle East must be designed for the high ambient temperatures of the region.
- Confirm what distribution voltage is supplied to location. There are many industry standard distribution voltages possible.
- Are there any special requirements for non standard insulation oil? Typically only an issue with offshore installations.
- Are there any special requirements for controls, instrumentation or remote monitoring?
- Is there any special requirement for non standard terminals, terminations or bushing chambers etc.?
- What range of secondary voltages are required?



6 Example of ESP Equipment Design with Fixed Speed

6.1 Well Bore and Reservoir Information:

Mechanical data:	Well Total Depth (H_w):	7500 feet (Vertical Well)
	Pump Seating Depth (H_{sd}):	7000 feet
	Perforations Depth (H_{perms}):	7250 feet
	Casing Size and Weight:	5-1/2" 17.0 lb/ft
	Tubing Size and Weight	2-7/8" 6.5 lb/ft
Production data:	Test Flow Rate (Q_1):	900 BPD
	Wellhead Tubing Pressure ($P_{surface}$):	120 psi
	Test Bottom hole Flowing Pressure (P_{wf1}):	1900 psi
	Reservoir Static Pressure (P_{ws}):	2500 psi
	Bottom hole Temperature (BHT):	180 °F
	Gas-Oil Ratio (GOR):	150 scf/bl
	Water Cut (WC):	65%
	Desired Production Rate (Q_2):	2000 BPD
Fluid Data:	Specific Gravity of Water (γ_w):	1.05
	Gravity of Oil (API): 30 °API (γ_o)	0.876
	Specific Gravity of Gas (γ_g)	0.7
	Bubble Point Pressure (P_b):	2500 psi
	Viscosity of Oil (μ_o)	10 cp
Power Supply:	Available Primary Voltage ($V_{primary}$):	7200/12470Y
	Supplied Frequency (F):	60 Hertz

6.2 Reservoir Inflow Capacity:

Reviewing the data, we have a reservoir flowing below the bubble-point pressure so the method of Vogel should be used to determine bottom hole flowing pressure for the desired flow rate of 2000 BFPD:

First, we estimate reservoir maximum flow rate using the flowing data we have. (maximum drawdown condition):

$$Q_{max} = \frac{Q}{1 - 0.2 \left(\frac{P_{wf}}{P_{ws}} \right) - 0.8 \left(\frac{P_{wf}}{P_{ws}} \right)^2} = \frac{900 bpd}{1 - 0.2 \left(\frac{1900 \text{ psi}}{2500 \text{ psi}} \right) - 0.8 \left(\frac{1900 \text{ psi}}{2500 \text{ psi}} \right)^2}$$

$$Q_{max} = 2332 bpd$$

Now, knowing Q_{max} , we calculate flow rate for different bottom hole pressures in order to build the IPR Curve:

Pwf (psi)	0	250	500	750	1000	1250	1500	1750	2000	2250	2500
Q (BPD)	2332	2267	2164	2024	1847	1632	1381	1091	765	401	0



Interpolating in above table, we get the corresponding P_{wf} for the desired condition:

$$P_{wf} = 786 \text{ psi}$$

6.3 Pump Intake Pressure:

To determine Pump Intake Pressure (PIP) first we must determine the liquid specific gravity below the pump:

$$\gamma_l = \left(\frac{100 - WC}{100} \cdot \gamma_o \right) + \left(\frac{WC}{100} \cdot \gamma_w \right) = \left(\frac{100 - 65}{100} \cdot 0.876 \right) + \left(\frac{65}{100} \cdot 1.05 \right)$$

$$\gamma_l = 0.989$$

Now, we can determine PIP :

$$PIP = P_{wf} - 0.433 \cdot \gamma_l \cdot (H_{perfs} - H_{sd}) = 786 \text{ psi} - 0.433 \cdot 0.989 \cdot (7250 \text{ ft} - 7000 \text{ ft})$$

$$PIP = 679 \text{ psi}$$

Total Fluid Volume at Pump Intake Conditions:

Determine Gas Solubility, Rs:

$$Rs = 0.7 \left[\frac{679 \text{ psi} + 14.7}{18 \times 10^{(0.00091 \cdot (180^\circ F + 460) - 0.0125 \cdot 30^\circ \text{ API})}} \right]^{1.204}$$

$$Rs = 32 \text{ scf/sbl}$$

Determine Oil Volumetric Factor, Bo:

$$Bo = 0.972 + 0.000147 \cdot \left(32 \text{ scf/sbl} \cdot \left(\frac{0.7}{0.876} \right)^{0.5} + 1.25 \cdot 180^\circ F \right)^{1.175}$$

$$Bo = 1.0702 \text{ bl/sbl}$$

Determine Oil Volume at Pump Intake Conditions, Vo:

$$Vo = \left(2000 \text{ bpd} \cdot \left(\frac{100 - 65}{100} \right) \right) \cdot 1.0702 \text{ bl/sbl}$$

$$Vo = 749 \text{ bpd}$$

Determine Water Volume at Pump Intake Conditions, Vw:

$$Vw = Qw$$

$$Vw = 1300 \text{ bpd}$$

Determine Gas Compressibility Factor, z:



$$Tr = 1.6414$$

$$Pr = 1.0383$$

$$A = 0.4868$$

$$B = 0.0641$$

$$C = 0.3150$$

$$D = 0.2545$$

$$E = 0.0471$$

$$F = 1.1602 \times 10^{-6}$$

$$H = 8.6785 \times 10^{-5}$$

$$z = 0.4868 + 0.0641 \cdot 1.0383 + (1 - 0.4868) \cdot e^{-0.3150} - 8.6785 \times 10^{-5} \cdot \left(\frac{1.0383}{10} \right)^4$$

$$z = 0.928$$

Determine Gas Volumetric Factor, Bg:

$$Bg = \frac{0.0283 \cdot 0.928 \cdot (180^\circ F + 460)}{679 \text{ psi} + 14.7}$$

$$Bg = 0.0242 \text{ cf/scf}$$

Determine Free Gas Volume at Intake, Vg:

$$Vg = 0.17811 \cdot 700 \text{ bpd} \cdot (150 \text{ scf/sbl} - 32 \text{ scf/sbl}) \cdot 0.0242 \text{ cf/scf}$$

$$Vg = 356 \text{ bpd}$$

This volume of free gas corresponds to the total free gas produced at surface calculated at pump intake conditions. Experience indicates that for free tubing well configuration (standard for ESP Systems) there is an average natural separation of 35% which means that only 65% of Vg will be handle by the pump. So:

$$Vg_{\text{pump}} = 231 \text{ bpd}$$

Determine Total Volume at Intake, Vt:

$$Vt = 749 \text{ bpd} + 1300 \text{ bpd} + 231 \text{ bpd}$$

$$Vt = 2280 \text{ bpd}$$

Determine Free Gas Content at Intake, %Gas:

$$\%Gas = \frac{231 \text{ bpd}}{2280 \text{ bpd}} \times 100$$

$$\%Gas = 10.1\%$$



6.4 Total Dynamic Head:

Determine Average Fluid Specific Gravity at the tubing, γ_f :

$$\gamma_f = \frac{(0.876 \cdot 749 \text{ bpd}) + (1.05 \cdot 1300 \text{ bpd}) + \left(0.7 \cdot \frac{28.6}{64.2} \cdot 231\right)}{2280 \text{ bpd}}$$

$$\gamma_f = 0.918$$

The factor (28.6/64.2) is just the conversion factor from gas density to water density in order to work with same relative specific gravities for liquids and gases.

Determine Pumping Fluid Level, L_p :

$$L_p = \frac{679 \text{ psi}}{(0.433 \text{ psi/feet} \cdot 0.918)}$$

$$L_p = 1708 \text{ feet}$$

Determine Equivalent Vertical Head, H_d :

$$H_d = 7000 \text{ feet} - 1708 \text{ feet}$$

$$H = 5292 \text{ feet}$$

Determine Equivalent Surface Back-Head, P_d :

$$P_d = \frac{120 \text{ psi}}{(0.433 \text{ psi/feet} \cdot 0.918)}$$

$$P_d = 302 \text{ feet}$$

Determine Friction Losses, F_t :

From Table 13 of the "Engineering Tables" of this manual, begin with a flow rate of 2280 BPD from the horizontal axis. Go vertically and then cut the line corresponding to 2-7/8" OD Tubing. Read the respective value on vertical axis.

$$F_t = 45 \text{ ft/1000 ft} \times \frac{7000 \text{ feet}}{1000 \text{ feet}}$$

$$F_t = 315 \text{ feet}$$

Determine Total Dynamic Head, TDH:

$$TDH = 5292 \text{ feet} + 302 \text{ feet} + 315 \text{ feet}$$

$$TDH = 5909 \text{ feet „}$$



Selecting Pump, Motor and Seal Section:

Taking as a reference the Table 10 of the “Engineering Tables” section, we look at the different models that we can use for this application.

First of all, we are limited by a casing size of 5-1/2” 17.0 lb/ft (which ID is 4.892” and drift 4.767”), so only the 400 (4.00” OD Pumps) can be used.

Refer to the 400 series stages available that might be suitable for this application. The 400-2200 (ROR of 1550-2650 BPD) and 400-3000a (ROR of 2100-3900). Remember that target rate to be handled by the pump is 2280 BPD.

We will select the 400-2200 for two reasons. The pump efficiency of this stage is 66% at the target flow rate versus 58% for the 400-3000a. In addition the design flow rate is centered in the ROR. The design flow rate is very near to the left of the ROR for the 400-3000a stage. If the well PI is lower than calculated, or reservoir conditions change we would quickly move out of the ROR for a 400-3000a. In addition the selection of the more efficient pump will reduce the client capital cost of equipment (less installed HP) and operating cost (less power consumed). On its performance curve at 3500 RPM, read a lift per stage of 24.8 feet, a brake horsepower of 0.59 HP per stage, and an efficiency of 67%.

Determine Number of Stages, N_{stages} :

$$N_{stages} = \frac{5909 \text{ feet}}{24.8 \text{ feet/stage}}$$

$$N_{stages} = 238 \text{ stg}$$

See Table 2 and check review the available housings for the 400-2200 (floater construction, standard pump), combine two pump sections Qty one 150 Hsg (124 stages) and Qty one 140 Hsg. (115 stages) for a total of 239 stages. It is seldom practical to supply the exact stage count desired. It is common industry practice to utilize the closest combination of full housing pumps that will met or exceed the desired number of stages.

Determine Brake Horsepower Required, BHP:

$$BHP = 0.59 \text{ HP/stage} \cdot 239 \text{ stg} \cdot 0.918$$

$$BHP = 129.5 \text{ HP}$$

Looking at Table 8 (we are limited to 456 motors by the casing size) of the “Engineering Tables” section, it is apparent that two 70 HP motors must be used in tandem to get a total horsepower capacity of 140 HP. In order to minimize current, the higher voltage option is selected. Qty two 1134 Volt each (total of 2238 Volts.). Current is 39 Amps.

The Seal Section will be a two-labyrinth type in tandem, series 400, because the horsepower requirement will be relatively high and the well is vertical. We could also apply a bag type motor seal, or a combination bag & labyrinth seal for this application.



6.5 Physical limits of the DHE.

6.5.1 Shaft Ratings

Checking the limits of all five sections of down hole equipment (DHE) selected. The pump HP load is within the published limits for the pump (125 HP), intake (256 HP), motor seal (256 HP) and motor (300 HP).

6.5.2 Housing Burst Pressure

Determine Maximum Pressure to be supported by Housing, P_{hsg} :

$$P_{hsg} = 0.433 \cdot \gamma_f \cdot SO_{head/stg} \cdot N_{stages} = 0.433 \cdot 0.918 \cdot 33.1 \frac{feet}{stage} \cdot 239stg$$

$$P_{hsg} = 3,145 psi$$

We can use a standard housing with a burst rating of 5000 psi.

6.5.3 Motor Cooling / Fluid Velocity

Read the fluid velocity passing the motor on table 12 of "Engineering Tables" section. Using 2280 BPD cut the curve "456 Series Motor in 5-1/2" casing" and we see that the fluid velocity is approximately 5 FPM. Remember that this value should be greater than 1 feet/sec. NOTE: The 1 FPS is only a guideline. Some applications with a high oil cut or high BHT may require a higher velocity and some applications with a high water cut or low BHT may be suitable for a lower fluid velocity. If in doubt contact T&SSG who will review the application and advise if motor heat rise is within acceptable limits.

6.5.4 Selecting Downhole Power Cable

First, check for application ranges on Table 14 of the "Engineering Tables" and find that SubLine SL-285 meets the requirement of this application. SubLine SL-212 appears to meet the requirements but remember that the cable rating is based on conductor operating temperature and not wellbore temperature. If the operating current of the motor increases to even 36 amps then cable conductor temperature will exceed maximum rating of the SL-212. Therefore, select SubLine SL-285 as the cable for this application.

On Table 15, estimate the voltage drop that will occur at the cable string. Using a operating current of 35 Amps (motor nameplate current), read that voltage drop for cable #6 AWG is 27 Volts/1000ft, and for cable #4 AWG is 18 Volts/1000ft. Both of them are below the limit of 30 Volts/100ft. .

Using Table 16, correct above value for temperature. First, use Table 17 to estimate conductor temperature for the #4 AWG Parallel cable, which is 190°F. Now, use this value on Table 16 and find the equivalent correction factor, which is 1.18.

6.5.5 Calculating Required Surface Voltage – Operating Conditions

$$\Delta V_{1000 ft} = 18 \frac{Volts}{1000 ft} \cdot 1.18 = 21.24 \frac{Volts}{1000 ft}$$

$$\Delta V = 21.24 \frac{Volts}{1000 ft} \times \frac{7000 ft}{1000 ft}$$

$$\Delta V = 149 Volts$$

Therefore, surface voltage requirement will be:

$$V_{surface} = 2620 Volts + 149 Volts$$



$$V_{surface} = 2769 \text{ Volts}$$

Finally, verify if starting voltage (voltage at the motor terminals) meets the minimum requirement of 50% of motor nameplate voltage.
NOTE: If the voltage available at the motor drops below 50% of normal operating voltage there is a risk that the motor will not start.

During startup the ESP system will draw 5 to 6 times nameplate current, typically for 6 to 10 cycles. The current will then quickly drop to operating current based on motor HP load. Starting voltage at the motor terminal is calculated in the following manner.

6.5.6 Calculating Motor Terminal Voltage – Startup Conditions

$$\Delta V_{1000 \text{ ft}} = 18 \text{ Volts}/1000 \text{ ft} \cdot 1.18 = 21.24 \text{ Volts}/1000 \text{ ft}$$

$$\Delta V = 21.24 \text{ Volts}/1000 \text{ ft} \times \frac{7000 \text{ ft}}{1000 \text{ ft}}$$

$$\Delta V = 149 \text{ Volts} \times 5 = 745 \text{ Volts}$$

Therefore, motor terminal voltage at startup will be:

$$V_{motor \text{ at start}} = 2620 \text{ Volts} - 745 \text{ Volts}$$

$$V_{motor \text{ at start}} = 1875 \text{ Volts}$$

$$\% V_{motor \text{ terminal at start}} = \frac{1875 \text{ volts}}{2620 \text{ volts}} \times 100 = 71\%$$

6.5.7 Selecting Switchboard:

From the catalog, select a 3600 Volts, Vacuum Contactor, 35-70 Amps Switchboard.

6.5.8 Selecting Transformers:

Determine System Required KVA, KVA:

$$KVA = \frac{2769 \text{ Volts} \cdot 35 \text{ Amps} \cdot \sqrt{3}}{1000}$$

$$KVA = 167 \text{ KVA}$$

Select either one three-phase, 250 KVA, 7200/12470Y volt primary, 2200-3810 volt secondary transformer or three single-phase 75 KVA (total of 225 KVA), 720/12470 volt primary, 831-3325 volt secondary transformers.



7 Design with Variable Speed Drive

Figure 26a and 26 b show a typical pump multi-frequency performance curve.

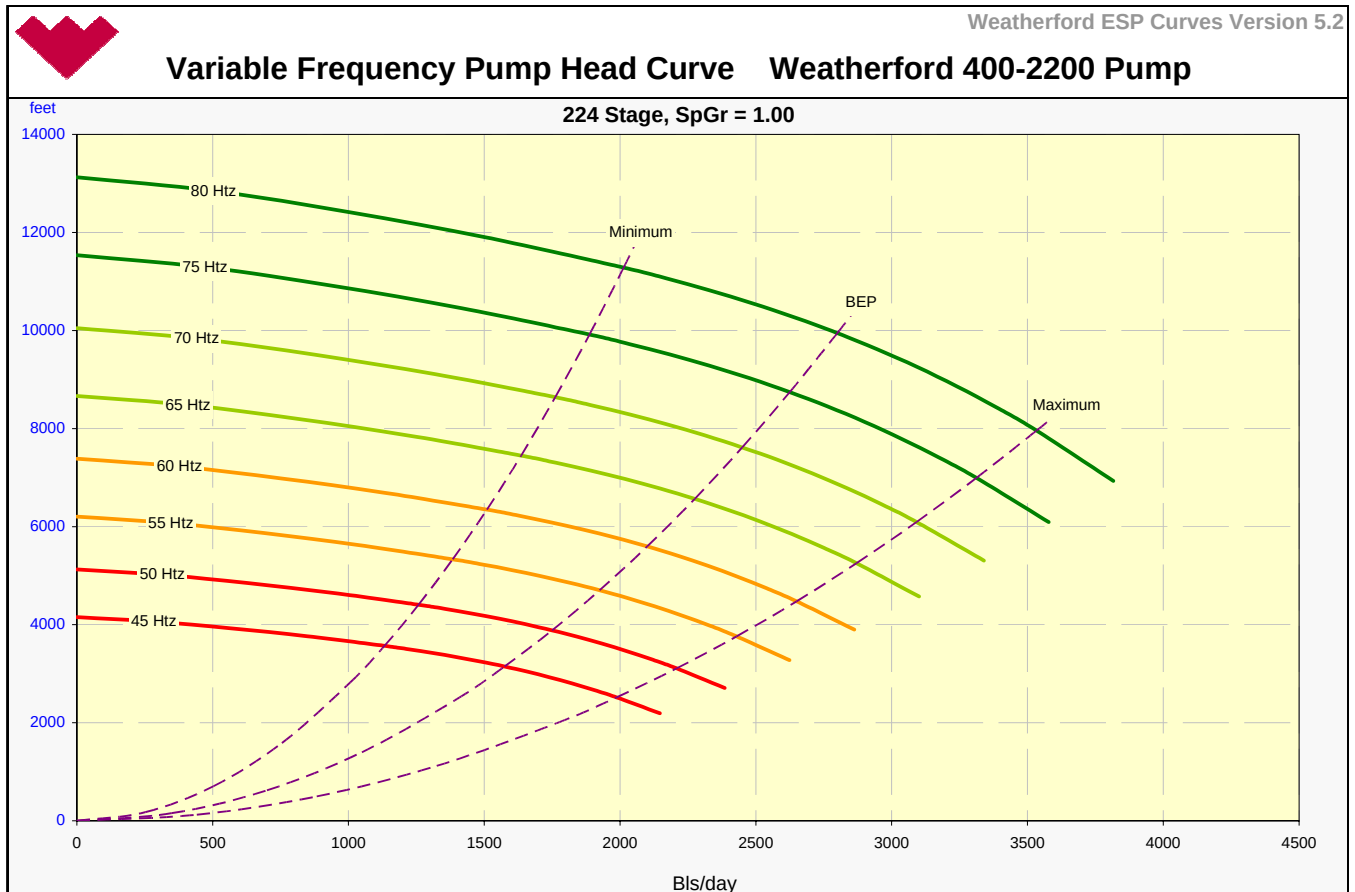


Figure 26a Typical Multi Frequency
Head- Flow Performance Curve

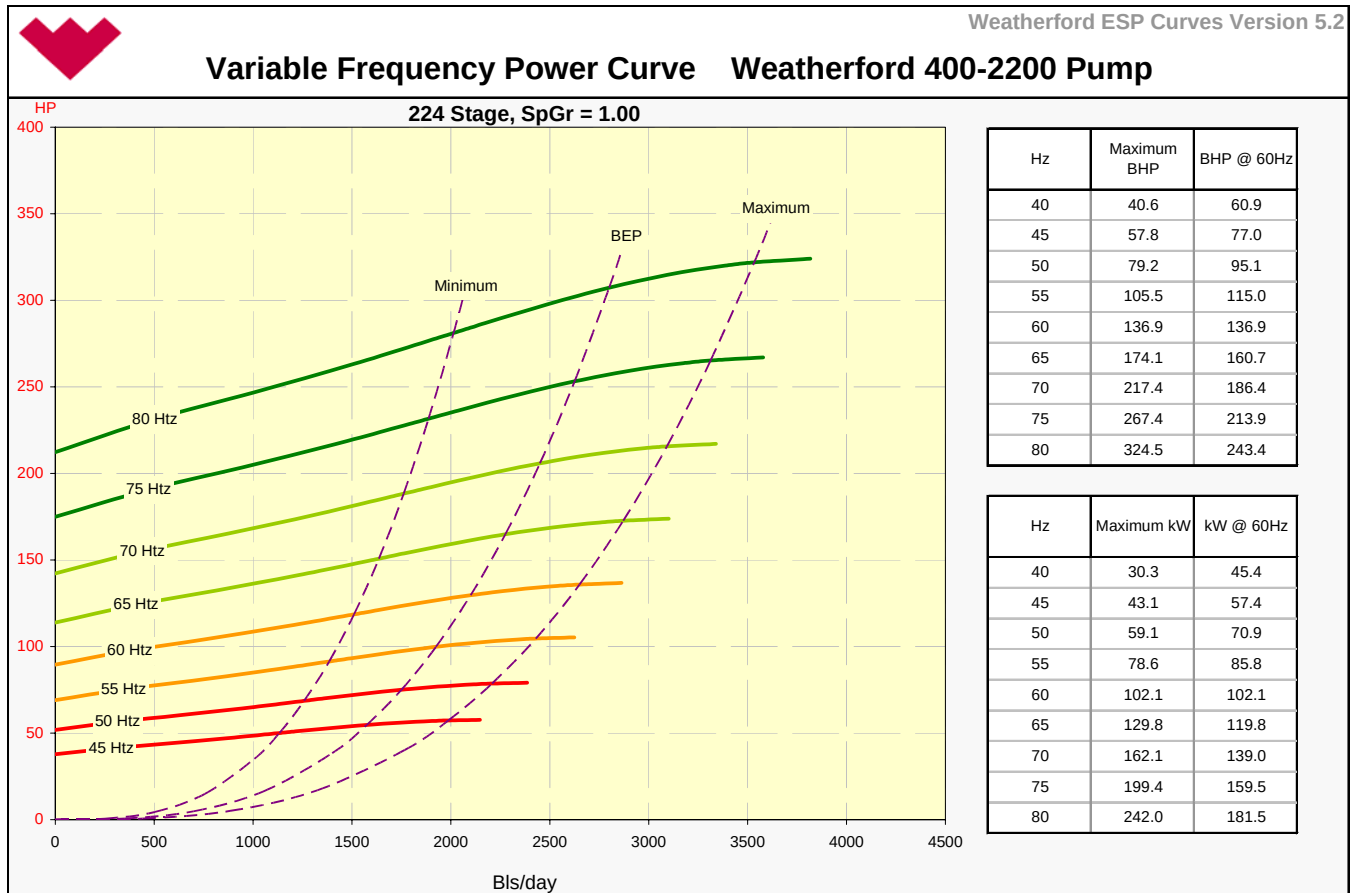


Figure 26b Typical Multi Frequency HP
Performance Curve

7.1 Pump Performance:

When applying a VFD to a submersible pump installation, it is first necessary to understand the effects of varying the pump speed. Pump performance is affected by changes in rotational speed (known as centrifugal pump's affinity laws).

When the speed (N) is changed, the flow (Q) varies directly as the speed change:

$$\frac{Q_1}{Q_2} = \frac{N_1}{N_2}$$

When the speed is changed, the head (H) varies directly as the square of the speed change:

$$\frac{H_1}{H_2} = \left(\frac{N_1}{N_2} \right)^2$$

When the speed is changed, the brake horsepower required by the pump (BHP) varies directly as the cube of the speed change:

$$\frac{BHP_1}{BHP_2} = \left(\frac{N_1}{N_2} \right)^3$$



7.2 Motor Performance:

The rotational speed of an AC motor varies directly with the change in frequency or cycles per second. A normal 60 cycles input to a two-pole electric motor produces a rotational synchronous speed of 3500 RPM. Any other frequency (F) will produce a proportional change in rotational speed; i.e. 30 cycles equals approximately 1750 RPM, 90 cycles equal approximately 5250 RPM. This statement can be simplified with the formula:

$$\frac{F_1}{F_2} = \frac{N_1}{N_2}$$

When the frequency (F) is changed, the motor output brake horsepower (BHP_{motor}) varies directly as the frequency; provided a constant voltage (V) to frequency ratio is maintained:

$$\frac{BHP_{motor1}}{BHP_{motor2}} = \frac{F_1}{F_2} \quad \text{provided} \quad \frac{F_1}{F_2} = \frac{V_1}{V_2}$$

Assuming the voltage to frequency ratio is constant, the motor full current load (I) will remain approximately constant:

$$I_1 = I_2 \quad \text{provided} \quad \frac{F_1}{F_2} = \frac{V_1}{V_2}$$

It is important to note that variable speed drives generally maintain a constant voltage to frequency ratio over a limited frequency range. Therefore, the transformer located between the drive and the motor may require a change to the transformer ratio to maintain the required constant voltage to frequency ratio (maintain constant flux density). Be sure to consult the VFD manufacturer to obtain the equipment limitations and select a VFD that is capable of operating within the required range.

7.3 VFD Output Transformer:

It is important to note that special output transformers are required on the output of a VFD. A conventional 3 phase dual wound transformer is not suitable for use on a VFD. The VFD output transformer is designed for operation across a wide range of frequencies and has additional iron in the transformer core to allow for the high primary current draw at startup. Insure that the output / step-up transformer is rated for VFD operation.

7.4 Operating Range:

Since the pump will operate at a head/capacity that intersects the system required head/capacity, definite speed limitations (both high and low) need to be established to prevent premature failures insure the speed range allows the pump to perform within the recommended pump operating range.

A speed that is too low may result in an insufficient flow past the motor to maintain adequate cooling (a minimum of 1 ft/sec is recommended).

A speed that is too low may also result in the unit operating at shut-in (zero flow). Operating in this condition will destroy the DHE in a very short period of time. In addition to the motor having insufficient flow past it for cooling, the pump is adding energy to the fluid in the form of heat that compounds the problem.

A speed that is too high can result in a motor overload condition. Since the pump brake horsepower varies as the cube of the speed, care must be taken when selecting the required motor horsepower rating (the motor must be sized for the largest anticipated load).



8 Example of ESP System Design with Variable Speed

Using the same example as the one for the Fixed Speed scenario and assuming use of an ESP system to produce the range from 1200 BPD to 2000 BPD, calculate the hydraulic requirement (volume and total dynamic head) for each condition, following exactly the same procedure (steps “a” to “e”). The next chart summarizes results of such calculations:

	CASE 1	CASE 2
a) Data Required:		
Fluid Rate, bpd:	1200	2000
Rest of Conditions:	Same	Same
b) Reservoir Inflow Capacity:		
Bottom hole Flowing Pressure, psi:	1524	686
c) Pump Intake Pressure:		
Liquid Specific Gravity:	0.91	0.916
Pump Intake Pressure, psi:	1524	782
d) Total Fluid Volume at Pump Intake Conditions:		
Gas Solubility, scf/sbl:	440.4	440.4
Oil Volumetric Factor, bl/sbl:	1.255	1.255
Oil Volume at Pump Intake conditions, bpd:	498	790
Water Volume at Pump Intake Conditions, bpd:	806	1343
Gas Compressibility Factor, z:	0.855	0.9268
Gas Volumetric Factor, cf/scf:	0.0099	0.094
Free Gas Volume at Pump Intake, bpd:	149	1035
Total Volume at intake, bpd:	1452	3169
Free Gas Content, %:	10.24	32.6
e) Total Dynamic Head:		
Average Fluid Specific Gravity:	0.915	0.916
Pump Net Suction Head, feet:	3998	2037
Equivalent Vertical Head, feet:	2150	4674
Surface Back-Head, feet:	302	302
Friction Losses, feet:	122	315
Total Dynamic Head, feet:	2574	5291

8.1 Selecting Pump, Motor and Seal Section:

A pump must be selected that can meet both production conditions, keeping them within the recommended range of application. Considering the same model, the 400-2200, condition 1 can be reach working at 38.3 Hz and the condition 2, as in the last example, at 60 Hz. Both conditions are close to the BEP corresponding to each frequency, so the selected model, 400-2200, is suitable for both case 1 and case 2.

	CONDITION 1	CONDITION 2
Total Volume at Pump Intake Conditions, bpd:	1337.47	2197.66
Total Volume at Surface, bdp:	1210.55	1989.10
Total Dynamic Head Required, feet:	2352	5383
Frequency, Hz:	38.3	60
Lift per stage @ frequency, feet:	10.45	23.92
Selected Number of Stages:	225	225
BHP per stage @ Frequency, HP:	0.144	0.553
Motor BHP Required @ Frequency, HP:	32.3	122.8
Motor BHP Available @ 60 Hz, HP:	89.32	140
Motor Operating Voltage / Current @ Design Frequency	1452 / 22.6	2275 / 35.8



From Table 8, select two 70 HP motors, which must be used in tandem to get a total horsepower capacity of 140 HP. We will select the 39 amp 1134 volt motors for this application.

The seal section will be a two-labyrinth type in tandem, series 400, because horsepower requirement will be relatively high and the well is vertical. There are no issues with temperatures.

9 Selecting Downhole Power Cable:

Following the same procedure, select SubLine 285 cable type and calculate voltage drop as follows:

	CONDITION 1	CONDITION 2
Voltage Drop per 1000 feet, Volts	22.1	35.1
Total Voltage Drop, Volts:	155	246
Motor Voltage at Frequency, Volts:	1452	2275
Surface Voltage @ frequency, Volts:	1607	2521
Operating Current, Amps	22.6	35.8
Surface Required KVA:	63	153

A 4KV cable rating must be used and a 160 KVA, 480 Volts. Variable Frequency Drive. Transformer will be a three-phase, 165 VFD Rated step-up transformer.

10 ESP Installation Procedures

After a pump has been selected, assembled, and shipped to the well location for installation, the service company and the oil company representatives must ensure the equipment is installed correctly. Sometimes the job is rushed - a costly mistake. The equipment being installed is expensive, so care and time taken at assembly are good investments for the future.

Close cooperation between the representatives of both companies is the key to a successful installation. To ensure long - term, efficient, and reliable operation, several precautions should be taken during installation and day-to-day operation of the ESP system.

10.1 Equipment Transportation and Handling

The safety of company personnel is always a concern when heavy equipment is moved. Precautions should always be taken to prevent injury. Follow these recommendations on transporting and handling ESPs to prevent injury to personnel or costly damage to components:

10.2 Transportation

- Always place equipment transported to and from the field location in proper shipping containers.
- Properly support and secure all components to prevent bouncing or bending during transport.
- Always chock cable reels and install tie-downs through the center of the reel on top of the hub.

10.3 Handling

- Do not drop shipping containers or handle them roughly to prevent damage to the components inside, some of which are extremely fragile. Damage cannot always be detected during normal installation or servicing.
- Remove equipment from shipping containers or place it into containers only when supervised by a qualified service technician.
- Always lift equipment with appropriate safety-approved lifting clamps under the supervision of a qualified service technician.



- Do not jar equipment against catwalks, the wellhead, etc. Equipment removed from shipping containers is very susceptible to damage.
- Always lift the motor controllers and transformers from the top with a spreader bar and slings, using the lifting lugs provided on the units.
- Lift the cable reel properly, using an approved bar inserted through the center of the reel that is long enough to attach a spreader bar with slings to the ends.
- Do not install any ESP equipment that has been dropped.

10.4 Well Preparation

- Review well logs to ensure a smooth transition from surface to pump setting depth. Run a bit and scraper, especially in small casing, to the pump setting depth to check for tight spots and to remove any sharp edges, scale, or paraffin from the casing.
- Before installing the ESP, circulate clean produced water through the well bore. This removes any solids that could plug the ESP.

10.5 Installing/Pulling the ESP Assembly

- Once the surface equipment is installed, maintenance is mostly electrical and reflects typical procedures for electrical control equipment. Troubleshooting the downhole components is very difficult from the surface; therefore, using a logical process of elimination helps identify what may be disrupting the system's performance.
- For each ESP installation, site/equipment preparation, installation procedures, and equipment handling must be addressed to ensure the installation proceeds as smoothly as possible. Following these procedures carefully will avoid premature failures.

10.6 Pre Installation Preparations

10.6.1 ESP System

- Rig time must be minimized, and a smooth installation will help ensure this happens. Before running the equipment, several checks must be completed. The following sections note the steps you can take to prepare the equipment.

10.6.2 ESP System

- Spot all shipping boxes / equipment at the location from which it will be picked up to RIH.
- Open all shipping boxes.
- Check all nameplates against the shipping documents to insure the correct equipment is on location.
- Check all equipment for free shaft rotation, that all couplings are present and correct.
- Record all equipment nameplate information and lengths.

10.6.3 Ancillary Equipment

- Check that all ancillary equipment (tubing check valve, tubing drain valve, Y-Tools etc.) are complete and ready for assembly.
- Confirm all crossovers / threads are correct, clean and ready to be made up.
- Confirm all tubing head, wellheads, tubing hangers, wellhead penetrators etc. are on hand, complete and correct.
- Record serial numbers, test certificate numbers, and lengths.

10.6.4 Electrical System



10.6.5

The electrical system is made up of the main cable; the motor leads, complete with penetrators; and wellhead penetrators and jump leads.

- Check all main cable and cable assemblies - both phase-to-phase and phase-to ground - for continuity and isolation resistance.
- Check phase rotation.
- Mark cable ends.
- Record serial numbers, test certificate numbers, and lengths.

10.6.6 Client / Rig Tooling

Have the following items available for the installation:

- Cavins or similar slips that have a provision for the power cable.
- Slip dies are clean, sharp and tubing will not move once the slips are set...
- Tubing tongs have a proper backup that will not slip.
- Tong dies are clean and sharp, especially the dies in the backup.
- Cable sheave complete with secondary safety line.
- Spooler

10.7 Installation and Servicing Procedures

It is not practical to cover installation procedures in this document. They are often job / equipment specific and are not general in nature. Detailed job procedure templates are available that can be modified to fit the specific workover / installation.

10.8 Start-up and Operating Procedures

Users must follow procedures for controlling ESP system start-ups to ensure pumps operate within their design parameters and maximize run life. Consult the following guidelines and checklists when commissioning an ESP system that uses an ESP powered by a variable-speed controller. Each procedure area is broken out by task and the party responsible for completing the task

10.9 Prestart-up Procedures

10.9.1 Responsible Party - ESP Technician

- Check phase rotation.
- Complete VFD (variable-speed control) start-up sheet.
- Ensure transformer tap settings are set to the required surface voltage. Perform a no-load test of the VSD.
- Ensure the VFD is programmed for the correct range of operating frequencies and for the correct setting of overload protection; function test if necessary.
- Ensure the local/remote switch on the VFD is in the Remote position.
- Check electrical shutdown systems for proper functioning.
- Ensure that amp charts are sized correctly and installed.



10.9.2 Responsible Party - Operations

- Function test all shutdown systems.
- Function test relevant screens and input on monitoring systems.
- Install gauges on the annulus and up stream of the choke.
- Function test the subsurface safety valve; ensure the valve is left in the Open position.
- Function test the choke valve; leave it in the closed position.
- Line up the well to the cleanup separator.
- Open the annulus valve.
- Test the port, upstream of the choke.
- Open the wing valve fully.

10.10 Initial Start-up Procedure

- Ensure the following personnel are at these locations:
- Production Operator -at the wellhead to open the surface choke valve and monitor wellhead pressure.
- Control Room Operator -in the control room to monitor the Production Control Center unit and to coordinate start-up.
- ESP Technician-at the VFD
- Record all start-up times and events on the amp.
- Program the VFD to accelerate to minimum frequency on
- Ensure the surface choke is cracked open.
- Note: The ESP system should not operate below 35 Hz.
- When all parties are ready, start the unit and open the choke steadily to about 50% open. (The actual choke reading will probably vary between wells, so operator experience is crucial.)
- Adjust the underload setting on the VFD to 80% of the observed running current.

Determine whether the unit is operating in the correct rotation by observing the following and comparing them to the ESP operating curve:

- Amps
- Drawdown (intake pressure)
- Motor speed
- Discharge pressure
- Wellhead pressure
- Flow rate



If necessary, modify the choke setting or frequency to provide a production rate close to the right side of the recommended operating range of the pump's performance curve

Note: Operate the ESP at the lowest possible frequency until the well has cleaned up and stabilized.

Note: If, while waiting for the unit to stabilize and/or clean up, the pump's performance monitoring parameters raise concern, shut down the unit and analyze the problem. The ESP Technician should discuss the situation with the Operator, Plant Supervisor, and/or Petroleum Engineer.

- Check the calibration of the ammeter.
- Obtain a fluid sample at the wellhead to determine the condition of the produced fluids (including solids content).
- When the well is stabilized, increase the frequency (pump speed) in 5-Hz increments, modifying the choke setting or frequency to maintain a production rate close to the right side of the recommended operating range of the pump's performance curve.
- Adjust the choke and/or VFD to provide a production rate close to the right side of the recommended operating range of the pump's performance curve. If in doubt, shut down the unit and analyze the problem.
- When the well is stabilized at the new frequency, check the running current. If it is acceptable, repeat step 11 until the desired production rate or the maximum current/frequency is reached.
- When the well is delivering the desired production rate and all frequency and choke settings have been finalized, verify the VFD underload and overload settings.

10.10.1 Routine Start-up Procedure

Use this procedure when the ESP unit has been shut down for less than 24 hours. If it has been shut down longer, refer to the Commissioning procedure detailed above.

Note: Depending on the well's productivity, the fluid level should reach equilibrium within a given time of the well shutting in, i.e., the ESP unit stops rotating in the reverse direction. However, to be totally certain, the pump inlet pressure should be monitored. Once it is stabilized, wait 5-10 minutes longer before starting the pump. Adopt 20 minutes total for this procedure initially. This can be modified if necessary once well performance and response are better understood.

Ensure start up personnel are at this location.

- Wellhead
- Control room
- VFD
- Ensure all start-up times and events are recorded on the amp chart.
- Line up the well to the test separator (if possible).
- Ensure the subsurface safety valve and all tree valves are open and the choke is closed.
- Vent the annulus if its pressure indicates zero (it may be under a vacuum).
- Start the unit and monitor the following continuously:
 - o Amperage Frequency
 - o Intake pressure and temperature Discharge pressure



- o Motor fluid temperature
- o Wellhead pressure and temperature Flow-line pressure
- o Choke setting
- o Test separator parameters
- o Open the choke slowly to the identical position prior to shutdown, always maintaining differential pressure between wellhead and flow line.
- o Note: The VFD ramps up automatically to its previous operating frequency.
- o Monitor the annulus; close the vent as required.
- o Ensure the choke setting is as before if the same rate is desired (done by Operations).
- o Monitor performance closely until the unit operates steadily.

10.11 Troubleshooting

If...	Then...
The test separator is not available	Adjust the choke to achieve steady operating conditions similar to previously agreed values of amperage, frequency, wellhead temperature and pressure, and intake/discharge pressures.
The test separator is available	Adjust the choke and/or VFD to provide the desired production rate, close to the right side of the recommended operating range of the pump's performance curve.
Monitoring parameters raise concern during the stabilization period	Shut down the unit and analyze the problem. Discuss the situation with the ESP Operator and Petroleum Engineer.

Take samples at the wellhead to determine the condition of the produced fluids.

10.11.1 Annulus Pressure Control

For maximum reliability of ESP components, both packer and wellhead penetrators should be subjected to minimum stress. Therefore, the magnitude and rate of change of the annulus pressure need to be monitored and controlled carefully.

There are two main sources of pressure imbalance, both created by free or solution gas present in the annulus:

Diffused gas present within the cable and penetrator materials at the wellhead.

Diffused gas present with the cable at depth (and at the equivalent hydrostatic pressure), which is able to migrate toward the wellhead penetrator. Gas can be conveyed along the central strand of the conductor cable.

If the annulus pressure around the wellhead penetrator decreases, then either of these sources of penetration imbalance - which are internal to the Penetrator - experiences stress to a greater degree than if the pressure had been held, even at a higher pressure.



10.11.2 Controlling Annulus Pressure

Depressurize the annulus for any pressure that occurs, including test pressures:

Reduce the pressure by half at 50 PSI/min maximum.

Follow with a dwell period of 30 min.

Repeat the process until the desired annulus pressure is reached.

Example: An 10 -in. annulus pressure of 600 PSI reduced to 40 PSI takes 1 hour, 42 minutes.

Time (min)	0	6	36	39	69	71	101	102
Pressure (PSI)	600	300	300	150	150	75	75	40

Overall, the longer the duration that can be allocated to the pressure schedule, the less stress placed on the penetrators and cable.

Should it be necessary to shut-in the well for an extended period, remember that as the annulus cools, the pressure in the annulus falls. To prevent negative pressure, vent the annulus until the pressure stabilizes.

10.11.3 Monitoring Performance

ESPs have limits to their production capabilities. If they operate outside these limits, performance is impaired and damage may occur. The primary reason for these limits is the multistage centrifugal pump.

A stage consists of a static diffuser and an impeller, rotated by a shaft connected to the electric motor. The impeller speeds the fluid and pushes it outward; the diffuser slows the flow and converts it to pressure energy before it enters the next stage and the process repeats. Two opposing forces act on the impeller: the pressure it generates and the force from the momentum of the fluid passing through it. When a pump operates within its correct range, the forces are approximately balanced. When the forces are unbalanced, wear accelerates and performance declines. If flow rate is low, the impellers press down onto the diffusers and down thrust occurs. If flow is high, the opposite happens and up thrust occurs. Depending on design, down thrust is taken by the diffusers or by a single thrust bearing housed in a seal assembly situated between the motor and the pump intake.

As well as causing pump wear, low flow can cause one other condition:

The electric motor can overheat if too little fluid flows past to help cool it. A high supply current to the motor indicates a large power demand from the pump. This can also shorten the life of the motor cable and the electrical penetrator system.

10.11.4 Monitoring Guidelines

When operating conditions change, some occurrence always initiates that change. The chart below lists possible reasons where process conditions may change because of operator intervention (e.g., changing choke position or pump speed) or outside conditions (e.g., increase in water cut).

Intake pressure	Rising	• Heavier fluid in the well. • Pump slows down. • Tubing retrievable subsurface safety valve (TRSSV) closed. • Blockage in flow line. • Wellhead valve closed. • Unit shut down. • Higher wellhead pressure. • Recirculation of downhole fluids. • Reservoir pressure increase. • Restriction at pump intake.
Intake pressure	Falling	• Lighter fluid in the well~ • Pump sped up.



		• Unit just restarted~ • Lower wellhead pressure. • Reservoir pressure decreased. • Blockage at perforations.
Intake pressure	No change at startups	• No flow from perforations. • Pump rotating in wrong direction. • Downhole fluids being recirculated. • TRSSV closed. • Blockage in line. • Pump intake plugged.
Downhole temperature	Rising	• Well warms after start-up • Insufficient rate to cool motor. • Recirculation of downhole fluids, e.g., through bypass or hole in tubing.
Downhole temperature	Falling	• Pump shuts down.
Amps	Rising	• Greater load on the motor. • Pumping more fluid. • Pumping heavier fluid. • Debris, solids, or sand entering the pump (current may be erratic).
	Falling	• Lighter load on the motor. • Pumping less fluid. • Pumping lighter fluid, e.g., gas breakout (current may be erratic). • Restriction in the flow line.
	Low	• No fluid flow. • TRSSV closed. • Wellhead valve closed. • Blockage in the tubing. • Downhole fluids being recirculated. • Broken shaft.
Wellhead pressure	Rising	• Lower flow rate through choke; restricted. Choke closed. • Surface line restriction. • Surface valve closed~ • Header pressure rising. • Pump speed increased. • Lighter fluid being pumped (higher flow).
Wellhead pressure	Falling	• Higher flow rate through the choke, e.g., worn. • Choke is opening. • Pump stopped. • Downhole fluids being recirculated. Pump speed decreased. • Header pressure falling. • Heavier fluids being pumped (lower flows).
Wellhead flowing temperature	Rising	• Well warming up after start-up. • More flow from the well.
Wellhead flowing temperature	Falling	• Less flow from the well. • Pump shut down.
Current leakage	Rising	• Temperature increase in the well. • Deterioration of electrical integrity of insulating material. • Increasing pressure in wellbore or annulus.
Motor fluid temperature	Rising	• Unit started. • Frequency (pump speed) increased. • Pump frequency decreased to a level where produced fluid does not cool sufficiently. • Downhole fluids recirculated, e.g., through bypass or hole in tubing. • Restriction at pump intake. • Scale buildup.



		• Secondary tap settings on transformer set incorrectly. • VFD voltage/frequency ratio set incorrectly. • Wellhead valves, TRSSV, choke closed.
--	--	---

10.12 Installation Maintenance and Troubleshooting

The below tables lists troubleshooting procedures, including the condition of the system, the apparent problem, possible causes, and corrective measures.

10.12.1 Troubleshooting Procedures Pump Running

Production greater than pump design capacity or range.	Well inflow (PI) greater than pump design capacity or range	• If fluid level is in acceptable operating range, increase tubing wellhead pressure to bring pump production rate within design range (close choke). • Check relevant data for possible future resizing.
	Change in fluid characteristics	• Increase tubing wellhead pressure to bring pump production rate within design range. • Resize pump considering the changes in fluid characteristics.
No production or production below pump design capacity or range	Total pump discharge head not sufficient for application	• Check pump design head in connection with the operating fluid level.
	Reverse rotation	• Caution. Verify no backspin before turning pump back on. Leave for at least 30 min. before restarting.
	Tubing leak	• Pressure-test tubing with downhole plugs or perform a spinner/temp. survey to determine if a leak exists. If it does, patch the tubing or pull it and replace faulty joints.
		• A high or low current may be noted, depending on the location of the leak, working fluid level, and size of the unit; however, this does not always indicate a tubing leak.
	Obstruction in flow line.	• Check pressure in flow line at the wellhead. If it is abnormally high, take appropriate measures to correct • Ensure the downhole safety valve is open.
	Restricted pump	• If well has scale, paraffin or salt problem, pump may be restricted. Take appropriate corrective action, e.g., acidize, solvent flush scale, dissolver soak. • Solids may be restricting pump intake. To clean, reverse flow through the pump (bullhead).
	Broken pump shaft	• Pull unit; replace failed equipment. • Where undercurrent relay is used, this condition usually stops pump undercurrent.
	Worn pump	• Obtain fluid level and BHP to determine pump submergence. • Check well load pressure for decreased reading, assuming choke position not altered. • If the points above confirm a worn pump, pull and replace the unit.
No production or production below pump design capacity or range	Flow line leak (surface)	• Check pump design head in connection with the operating fluid level.
	Change in fluid characteristics	• Check the pump design head in connection with the operating fluid level.
	Well productivity less than pump design capacity or range	• Determine the working fluid level; refer to "Well Pumped Off" in the following entry in this table.



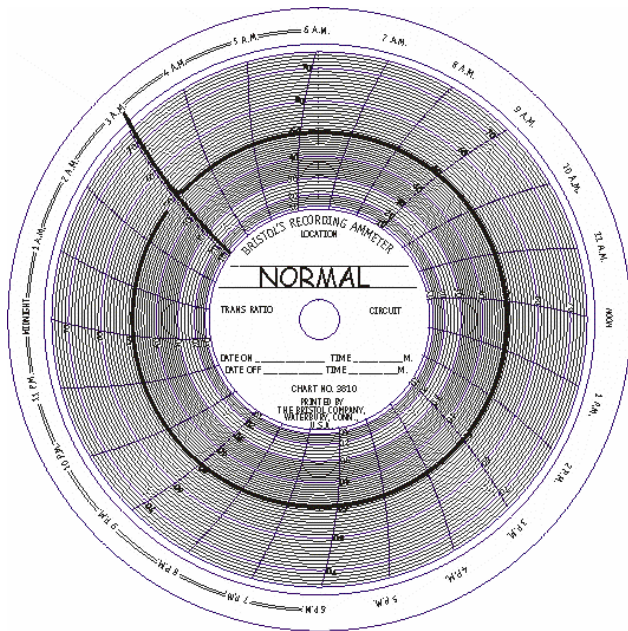
10.12.2 Pump not operating

Down on undercurrent	Pump is gas locked; well has been allowed to be drawn below the gas bubble point	- Do not allow downhole pressure to drop to the gas breakout point. - If the pump is pulled, include a gas separator at reinstallation with a high-set vented packer - If possible, set the pump deeper in future installs
	Well pumped off	- Obtain the fluid level to confirm the pumped-off condition. Possible actions: - If pump capacity is greater than well production, try choking back on production to continue operations. - Stimulate or clean the well to increase production
	Total pump head insufficient for application	- Check the pump design head's operating fluid level. - Consider installing VFD. - Resize the pump, considering the changes in fluid characteristics.
Down on undercurrent	Primary power surge or outage	If a repeated problem, use power system monitors to determine the cause. Correct as appropriate.
	Broken pump shaft	See 'Pump Running' at the beginning of this troubleshooting table.
	Tubing leak	See 'Pump Running' at the beginning of this troubleshooting table.
	Plugged pump	See 'Pump Running' at the beginning of this troubleshooting table.
	Worn pump	See 'Pump Running' at the beginning of this troubleshooting table.
	Reverse rotation	See 'Pump Running' at the beginning of this troubleshooting table.
Down on overload	Power system sag	- If problem repeats, use system-monitoring equipment. Investigate unusually heavy electrical loads that may have been added to the power system. - Upgrade power distribution system.
	Debris, solids, sand, scale, etc. in system	- Clean up annulus by reverse circulating up the annulus. - Review well treatment program with Petroleum Engineer.
	Change in fluid characteristics	Insufficient horsepower
Down on overload	Worn pump	Consider past running time of pump and well history, sand, mud, etc. Worn thrust washers and bearings may be causing unnecessary friction. Pull and replace unit.
	Locked pump	- Reverse rotate. - Pump clean fluid down the tubing and through the pump to remove debris. - Pump acid through the pump to dissolve scale.
	Unit in bind from a crooked place in the wellbore	Raise or lower the unit to a straight portion of the wellbore.
	Pump start-up tried while pump is back-spinning	Leave pump alone at least 30 min. before restarting.
	Blow fuses	Check incoming voltages on all three phases
	Fuses improperly spec'd or faulty	- Check and reset. - Check line fuses and motor control panel fuses. Repair or replace as necessary.
	Electrical fault in the system	- Disconnect the power cable at the junction box; check the downhole cable for shorts. - If a short is found, check wellhead penetration by removing the Christmas tree. Replace if faulty. Otherwise, fault is below tubing hanger; pull downhole equipment.
	Electrical fault in the system	If a short is not found, check the surface power system for shorts.
Motor control will not operate	No power to the motor control panel	- Call for a qualified electrician to investigate. Caution: leave this test to qualified personnel.



11 Basic Amp Chart Interpretation

NORMAL OPERATION

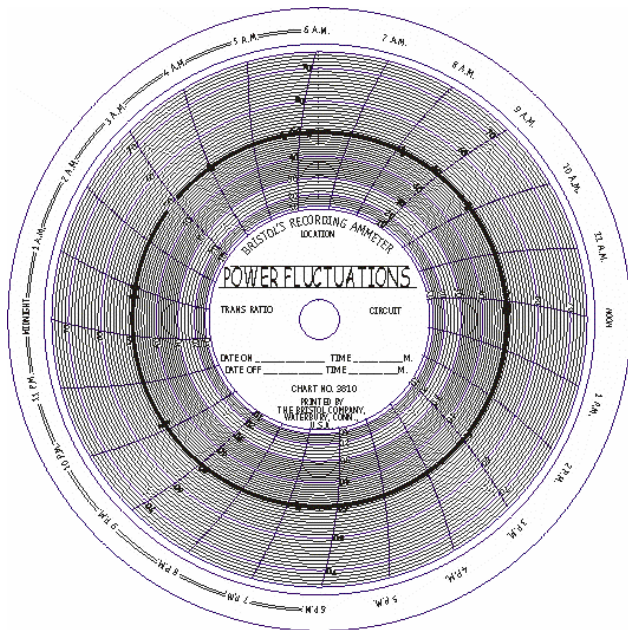


Amperage on this chart indicates normal type of operation. Unit is running continuously with smooth line, and pulling motor nameplate amps. Due to different types of oil well conditions, an ammeter chart could have a different amp line configuration and still be considered normal operation for that particular installation.

As long as amp chart line is symmetric on a day-to-day basis, ammeter chart can be considered normal.

It is important to use ammeter chart to detect and correct any deviations from a well's normal operation before abnormal operations cause a premature failure.

POWER FLUCTUATION



POWER FLUCTUATIONS

This chart indicates a normal operation, good production rate, and a smooth steady current of 52 amperes.

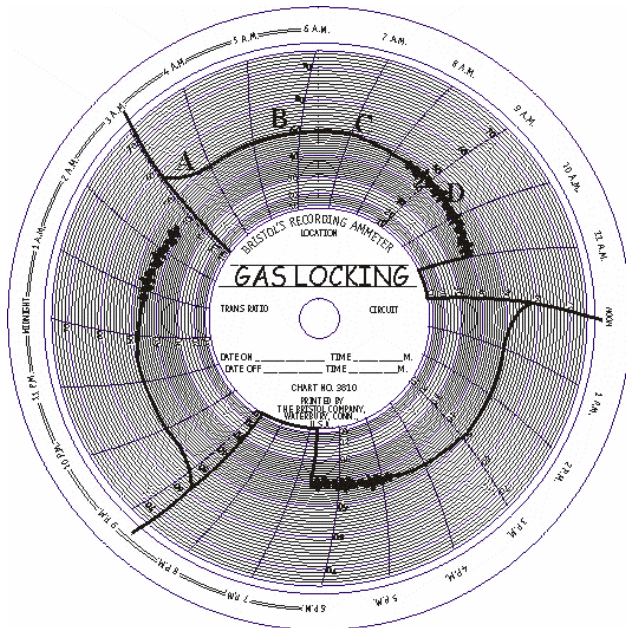
The "blips" or "kicks" shown at different intervals are caused by power fluctuations.

This could be caused by the periodical starting of some other heavy electrical load on the power system, such as an injection pump.

This type of operation is not detrimental as long as the kicks are not too severe or close together.

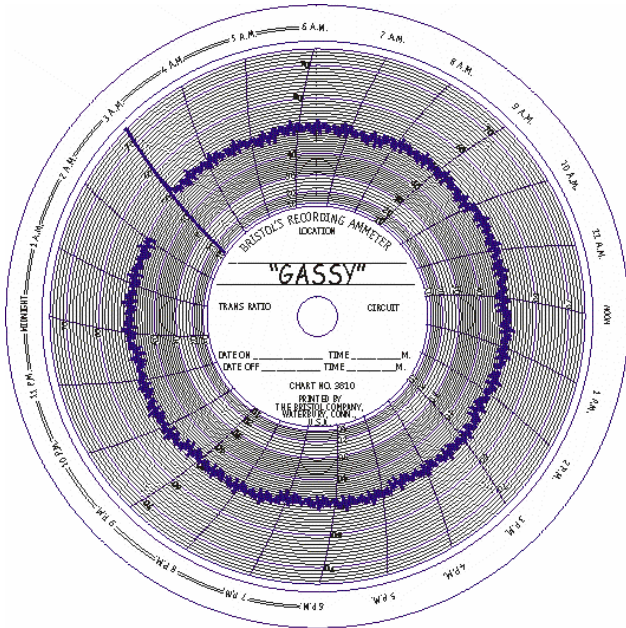


GAS LOCKING





GASSY WELL

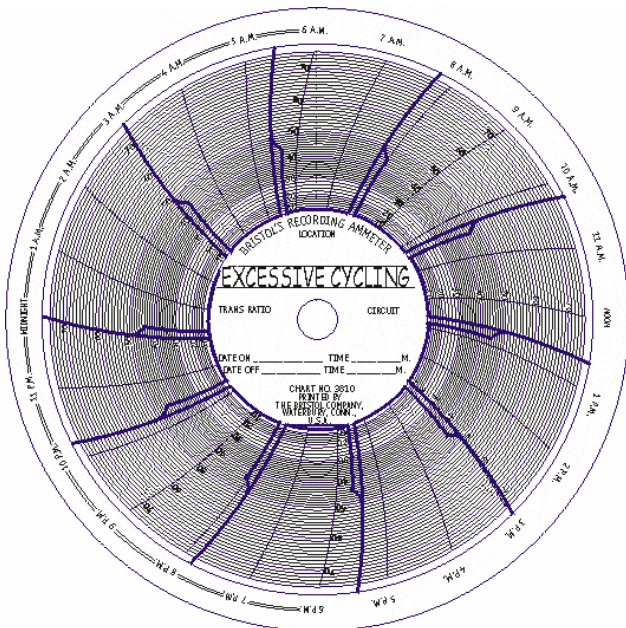


Amperage fluctuations on this chart indicate large amounts of gas going through the pump. Amperes change as pump handles heavy and light (gas cut) density fluid and pressures in pump shift.

This type of chart is considered normal for pumps handling large quantities of gas. Sometimes amp line can be smoothed out with a combination of increased casing and tubing pressures. Amount of change is dependent on well conditions that exist in each individual well.

This chart may result from a well that is very near pump off (low submergence) and is pulling in some air, causing cavitation; however, this is rare. This type of operation could also be caused by prevailing well conditions, such as volume being produced by type of pump installed, kind of fluid being produced, etc.

EXCESSIVE CYCLES



This chart indicates that unit is starting normally but amperage immediately begins to decline and unit goes down on undercurrent in 15 minutes.

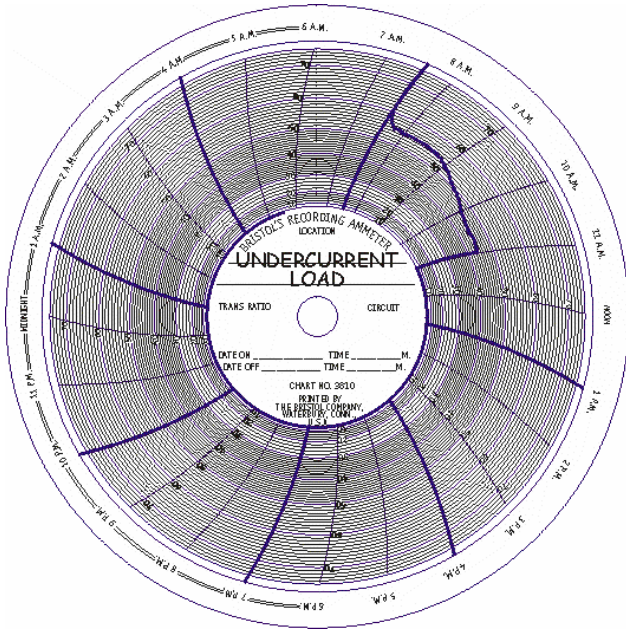
Unit times out and restarts automatically and repeats cycle.

There are several reasons for this type of operation, such as a highly oversized pump, pumping against a closed valve; a hole in tubing high up in string, etc.

Regardless, this type of operation is detrimental to good submersible pump operations and should be corrected as soon as possible.



UNDERLOAD CURRENT

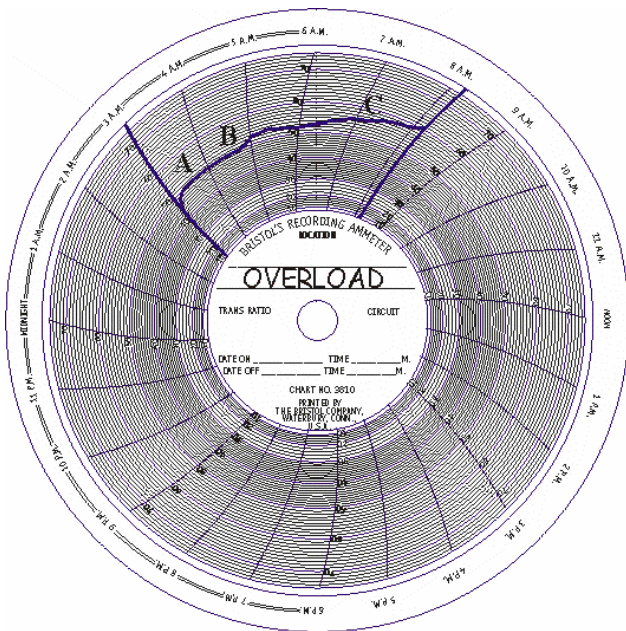


This chart configuration sometimes occurs a few days after a new installation. Shows unit starting at 7:30 am. (Heavy line pegs ammeter chart scale - this is normal movement of ammeter marker at start-up). Amperage drops to 47 amps, runs a few minutes and goes down on underload.

Chart indicates that pump is not handling fluid of sufficient density and/or volume to load unit above present undercurrent settings. Horsepower requirements is less than anticipated due possibly to pump handling a lighter gradient fluid or a smaller volume than designed for. If previous production rates indicate sufficient fluid is available, UL setting should be lowered. UL setting can be lowered as long as sufficient fluid is passing by motor to cool it and UL is not set below no-load amperage of motor.

NOTE: Unit is going down on UL because it is timing out and restarting automatically. This type of operation could also be caused by a defective relay in switchboard, and it must be corrected immediately.

OVERLOAD CURRENT



Current load started below its rating, as shown, then gradually built up to normal load (a normal occurrence with certain types and sizes of pumps). Ammeter chart indicates that unit pulled nameplate amps for approximately 1 1/2 hours and current began to climb and started leveling off at 59 amps (14% overload). This unit went down on OL at 8:15 am. This unit must be completely checked out electronically downhole and at surface before a restart is attempted. Cause of running overloaded condition must also be determined. It could be due to mechanical problems with pump, sand entry, emulsion, overheating, etc.

Note: Unit will not restart automatically as indicated on chart because of overload condition.



Engineering Tables

Table 1 Well Data Sheet



Weatherford

ESP Application Data Request Form

Based on the information provided, we offer the ability to model and select an Electrical Submersible Pumping System to meet certain well bore criteria. Not all wells are candidates for ESP Systems.

Instructions: Complete all sections including check boxes. Place the cursor on the "triangle" in the field box to obtain additional information.

Company Name		Contact	
Well Name		Phone	Fax
Location		e-mail	
Country		Date	
WELL DATA		IPR DATA	
Vertical: ☐ Horizontal: ☐ Slant: ☐ Survey: ☐		Vogel Corrected for Water Cut	
Use Griffin & Wallace Bubble Flow Equation ☐		Productivity Index - PI	
Wellbore Correlations		Static BHP	
Datum Point		Static Fluid Level	
Pump Setting Depth		Static Casing Pressure	
Reservoir Temperature		Test Fluid Rate	
Wellhead Temperature		Test BHP	
Casing-1	7" x 20.00 lb/ft	Casing-2	None
Top	MD ft	Top	MD ft
Bottom	MD ft	Bottom	MD ft
Tubing-1	2-3/8" x 4.70 lb/ft	Tubing-2	None
Top	MD ft	Top	MD ft
Bottom	MD ft	Bottom	MD ft
FLUIDS DATA		Design Conditions	
Oil Gravity-°API		Solve for Pump Intake Conditions	
Gas Gravity		Total Fluid Rate	
Water Gravity		Intake Pressure	
Water Cut - %		Surface Pressure	
GOR		Solve for Fluid Rate	
Bble Pt.		Pump Depth	
GLR		Intake Pressure	
Sol.GOR		Surface Pressure	
Gas Impurities - %		Solve for Pump Depth	
CO ₂		Total Fluid Rate	
N ₂		Pump Depth	
H ₂ S		General Comments	
OTHER INFORMATION			
Sand			
Y Tool			
Offshore			
Scale			
Please ensure that all production and well data information is completed. This information is critical to the surface, downhole equipment, and elastomer selection. Incomplete and inaccurate information may result in untimely delays in providing a commercial sales quote and the selection of surface and downhole components. Since the equipment selection and any commercial sales quote are based on the information provided, the results will only be as accurate as the information provided.			
ESP Technical & Sales Support Group			



Table 2 Catalog Section 400-2200 Pump

Pump 400-2200

Pump Data

Housing	Length [ft]	Weight [lb]	Length [m]	Weight [kg]	Standard Pump [SA]		Abrasion Resistant Pump [TA]	
					Stages		Stages	
					Floater	Compression	Floater	Compression
10	2.1	44	0.64	20.0	6	6	6	6
20	3.5	73	1.07	33.1	15	15	15	15
30	4.9	106	1.49	48.1	23	23	23	23
40	6.4	134	1.95	60.8	31	31	31	31
50	7.8	165	2.38	74.8	39	39	39	39
60	9.2	196	2.80	88.9	48	48	48	48
70	10.6	225	3.23	102.0	56	56	56	56
80	12.0	258	3.66	117.0	65	65	65	65
90	13.4	287	4.08	130.2	73	73	73	73
100	14.8	317	4.51	143.8	82	82	82	82
110	16.2	348	4.94	157.8	90	90	90	90
120	17.6	377	5.36	171.0	98	98	98	98
130	19.0	403	5.79	182.8	107		107	
140	20.4	438	6.22	198.6	115		115	
150	21.8	478	6.64	216.8	124		124	

Options

Type	Options			Pump Build		Elastomers	
Center Tandem CT	CS* Head & Base, CS Housing, CS Fasteners	CS		Floater	FP	HSN	Std
	SS** Head & Base, SS Housing, MONEL Fasteners	SS		Compression	CP	Aflas	HT
	CS Head & Base, CS Housing, MONEL Fasteners	MT					
	CS Head & Base, CS Housing, Flame Spray Coating, MONEL Fasteners	MTC					
	SS Head & Base, CS Housing, MONEL Fasteners	MTS					
	SS Head & Base, CS Housing, Flame Spray Coating, MONEL Fasteners	MTSC					

Shaft	Abrasion Handling			Internal Coating		Housing Thread	
MONEL® MS	TuC Head & Base	SA		No Coating	NC	Buttress	BT
INCONEL® HSS	TuC Head/Base & Stages	TA		Boronize Coating	BC		
	TuC Head/Base & X Stages	XA		Epoxy Coating	EC		

*CS = Carbon Steel / **SS = Stainless Steel

Performance Summary

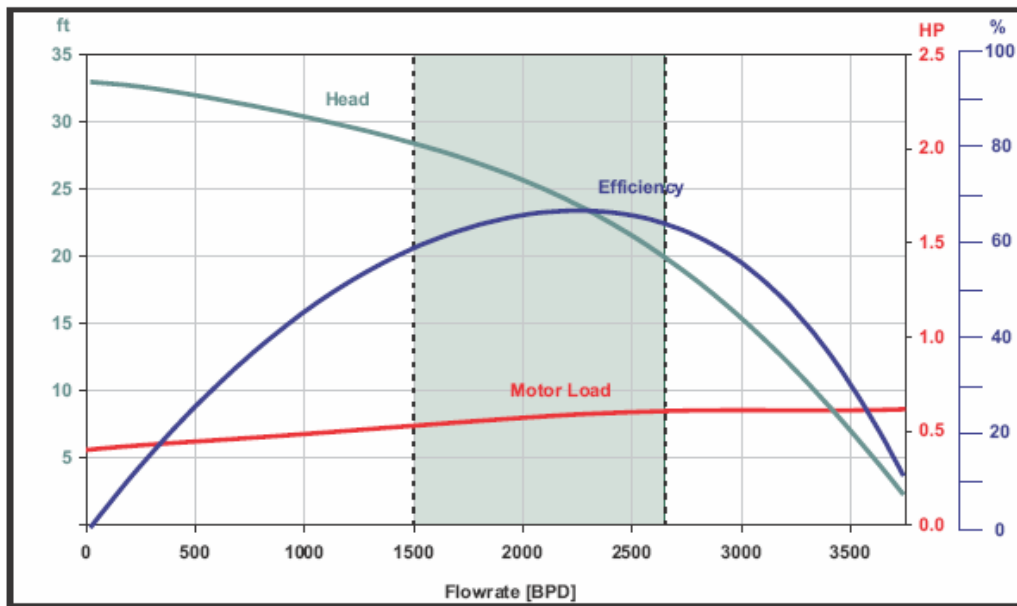
Best Efficiency Data - 60Hz				
Optimum Flow Rate [BPD]	Operating range [BPD]	Head per Stage [ft]	Power per Stage (HP)	Efficiency [%]
2245	1550-2650	23.9	0.59	67

Best Efficiency Data - 50Hz				
Optimum Flow Rate [M³/D]	Operating range [M³/D]	Head per Stage [m]	Power per Stage (kW)	Efficiency [%]
297	199-351	5.1	0.25	67



Pump 400-2200

60 Hz Data



Technical Data - 60 Hz			
Housing Diameter [in.]	4.00	Shaft BHP Limit -	Standard [HP] 125
Shaft Diameter [in.]	0.6875		High Strength [HP] 200
Housing Pressure Limit [psi]	5000	Rotational Direction	CCW

*An Ultra High-Strength shaft is also available upon special request

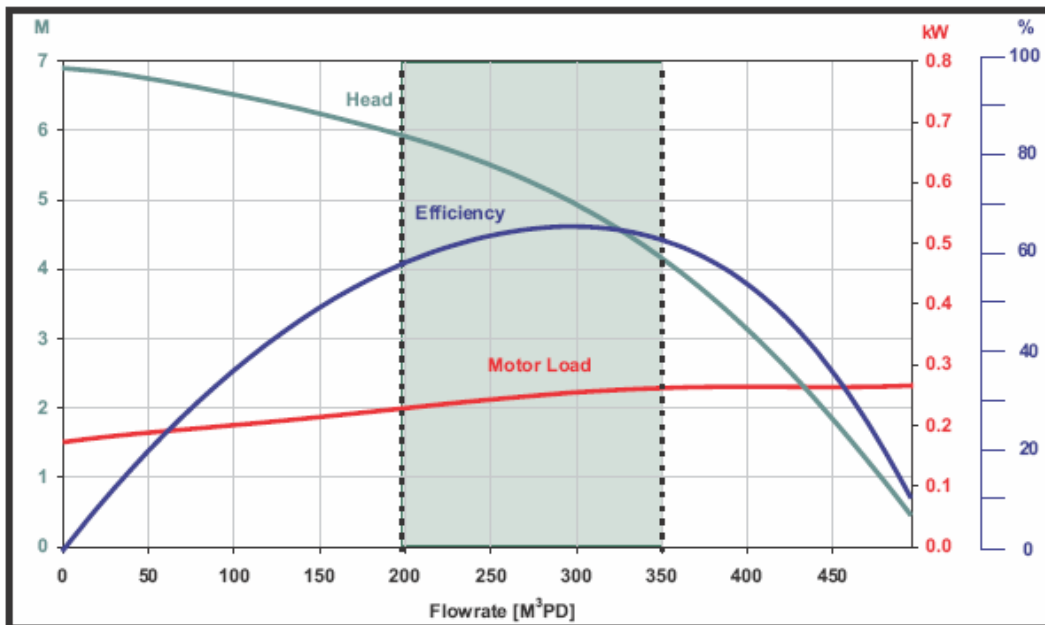


Mixed-Flow Construction Stage



Pump 400-2200

50 Hz Data



Technical Data - 50 Hz			
Housing Diameter [mm]	101.6	Shaft BHP Limit -	Standard [kW/HP] 78/104
Shaft Diameter [mm]	17.46		High Strength [kW/HP] 125/167
Housing Pressure Limit [bar]	345	Rotational Direction	CCW

*An Ultra High-Strength shaft is also available upon special request

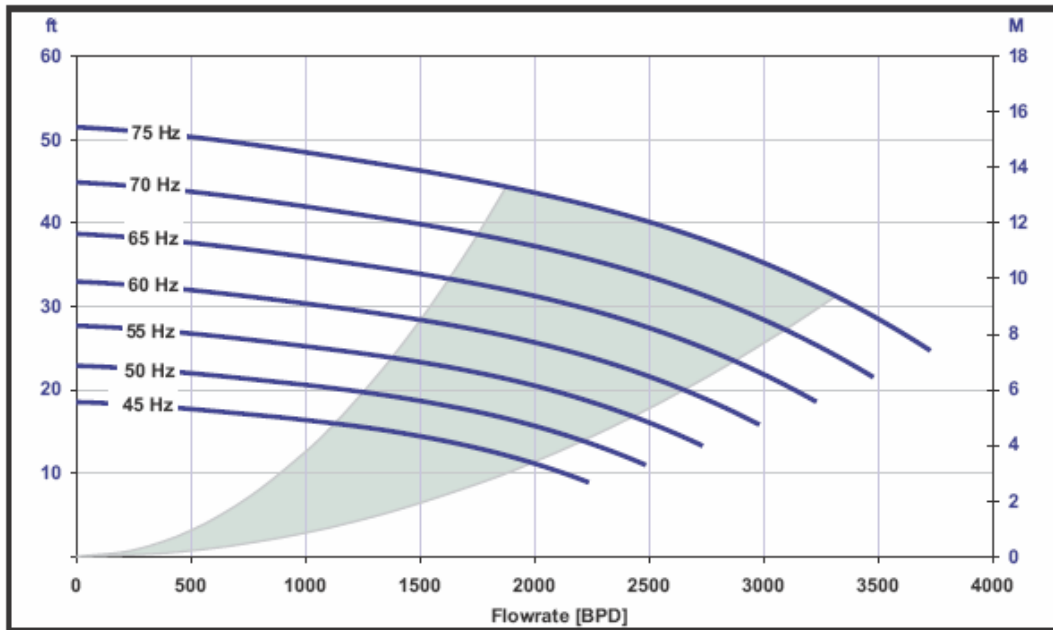


Mixed-Flow Construction Stage



Pump 400-2200

Multi-Frequency Pump Performance Curve - Head



Multi-Frequency Pump Performance Curve - Power

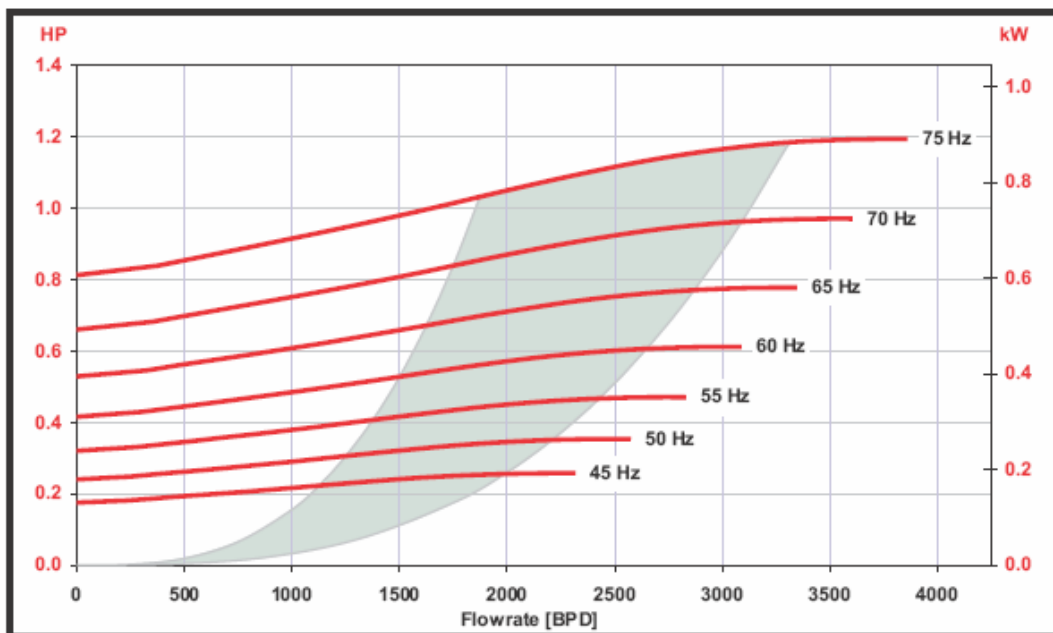




Table 3 Pump Shaft Ratings

Stage Name	Shaft Diameter		Shaft area		Shaft Material	60 Hz Rating		50 Hz Rating	
	Inch	mm	In ²	mm ²		HP	kW	HP	Kw
400-180	.625	15.875	.491	316.7	Monel	94	70	78	58
400-350					Inconel	150	112	125	93
400-450									
400-700									
400-950									
400-1250	.687	17.46	.540	348.1	Monel	125	93	104	78
400-1750a					Inconel	200	149	167	125
400-2200									
400-3000a	.875	22.2	.687	443.4	Monel	256	191	213	159
400-4500					Inconel	410	306	342	255
400-5800									
513-1600	.875	22.2	.687	443.4	Monel	256	191	213	159
513-2500					Inconel	410	306	342	255
513-3900	1.00	25.4	.785	506.7	Monel	375	280	313	234
513-6000a					Inconel	600	448	500	373
513-7500a									
513-10000	1.187	30.2	.932	601.5	Monel	637	475	531	396
					Inconel	1019	760	849	633
538-1900	.875	22.2	.687	443.4	Monel	256	191	213	159
538-2600					Inconel	410	306	342	255
538-3600									
538-4700	1.00	25.4	.785	506.7	Monel	375	280	313	234
538-7000					Inconel	600	448	500	373
538-9000									
538-12500	1.187	30.2	.932	601.5	Monel	637	475	531	396
					Inconel	1019	760	849	633
675-9000	1.187	30.2	.932	601.5	Monel	637	475	531	396
675-12000					Inconel	1019	760	849	633
862-18000	1.375	34.98	1.080	.696.7	Monel	800	596	667	497
862-25000					Inconel	1280	955	1067	796



Table 4 Pump Intake Shaft Ratings

Stage Name	Shaft Diameter		Shaft area		Shaft Material	60 Hz Rating		50 Hz Rating	
	Inch	mm	In ²	Mm ²		HP	kW	HP	kW
400 series	.875	22.2	.687	443.4	Monel	256	191	213	159
					Inconel	410	306	342	255
513 series	1.187	30.2	.932	601.5	Monel	637	475	531	396
					Inconel	1019	760	849	633
538 series	1.187	30.2	.932	601.5	Monel	637	475	531	396
					Inconel	1019	760	849	633

Table 5 Motor Seal Shaft Ratings

Stage Name	Shaft Diameter		Shaft area		Shaft Material	60 Hz Rating		50 Hz Rating	
	Inch	mm	In ²	Mm ²		HP	kW	HP	kW
400 series	.875	22.2	.687	443.4	Monel	256	191	213	159
					Inconel	410	306	342	255
513 series	1.187	30.2	.932	601.5	Monel	637	475	531	396
					Inconel	1019	760	849	633

Table 6 Motor Shaft Ratings

Stage Name	Shaft Diameter		Shaft area		Shaft Material	60 Hz Rating		50 Hz Rating	
	Inch	mm	In ²	Mm ²		HP	kW	HP	kW
456 series	1.187	30	.940	606.5	Standard	360	268	300	224
					High Strength	450	336	375	280
540 series	1.375?	35	1.10	709.6	Standard	700	522	583	435
					High Strength	800	597	667	497
562 series	1.375	35	1.10	709.6	Standard / 6T Spline	700	522	583	435
					Standard / Involute Spline	900	671	750	559
					High Strength	1250	932	1042	777



Table 7 IL-150 456 Motor Table

60Hz				50Hz				Available Configurations		
HP	kW	Volts	Amps	HP	kW	Volts	Amps	Single	UT	CT
10	7.5	436	15	8.3	6	363	15	x	x	
15	11	436	23	12.5	9	363	23	x	x	
15	11	655	16	12.5	9	546	16	x	x	
20	15	450	28.5	16.7	12	375	28.5	X	x	
20	15	750	17	16.7	12	625	17	X	x	
25	19	410	39	20.8	16	342	39	X	x	
25	19	690	22	20.8	16	575	22	X	x	
30	22	426	44.5	25	19	355	44.5	X	x	
30	22	750	25.5	25	19	625	25.5	X	x	
30	22	1260	15	25	19	1050	15	X	x	
35	26	385	57	29.2	22	321	57	X	x	
35	26	785	28	29.2	22	654	28	X	x	
40	30	431	59	33.3	25	359	59	X	x	
40	30	880	29	33.3	25	733	29	X	x	
40	30	1340	19	33.3	25	1117	19	X	x	
50	37	674	47	41.7	31	562	47	X	x	
50	37	815	39	41.7	31	679	39	X	x	
50	37	1390	23	41.7	31	1158	23	X	x	
60	45	640	59	50	37	533	59	X	x	
60	45	745	52	50	37	621	52	X	x	
60	45	810	47	50	37	675	47	X	x	
60	45	970	39	50	37	808	39	X	x	
60	45	1330	29	50	37	1108	29	X	x	
70	52	540	82.5	58.3	44	450	82.5	x	x	x
70	52	750	60	58.3	44	625	60	x	x	x
70	52	946	47	58.3	44	788	47	x	x	x
70	52	1134	39	58.3	44	945	39	x	x	x
80	60	635	80	66.7	50	529	80	x	x	x
80	60	860	60	66.7	50	717	60	x	x	x
80	60	1085	46	66.7	50	904	46	x	x	x
80	60	1310	39	66.7	50	1092	39	x	x	x
80	60	2155	24	66.7	50	1796	24	x	x	
90	67	710	81	75	56	592	81	x	x	x
90	67	960	59	75	56	800	59	x	x	x
90	67	1135	50	75	56	946	50	x	x	x
90	67	1220	46	75	56	1017	46	x	x	x
90	67	1460	39	75	56	1217	39	x	x	x
90	67	1960	29	75	56	1633	29	x	x	
100	75	790	80	83.3	62	658	80	x	x	x
100	75	920	70	83.3	62	767	70	x	x	x
100	75	1075	59	83.3	62	896	59	x	x	x
100	75	1355	46	83.3	62	1129	46	x	x	x
100	75	2205	28.5	83.3	62	1838	28.5	x	x	
110	82	1190	60	91.7	68	992	60	x	x	x
110	82	1488	46	91.7	68	1240	46	x	x	x
110	82	2380	30	91.7	68	1983	30	x	x	
120	90	945	81	100	75	788	81	x	x	x
120	90	1125	70	100	75	938	70	x	x	x
120	90	1295	59	100	75	1079	59	x	x	x
120	90	1626	46	100	75	1355	46	x	x	
120	90	2245	35	100	75	1871	35	x	x	



Table 8 IL-150 540 Motor Table

60Hz				50Hz				Configuration		
HP	kW	Volts	Amps	HP	kW	Volts	Amps	Single	UT	CT
20	15	440	29	17	12	367	29	x	x	
20	15	755	17	17	12	629	17	x	x	
30	22	435	45	25	19	363	45	x	x	
30	22	710	28	25	19	592	28	x	x	
30	22	1215	16	25	19	1013	16	x	x	
40	30	435	60	33	25	363	60	x	x	
40	30	660	40	33	25	550	40	x	x	
40	30	730	36	33	25	608	36	x	x	
40	30	880	30	33	25	733	30	x	x	
40	30	1325	20	33	25	1104	20	x	x	
50	37	450	72	42	31	375	72	x	x	
50	37	725	45	42	31	604	45	x	x	
50	37	905	34	42	31	754	34	x	x	
50	37	1375	22	42	31	1146	22	x	x	
60	45	425	91	50	37	354	91	x	x	
60	45	460	82	50	37	383	82	x	x	
60	45	645	60	50	37	538	60	x	x	
60	45	870	45	50	37	725	45	x	x	
60	45	970	40	50	37	808	40	x	x	
60	45	1320	30	50	37	1100	30	x	x	
70	52	755	60	58	43	629	60	x	x	
70	52	1015	45	58	43	846	45	x	x	
70	52	1545	30	58	43	1288	30	x	x	
80	60	865	60	67	50	721	60	x	x	
80	60	1160	45	67	50	967	45	x	x	
90	67	755	76	75	56	629	76	x	x	
90	67	965	60	75	56	804	60	x	x	
90	67	1300	45	75	56	1083	45	x	x	
90	67	1630	35	75	56	1358	35	x	x	
90	67	1980	30	75	56	1650	30	x	x	
100	75	710	89	83	62	592	89	x	x	
100	75	835	76	83	62	696	76	x	x	
100	75	1070	60	83	62	892	60	x	x	
100	75	1450	44	83	62	1208	44	x	x	
100	75	2170	29	83	62	1808	29	x	x	
110	82	920	76	92	68	767	76	x	x	
110	82	1180	59	92	68	983	59	x	x	
110	82	1545	45	92	68	1288	45	x	x	
110	82	1985	35	92	68	1654	35	x	x	
110	82	2390	29	92	68	1992	29	x	x	
120	89	855	88	100	75	713	88	x	x	x
120	89	1030	73	100	75	858	73	x	x	x
120	89	1295	59	100	75	1079	59	x	x	x
120	89	1740	44	100	75	1450	44	x	x	
120	89	2165	33	100	75	1804	33	x	x	
130	97	925	88	108	81	771	88	x	x	x
130	97	1125	67	108	81	938	67	x	x	x
130	97	1896	44	108	81	1580	44	x	x	
140	104	1000	88	117	87	833	88	x	x	x



140	104	1170	76	117	87	975	76	x	x	x
140	104	1510	59	117	87	1258	59	x	x	x
140	104	1965	45	117	87	1638	45	x	x	
140	104	2525	35	117	87	2104	35	x	x	
150	112	1075	87	125	93	896	87	x	x	x
150	112	2105	44	125	93	1754	44	x	x	
160	119	825	122	133	99	688	122	x	x	x
160	119	1115	88.5	133	99	929	88.5	x	x	x
160	119	2185	46	133	99	1821	46	x	x	
170	127	880	120	142	106	733	120	x	x	x
170	127	1210	88	142	106	1008	88	x	x	x
170	127	1840	59	142	106	1533	59	x	x	
170	127	2390	44	142	106	1992	44	x	x	
180	134	945	120	150	112	788	120	x	x	x
180	134	1275	89	150	112	1063	89	x	x	x
180	134	1945	59	150	112	1621	59	x	x	
190	142	1000	120	158	118	833	120	x	x	x
190	142	1345	89	158	118	1121	89	x	x	x
190	142	2055	59	158	118	1713	59	x	x	
190	142	2595	46	158	118	2163	46	x	x	
200	149	1100	115	167	124	917	115	x	x	x
200	149	1416	90	167	124	1180	90	x	x	x
200	149	2140	54	167	124	1783	54	x	x	
225	168	1135	127	187	140	946	127	x	x	x
225	168	1470	97	187	140	1225	97	x	x	x
225	168	2235	62	187	140	1863	62	x	x	



Table 9 IL-150 562 Motor Table

60Hz				50Hz				Configurations Available		
HP	kW	Volts	Amps	HP	kW	Volts	Amps	Single	UT	CT
35	26	510	43	29	22	425	43	x	x	
50	37	765	41	42	31	638	41	x	x	
70	52	850	50	59	44	708	50	x	x	
70	52	1295	34	59	44	1079	34	x	x	
85	64	1065	49	71	53	888	49	x	x	
100	75	1270	48	84	63	1058	48	x	x	
120	90	1500	49	100	75	1250	49	x	x	x
140	105	1345	64	117	87	1120	64	x	x	x
140	105	1715	50	117	87	1429	50	x	x	
155	116	1510	63	129	96	1258	63	x	x	x
155	116	1905	50	129	96	1588	50	x	x	
170	127	1065	97	142	106	888	97	x	x	x
170	127	1690	62	142	106	1408	62	x	x	
170	127	2130	49	142	106	1775	49	x	x	
190	142	1150	99	159	119	958	99	x	x	x
190	142	1835	63	159	119	1529	63	x	x	
190	142	2300	51	159	119	1917	51	x	x	
205	153	1285	97	171	128	1071	97	x	x	x
205	153	2025	62	171	128	1688	62	x	x	
205	153	2565	49	171	128	2138	49	x	x	
220	164	1350	99	184	137	1125	99	x	x	x
220	164	2160	62	184	137	1800	62	x	x	
220	164	2725	49	184	137	2271	49	x	x	
240	179	1470	99	200	149	1225	99	x	x	x
240	179	2340	62	200	149	1950	62	x	x	
255	190	1595	97	213	159	1329	97	x	x	
255	190	2520	62	213	159	2100	62	x	x	
275	205	1680	99	229	171	1400	99	x	x	
275	205	2665	63	229	171	2221	63	x	x	
290	217	1775	99	242	181	1479	99	x	x	
290	217	2820	63	242	181	2350	63	x	x	
310	232	1890	99	259	193	1575	99	x	x	
310	232	3000	62	259	193	2500	62	x	x	
325	243	1980	99	271	202	1650	99	x	x	
325	243	3155	63	271	202	2629	63	x	x	
340	254	2125	97	284	212	1771	97	x	x	
340	254	3360	61	284	212	2800	61	x	x	



Table 10 Pump Information, 60 Hz – 3500 RPM

Stage Name	Series	Construction		Stage Type		Recommended Operating Range (ROR) (BPD)	Flow at (BEP) (BPD)	Pump only efficiency at BEP	BHP per Stage at BEP	Shut Off Head per Stage (feet)	Housing Pressure Limit (psi)	
		Flt.	Comp.	Radial Flow	Mixed Flow						V Thread	Buttress Thread
400-180	400	X	x	X		75-300	216	41%	0.07	22	5000	6000
400-350		X	x	X		203-320	324	42%	0.11	23.6		
400-450		X	x	X		300-555	440	47%	0.14	29.7		
400-700		X	x	X		500-900	700	57%	0.24	31.5		
400-950		X	x	X		650-1200	930	59%	0.36	35		
400-1250		X	x	X		800-1600	1240	63%	0.41	32.4		
400-1750a		x	x		x	1200-2050	1695	66%	0.39	29.5		
400-2200		x	x		x	1550-2650	2245	67%	0.59	33		
400-3000a		x	x		x	2100-3900	2900	64%	0.60	28.2		
400-4500		x	x		x	3000-5400	4410	68%	1.0	28.3		
400-5800		x	x		x	4250-7500	5335	68%	1.33	29.1		
513-1600	513	x	x	X		1200-2000	1600	59%	0.90	56.9	5000	n/a
513-2500		x	x	X		1860-3000	2460	65%	1.17	54		
513-3900		x	x		x	3180-5000	4010	66%	1.57	54.5		
513-6000a		x	x		x	4080-7200	5990	65%	1.8	44.5		
513-7500a		x	x		x	5080-9100	6616	62%	2.46	48		
513-10000		x	x		x	1060-1590	1272	66%	3.27	43.3		
538-1900	538	x	x	X		1050-2300	1960	61%	1.03	59.9	n/a	6000
538-2600		x	x	X		1600-3200	2600	63%	1.4	60		
538-3600		x	x	X		2400-4600	3600	70%	1.9	61.9		
538-4700		x	x		x	3500-6000	4700	67%	2.3	69.9		
538-7000		x	x		x	4000-10000	7000	72%	3.15	64.5		
538-9000		x	x		x	5500-11000	9000	71%	3.48	63.6		
538-12500		x	x		x	8000-16000	12500	75%	4.50	52.7		
675-9000	675		x		x	5000-13000	9000	71%	8.19	107.3	n/a	3000
675-12000			x		x	7500-18000	12000	76%	9.93	102		
862-18000	862		x		x	12000-23500	18000	75%	17.70	139.9	n/a	2000
862-25000			x		x	19000-32500	25000	71%	29.0	145.3		



Table 11 Pump Information, 50 Hz – 2917 RPM

Stage Name	Series	Construction		Stage Type		Recommended Operating Range (ROR) (m3/day)	Flow at (BEP) (m3/day)	Pump only efficiency at BEP	kW per Stage at BEP	Shut Off Head per Stage (meter)	Housing Pressure Limit (bar)	
		Flt.	Comp.	Radial Flow	Mixed Flow						V Thread	Buttress Thread
400-180	400	X	x	X		10-40	29	41%	0.03	6.7	346	415
400-350		X	x	X		27-56	43	42%	0.05	7.2		
400-450		X	x	X		40-74	58	47%	0.06	9		
400-700		X	x	X		66-119	93	57%	0.10	9.6		
400-950		X	x	X		86-159	123	59%	0.16	10.7		
400-1250		X	x	X		106-212	164	63%	0.18	9.9		
400-1750a		x	x		x	159-272	225	66%	0.17	9		
400-2200		x	x		x	199-351	297	67%	0.25	10.1		
400-3000a		x	x		x	278-516	384	64%	0.26	8.6		
400-4500		x	x		x	397-715	584	68%	0.43	8.6		
400-5800		x	x		x	563-994	707	68%	0.57	8.9		
513-1600	513	x	x		x	159-265	212	59%	0.39	17.4	345	n/a
513-2500		x	x		x	246-397	326	65%	0.50	16.5		
513-3900		x	x		x	421-662	531	66%	0.68	16.5		
513-6000a		x	x		x	591-954	794	65%	0.77	13.6		
513-7500a		x	x		x	673-1206	877	62%	1.06	14.6		
513-10000		x	x		x	1060-1590	1272	66%	1.40	13.2		
538-1900	538	x	x	X		139-305	260	61%	0.45	18.1	n/a	414
538-2600		x	x	X		212-424	345	63%	0.59	18.3		
538-3600		x	x	X		318-609	477	70%	0.80	18.9		
538-4700		x	x		x	464-795	623	67%	0.98	21.3		
538-7000		x	x		x	530-1325	928	72%	1.36	19.7		
538-9000		x	x		x	729-1457	1193	71%	1.50	19.4		
538-12500		x	x		x	1060-2120	1656	75%	1.94	16.1		
675-9000	675		x		x	662-1722	1193	71%	3.54	32.7	n/a	207
675-12000			x		x	994-2385	1590	76%	4.3	31.1		
862-18000	862		x		X	1590-3114	2385	75%	7.62	42.6	n/a	138
862-25000			x		x	2517-4305	3313	71%	12.5	44.3		



Table 12 Fluid Velocity Past the Motor

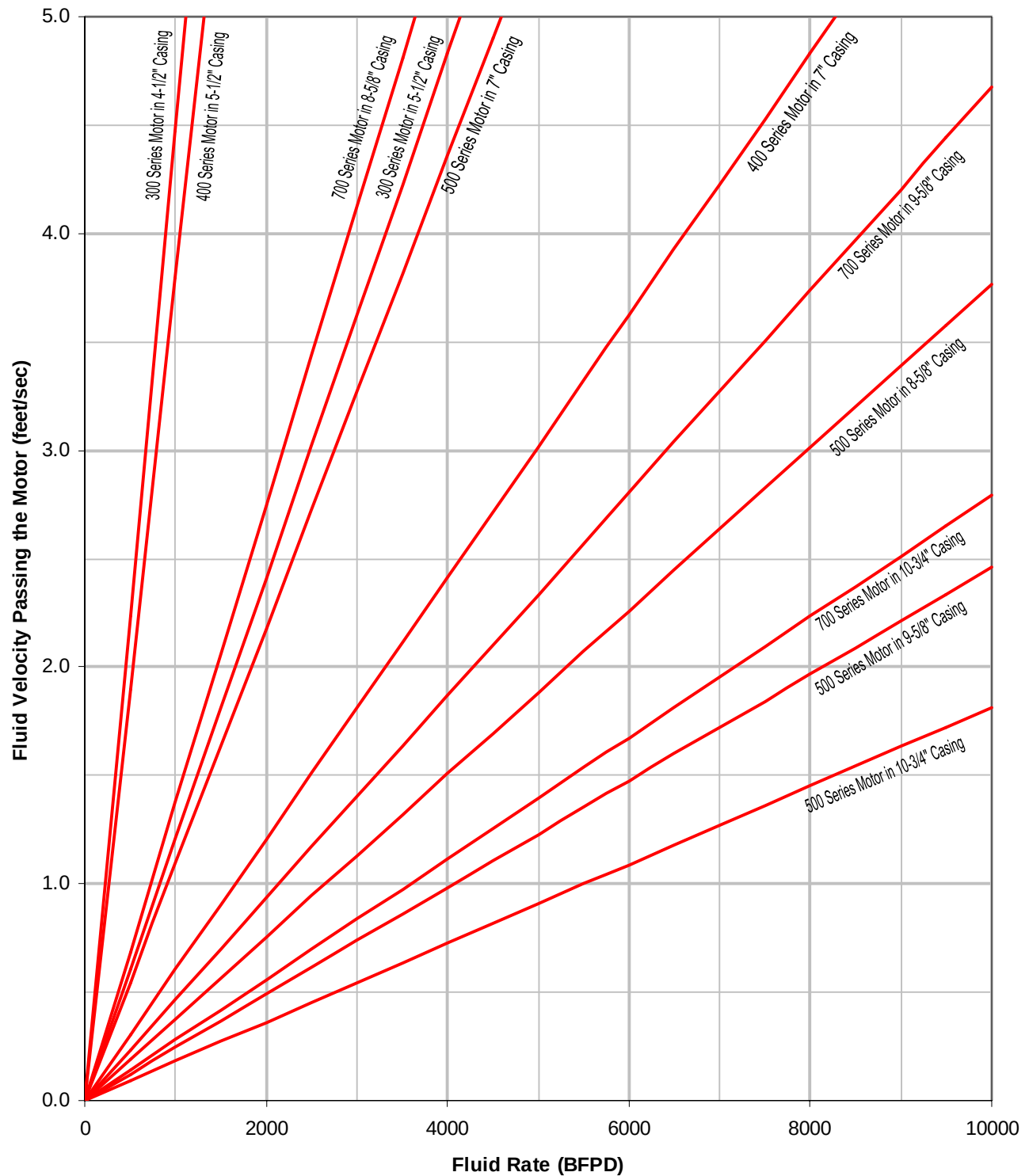
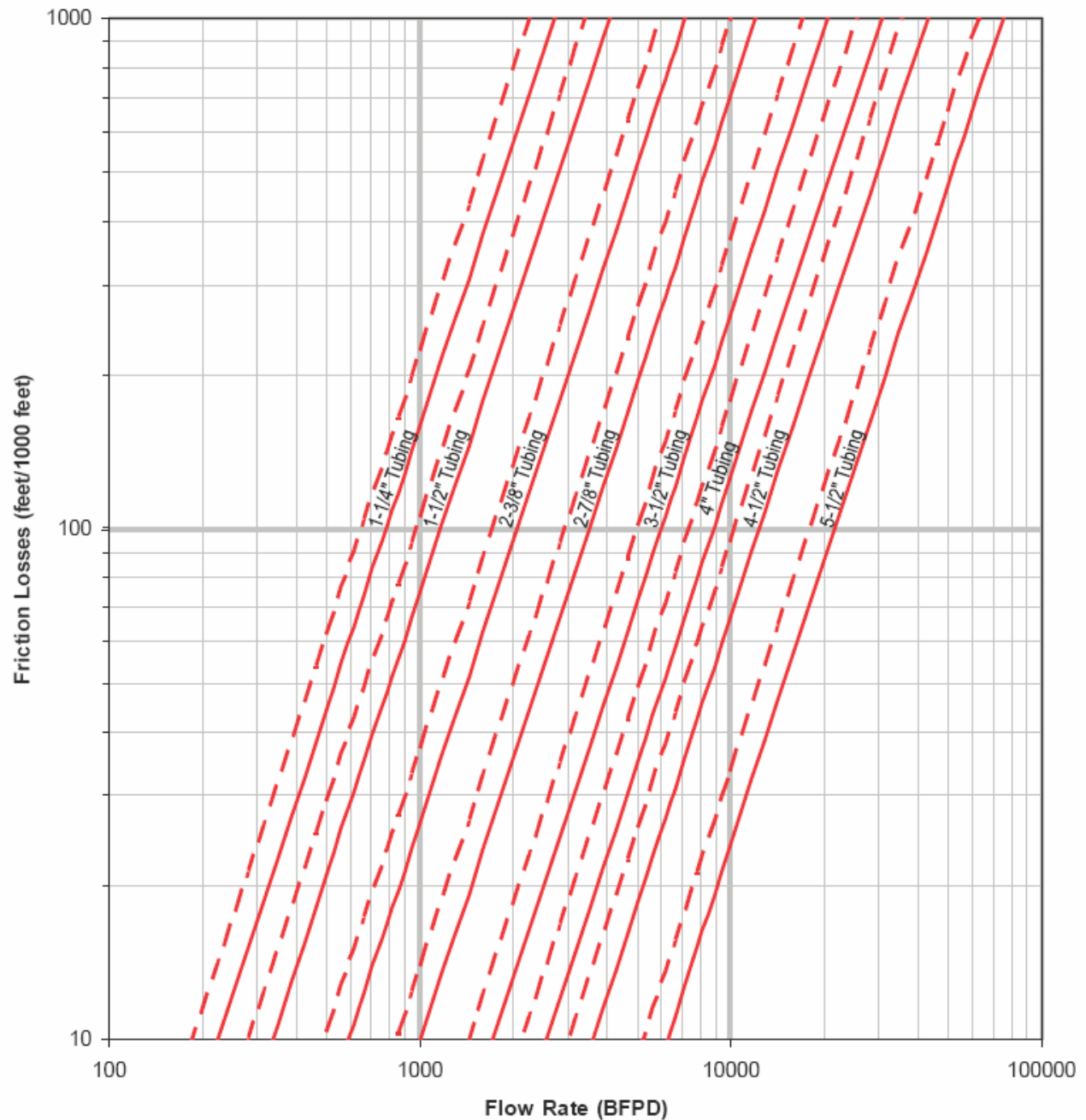




Table 13 Tubing Friction Loss



Based on Hazen William Formula:

$$F = \frac{2.083 \left(\frac{100}{C} \right)^{1.85} \left(\frac{Q}{34.3} \right)^{1.85}}{ID^{4.8655}}$$

Where:

F: Friction Losses (feet/1000 feet)
C: Pipe Coefficient (Old=100, New=120)
Q: Flow Rate (BFPD)
ID: Tubing Inside Diameter (inches)



Table 14 Power Cable Information

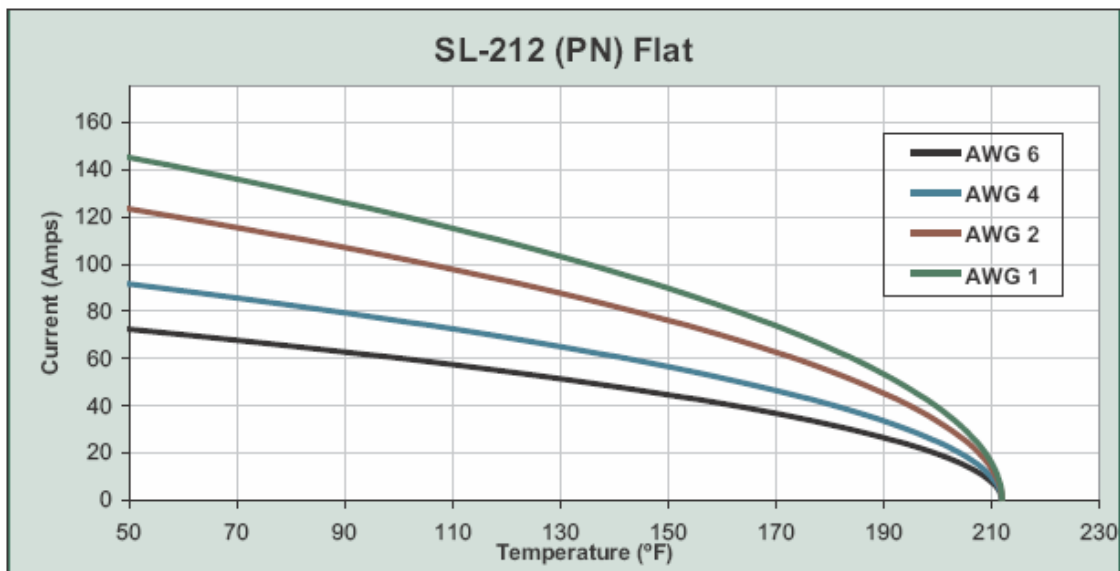
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance r/k -ft@20°C	Overall Dimensions		Weight	
	Strands	Wire		in	mm	[lb/ft]	[kg/m]

5kV SL-212 [PN]-Flat							
1	7	Stranded	0.137	0.76 x 2.04	19.3 x 51.8	1.92	2.9
1	7	Compacted	0.137	0.74 x 1.95	18.8 x 49.6	1.85	2.8
2	7	Stranded	0.172	0.73 x 1.93	18.6 x 49.1	1.65	2.5
2	7	Compacted	0.172	0.71 x 1.88	18.0 x 47.8	1.58	2.4
4	1	Solid	0.254	0.65 x 1.67	16.4 x 42.4	1.18	1.8
6	1	Solid	0.410	0.60 x 1.54	15.3 x 39.1	0.92	1.4

4kV SL-212 [PN] – Flat							
1	7	Stranded	0.137	0.76 x 2.01	19.2 x 50.9	1.85	2.8
1	7	Compacted	0.137	0.72 x 1.91	18.4 x 48.4	1.79	2.7
2	7	Stranded	0.172	0.72 x 1.90	18.3 x 48.2	1.58	2.4
2	7	Compacted	0.172	0.70 x 1.83	17.8 x 46.6	1.52	2.3
4	1	Solid	0.254	0.62 x 1.60	15.8 x 40.6	1.12	1.7
6	1	Solid	0.410	0.58 x 1.47	14.7 x 37.3	0.87	1.3

Ampacity Chart



SubLine SL-212 (PN) Parallel



SubLine SL-212 (PN) Round

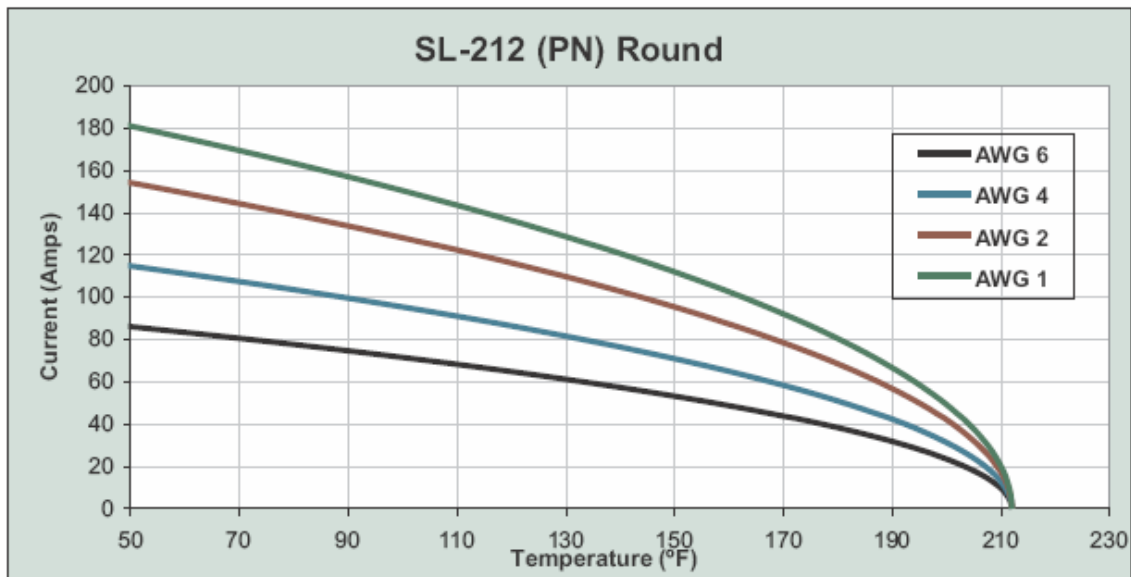
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance	Overall Dimensions		Weight	
	Strands	Wire	r/k-ft@20°C	in	mm	[lb/ft]	[kg/m]

5kV SL-212 [PN] – Round							
1	7	Stranded	0.137	1.44	36.4	1.82	2.7
1	7	Compacted	0.137	1.37	34.8	1.75	2.6
2	7	Stranded	0.172	1.36	34.5	1.58	2.3
2	7	Compacted	0.172	1.32	33.6	1.51	2.2
4	1	Solid	0.254	1.17	29.6	1.17	1.7
6	1	Solid	0.410	1.08	27.3	0.96	1.4

4kV SL-212 [PN] – Round							
1	7	Stranded	0.137	1.41	35.8	1.73	2.6
1	7	Compacted	0.137	1.41	34.0	1.66	2.5
2	7	Stranded	0.172	1.32	33.6	1.50	2.2
2	7	Compacted	0.172	1.29	32.7	1.43	2.1
4	1	Solid	0.254	1.12	28.3	1.10	1.6
6	1	Solid	0.410	1.02	26.0	0.88	1.3

Ampacity Chart





SubLine SL-285 (EN) Parallel

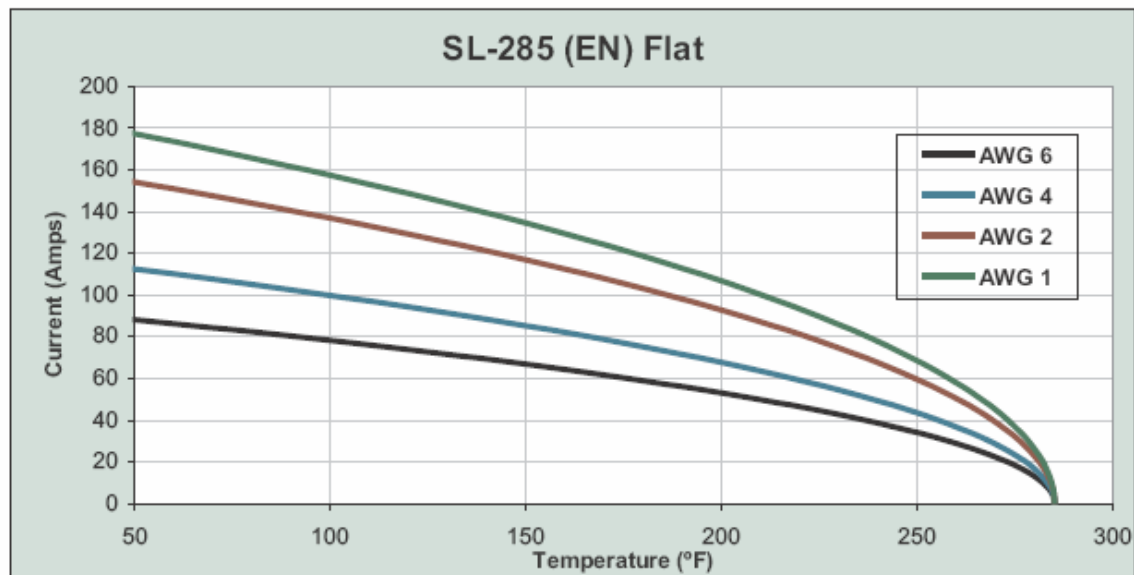
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance	Overall Dimensions		Weight	
	Strands	Wire	r/k -ft@20°C	in	mm	[lb/ft]	[kg/m]

5kV SL-285 [EN] – Flat							
1	7	Stranded	0.137	0.76 x 2.04	19.3 x 51.8	1.97	2.9
1	7	Compacted	0.137	0.74 x 1.95	18.8 x 49.6	1.90	2.8
2	7	Stranded	0.172	0.73 x 1.93	18.6 x 49.1	1.69	2.5
2	7	Compacted	0.172	0.71 x 1.88	18.0 x 47.8	1.62	2.4
4	1	Solid	0.254	0.65 x 1.67	16.4 x 42.4	1.21	1.8
6	1	Solid	0.410	0.60 x 1.54	15.3 x 39.1	0.95	1.4

4kV SL-285 [EN] – Flat							
1	7	Stranded	0.137	0.76 x 2.01	19.2 x 50.9	1.90	2.8
1	7	Compacted	0.137	0.72 x 1.91	18.4 x 48.4	1.83	2.7
2	7	Stranded	0.172	0.72 x 1.90	18.3 x 48.2	1.62	2.4
2	7	Compacted	0.172	0.70 x 1.83	17.8 x 46.6	1.55	2.3
4	1	Solid	0.254	0.62 x 1.60	15.8 x 40.6	1.16	1.7
6	1	Solid	0.410	0.58 x 1.47	14.7 x 37.3	0.89	1.3

Ampacity Chart





SubLine SL-285 (EN) Round

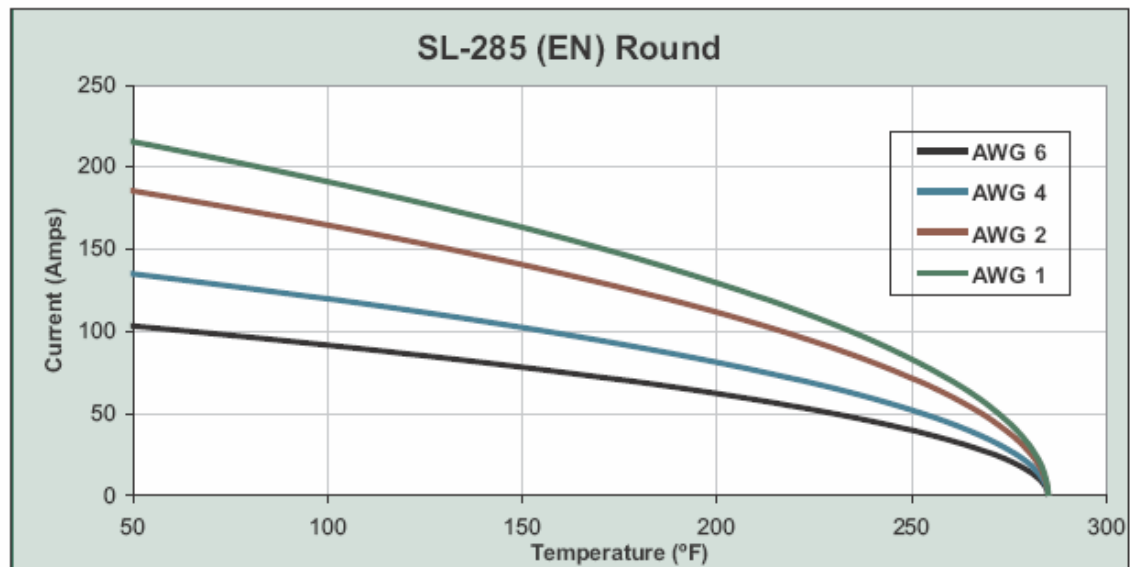
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance r/k-ft@20°C	Overall Dimensions		Weight	
	Strands	Wire		in	mm	[lb/ft]	[kg/m]

5kV SL-285 [EN] - Round							
1	7	Stranded	0.137	1.44	36.4	1.88	2.8
1	7	Compacted	0.137	1.37	34.8	1.81	2.7
2	7	Stranded	0.172	1.36	34.6	1.65	2.5
2	7	Compacted	0.172	1.32	33.6	1.58	2.4
4	1	Solid	0.254	1.17	28.5	1.18	1.8
6	1	Solid	0.410	1.08	27.3	0.96	1.4

4kV SL-285 [EN] - Round							
1	7	Stranded	0.137	1.41	35.8	1.80	2.7
1	7	Compacted	0.137	1.34	34.0	1.73	2.6
2	7	Stranded	0.172	1.32	33.6	1.60	2.4
2	7	Compacted	0.172	1.29	32.7	1.53	2.3
4	1	Solid	0.254	1.12	28.3	1.13	1.7
6	1	Solid	0.410	1.02	26.0	0.92	1.4

Ampacity Chart





SubLine SL-450 (EE) Parallel

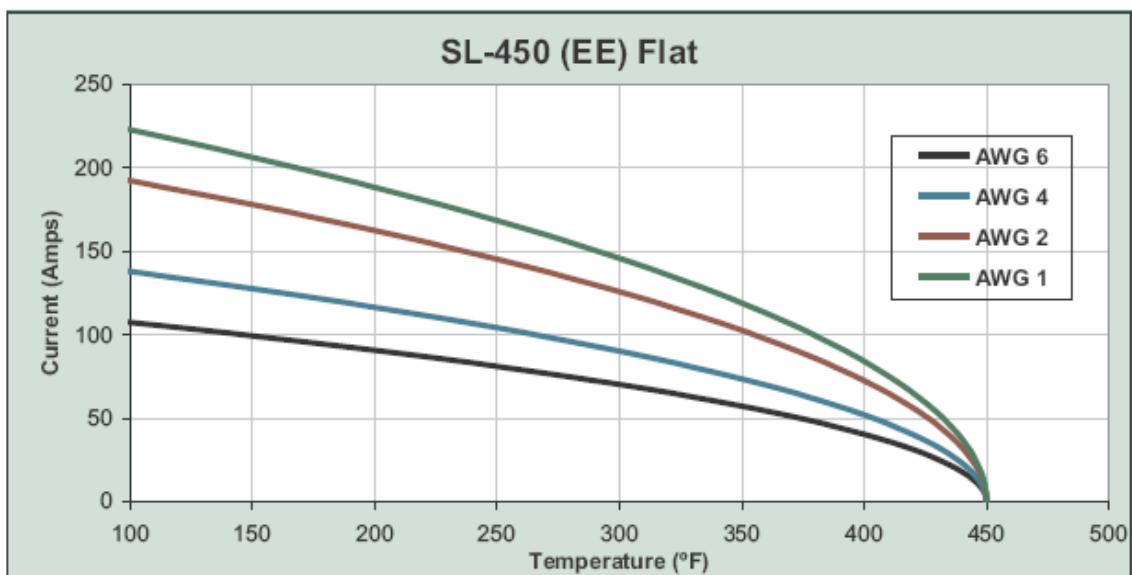
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance	Overall Dimensions		Weight	
	Strands	Wire	r/k-ft@20°C	in	mm	[lb/ft]	[kg/m]

5kV SL-450 [EE] - Flat							
1	7	Stranded	0.137	0.76 x 2.04	19.3 x 51.8	1.97	2.9
1	7	Compacted	0.137	0.74 x 1.95	18.8 x 49.6	1.90	2.8
2	7	Stranded	0.172	0.73 x 1.93	18.6 x 49.1	1.69	2.5
2	7	Compacted	0.172	0.71 x 1.88	18.0 x 47.8	1.62	2.4
4	1	Solid	0.254	0.65 x 1.67	16.4 x 42.4	1.21	1.8
6	1	Solid	0.410	0.60 x 1.54	15.3 x 39.1	0.95	1.4

4kV SL-450 [EE] - Flat							
1	7	Stranded	0.137	0.76 x 2.01	19.2 x 50.9	1.90	2.8
1	7	Compacted	0.137	0.72 x 1.91	18.4 x 48.4	1.83	2.7
2	7	Stranded	0.172	0.72 x 1.91	18.3 x 48.2	1.62	2.4
2	7	Compacted	0.172	0.70 x 1.83	17.8 x 46.6	1.55	2.3
4	1	Solid	0.254	0.62 x 1.60	15.8 x 40.6	1.16	1.7
6	1	Solid	0.410	0.58 x 1.47	14.7 x 37.3	0.89	1.3

Ampacity Chart





SubLine SL-450 (EE) Round

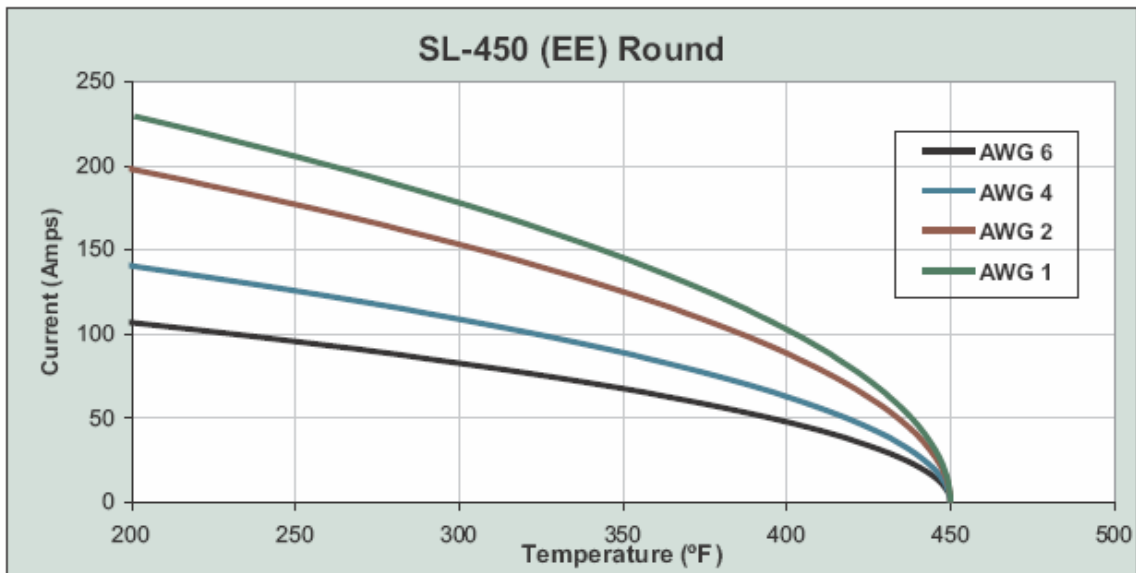
Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance	Overall Dimensions		Weight	
	Strands	Wire	r/k-ft@20°C	in	mm	[lb/ft]	[kg/m]

5kV SL-450 [EE] – Round							
1	7	Stranded	0.137	1.44	36.4	1.87	2.8
1	7	Compacted	0.137	1.37	34.8	1.80	2.7
2	7	Stranded	0.172	1.36	34.6	1.65	2.5
2	7	Compacted	0.172	1.32	33.6	1.58	2.4
4	1	Solid	0.254	1.12	29.6	1.18	1.8
6	1	Solid	0.410	1.08	27.3	0.96	1.4

4kV SL-450 [EE] – Round							
1	7	Stranded	0.137	1.41	35.8	1.80	2.7
1	7	Compacted	0.137	1.34	34.0	1.73	2.6
2	7	Stranded	0.172	1.32	33.6	1.55	2.3
2	7	Compacted	0.172	1.29	32.7	1.48	2.2
4	1	Solid	0.254	1.12	28.3	1.13	1.7
6	1	Solid	0.410	1.02	26.0	0.92	1.4

Ampacity Chart





SubLine SL-450 (E-Lead) Parallel

Mechanical and Electrical Properties

Size (AWG)	Conductor		Resistance r/k-ft@20°C	Overall Dimensions		Weight	
	Strands	Wire		in	mm	[lb/k ft]	[kg/km]

5kV SL-450 [E-Lead] - Flat							
1	7	Stranded	0.137	0.71 x 1.87	17.9 x 47.4	2.50	3.7
1	7	Compacted	0.137	0.68 x 1.78	17.2 x 45.2	2.42	3.6
2	7	Stranded	0.172	0.67 x 1.76	17.0 x 44.6	2.24	3.3
2	7	Compacted	0.172	0.65 x 1.7	16.5 x 43.1	2.00	3.0
4	1	Solid	0.254	0.58 x 1.49	14.8 x 38.0	1.74	2.6
6	1	Solid	0.410	0.54 x 1.37	13.7 x 34.7	1.42	2.1

4kV SL-450 [E-Lead] - Flat							
1	7	Stranded	0.137	0.69 x 1.82	17.5 x 46.2	2.39	3.6
1	7	Compacted	0.137	0.66 x 1.73	16.8 x 44.0	2.31	3.4
2	7	Stranded	0.172	0.65 x 1.71	16.6 x 43.5	2.14	3.2
2	7	Compacted	0.172	0.63 x 1.63	15.9 x 41.3	1.93	2.9
4	1	Solid	0.254	0.56 x 1.43	14.2 x 36.2	1.55	2.3
6	1	Solid	0.410	0.52 x 1.30	13.1 x 32.9	1.31	1.9

Ampacity Chart

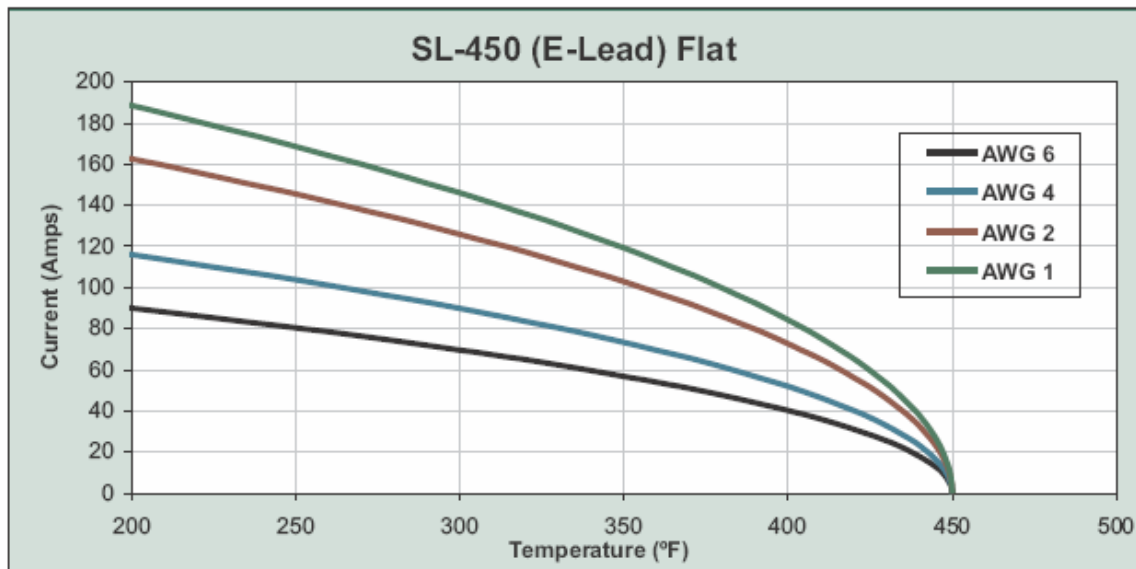




Table 15 Cable Voltage Drop Chart

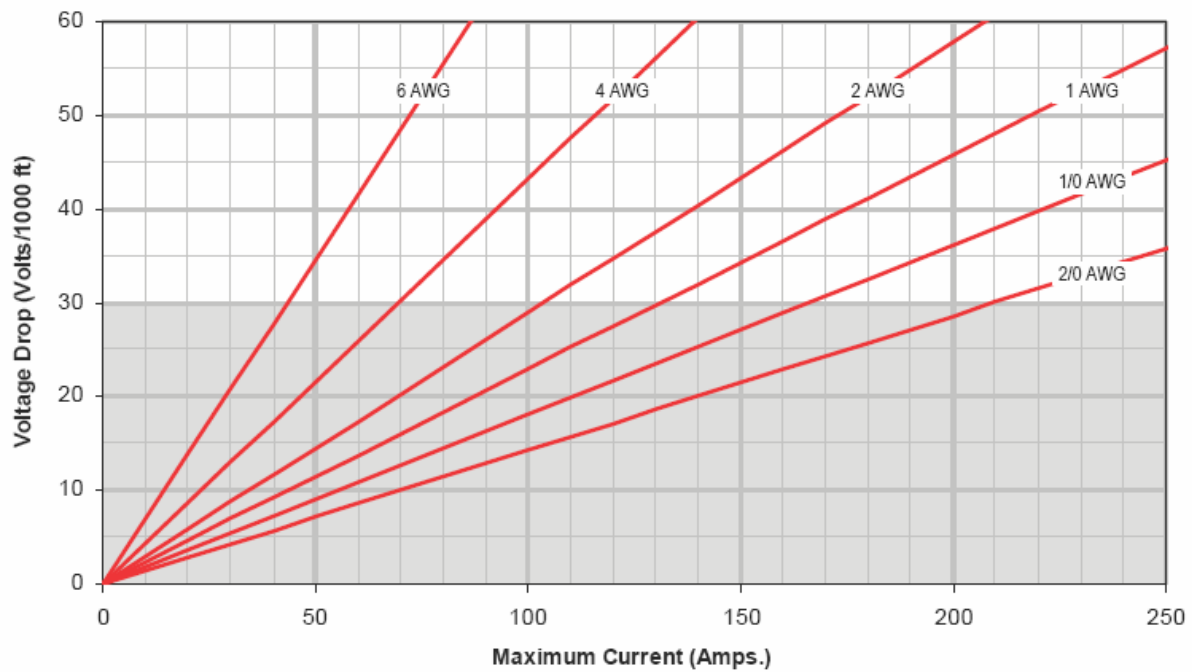


Table 16 Cable Voltage Drop Temperature Correction Chart

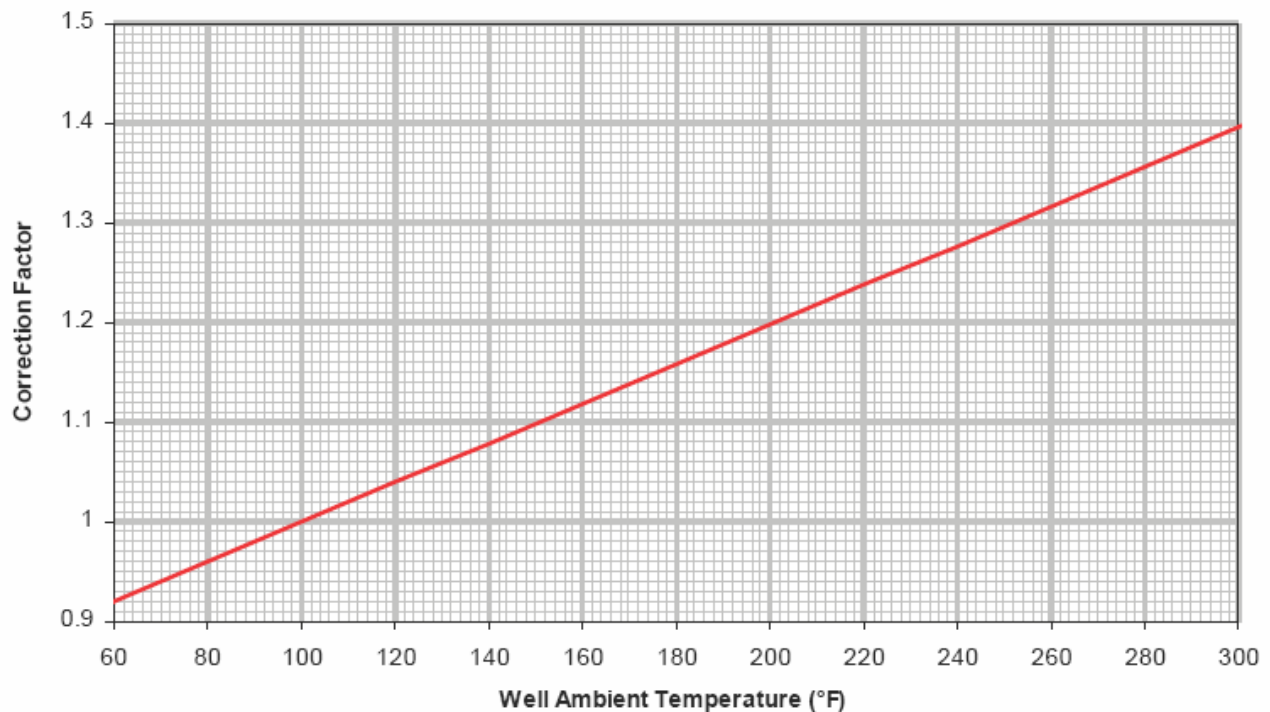




Table 17 API Tubular Goods

Size		Thread	Weight (lbs/ft)	Diameter (inches)			Coupling OD (inches)
Nominal	OD			OD	ID	Drift	
API Line Pipe							
1-1/4"		11-1/2 V	2.30	1.660	1.380		2.054
1-1/2"		11 to 11-1/2 V	2.75	1.900	1.610		2.200
2"	2-3/8"	11-1/2 V	3.75	2.375	2.067		2.875
3"	3-1/2"	8 V	7.70	3.500	3.068		4.000
4"	4-1/2"		11.00	4.500	4.026		5.200
6"	6-5/8"		19.45	6.625	6.065		7.390
8"	8-5/8"		25.55	8.625	8.071		9.625
API Tubing (Non Upset)							
1-1/2"		10rd	2.75	1.900	1.610	1.516	2.200
2"	2-3/8"		4.00	2.375	2.041	1.947	2.875
2-1/2"	2-7/8"		6.40	2.875	2.441	2.347	3.500
3"	3-1/2"		7.70	3.500	3.068	2.943	4.250
3-1/2"	4"	8rd	9.50	4.000	3.548	3.423	4.750
API Tubular (External Upset)							
1-1/2"		10rd	2.90	1.900	1.610	1.516	2.500
2"	2-3/8"	8rd	4.70	2.375	1.995	1.901	3.063
2-1/2"	2-7/8"		6.50	2.875	2.441	2.347	3.668
3"	3-1/2"		9.30	3.500	2.992	2.867	4.500
3-1/2"	4"		11.00	4.000	3.476	3.351	5.000
4"	4-1/2"		12.75	4.500	3.958	3.833	5.563
API Regular Casing							
	4-1/2"	8rd	9.50	4.500	4.090	3.965	5.000
			11.60		4.000	3.875	
	5-1/2"		14.00	5.500	5.012	4.887	6.050
			17.00		4.892	4.767	
			20.00		4.778	4.653	
	6-5/8"		17.00	6.625	6.135	6.010	7.390
			24.00		5.921	5.796	
	7"		20.00	7.000	6.456	6.331	7.656
			23.00		6.366	6.241	
			26.00		6.276	6.151	
	8-5/8"		28.00	8.625	8.017	7.892	9.625
			36.00		7.825	7.700	
	9-5/8"		36.00	9.625	8.921	8.765	10.625
			40.00		8.835	8.679	
	10-3/4"		40.50	10.750	10.050	9.894	11.750
			55.50		9.760	9.604	
	13-3/8"		48.00	13.375	12.715	12.559	14.375
			68.00		12.415	12.259	



Table 18 Gravity Correction Table

Degrees API	Specific Gravity	Weight (Density)			Fluid Head		Buoyancy Factor
		Gallon	Cubic Foot	Barrel	Pressure per Foot	Height per Pound	
		Pounds			Psi / ft²	Feet	
60.0	0.739	6.16	46.1	259	0.320	3.126	0.906
59.0	0.743	6.19	46.3	260	0.322	3.109	0.905
58.0	0.747	6.23	46.6	262	0.323	3.093	0.905
57.0	0.751	6.26	46.8	263	0.325	3.077	0.904
56.0	0.755	6.29	47.1	264	0.327	3.060	0.904
55.0	0.759	6.33	47.3	266	0.329	3.044	0.903
54.0	0.763	6.36	47.6	267	0.330	3.028	0.903
53.0	0.767	6.40	47.9	269	0.332	3.011	0.902
52.0	0.771	6.43	48.1	270	0.334	2.995	0.902
51.0	0.775	6.47	48.4	272	0.336	2.979	0.901
50.0	0.780	6.50	48.6	273	0.338	2.962	0.901
49.0	0.784	6.54	48.9	275	0.339	2.946	0.900
48.0	0.788	6.57	49.2	276	0.341	2.930	0.900
47.0	0.793	6.61	49.5	278	0.343	2.913	0.899
46.0	0.797	6.65	49.7	279	0.345	2.897	0.898
45.0	0.802	6.69	50.0	281	0.347	2.881	0.898
44.0	0.806	6.72	50.3	282	0.349	2.864	0.897
43.0	0.811	6.76	50.6	284	0.351	2.848	0.897
42.0	0.816	6.80	50.9	286	0.353	2.832	0.896
41.0	0.820	6.84	51.2	287	0.355	2.815	0.896
40.0	0.825	6.88	51.5	289	0.357	2.799	0.895
39.0	0.830	6.92	51.8	291	0.359	2.783	0.894
38.0	0.835	6.96	52.1	292	0.361	2.766	0.894
37.0	0.840	7.00	52.4	294	0.364	2.750	0.893
36.0	0.845	7.05	52.7	296	0.366	2.734	0.892
35.0	0.850	7.09	53.0	298	0.368	2.718	0.892
34.0	0.855	7.13	53.4	299	0.370	2.701	0.891
33.0	0.860	7.17	53.7	301	0.372	2.685	0.891
32.0	0.865	7.22	54.0	303	0.375	2.669	0.890
31.0	0.871	7.26	54.3	305	0.377	2.652	0.889
30.0	0.876	7.31	54.7	307	0.379	2.636	0.889
29.0	0.882	7.35	55.0	309	0.382	2.620	0.887
28.0	0.887	7.40	55.4	311	0.384	2.603	0.887
27.0	0.893	7.45	55.7	313	0.387	2.587	0.886
26.0	0.898	7.49	56.1	315	0.389	2.571	0.885
25.0	0.904	7.54	56.4	317	0.391	2.554	0.885
24.0	0.910	7.59	56.8	319	0.394	2.538	0.884
23.0	0.916	7.64	57.1	321	0.397	2.522	0.883
22.0	0.922	7.69	57.5	323	0.399	2.505	0.883
21.0	0.928	7.74	57.9	325	0.402	2.489	0.882



Degrees API	Specific Gravity	Weight (Density)			Fluid Head		Buoyancy Factor
		Gallon	Cubic Foot	Barrel	Pressure per Foot	Height per Pound	
		Pounds			Psi / ft²	Feet	
20.0	0.934	7.79	58.3	327	0.404	2.473	0.881
19.0	0.940	7.84	58.7	329	0.407	2.456	0.880
18.0	0.946	7.89	59.1	332	0.410	2.440	0.879
17.0	0.953	7.95	59.5	334	0.413	2.424	0.879
16.0	0.959	8.00	59.9	336	0.415	2.407	0.878
15.0	0.966	8.06	60.3	338	0.418	2.391	0.877
14.0	0.973	8.11	60.7	341	0.421	2.375	0.876
13.0	0.979	8.17	61.1	343	0.424	2.358	0.876
12.0	0.986	8.22	61.5	345	0.427	2.342	0.874
11.0	0.993	8.28	62.0	348	0.430	2.326	0.874
10.0 Degrees API or Fresh Water	1.000	8.34	62.4	350	0.433	2.309	0.873
	1.010	8.42	63.0	354	0.437	2.287	0.872
	1.030	8.59	64.3	361	0.446	2.242	0.869
	1.060	8.84	66.1	371	0.459	2.179	0.866
	1.080	9.01	67.4	378	0.468	2.138	0.862
Salt Water	1.100	9.17	68.6	385	0.476	2.100	0.860
	1.130	9.42	70.5	396	0.489	2.044	0.856
	1.150	9.59	71.8	403	0.498	2.008	0.853
	1.154	9.62	72.0	404	0.500	2.001	0.853
	1.180	9.84	73.6	413	0.511	1.957	0.850
	1.200	10.01	74.9	420	0.520	1.925	0.847
	1.220	10.17	76.1	427	0.528	1.893	0.844
	1.250	10.43	78.0	438	0.541	1.848	0.841
	1.270	10.59	79.2	445	0.550	1.818	0.838
	1.290	10.76	80.5	452	0.559	1.790	0.835
	1.320	11.01	82.4	462	0.572	1.750	0.832
	1.340	11.18	83.6	469	0.580	1.723	0.829
	0.137	1.14	8.5	48	0.059	16.857	0.826
	1.390	11.59	86.7	487	0.602	1.661	0.823
	1.410	11.76	88.0	494	0.611	1.638	0.820
	1.440	12.01	89.9	504	0.624	1.604	0.817
	1.460	12.18	91.1	511	0.632	1.582	0.814
	1.490	12.43	93.0	522	0.645	1.550	0.810
	1.510	12.59	94.2	529	0.654	1.529	0.808
	1.530	12.76	95.5	536	0.662	1.509	0.804
	1.560	13.01	97.3	546	0.675	1.480	0.801
	1.580	13.18	98.6	553	0.684	1.462	0.798
	1.610	13.43	100.5	564	0.697	1.434	0.795
	1.630	13.59	101.7	571	0.706	1.417	0.792
	1.650	13.76	103.0	578	0.714	1.400	0.789



Table 19 Conversion Factors

FLOW	Multiply	By	To Obtain
	Gallon / Min	34.286	Barrel / Day
	Gallon / Min	1.429	Barrel / Hour
	Gallon / Min	8.0208	Cubic Feet / Hour
	Gallon / Min	0.002228	Cubic Feet / Sec
	Gallon / Min	0.06309	Liter / Sec
	Cubic Feet / Min	10.689	Barrel / Hour
	Cubic Feet / Min	1.6957	Cubic Meter / Hour
	Cubic Meter / Hour	150.972	Barrel / Day
	Cubic Meter / Hour	4.4033	Gallon / Min
	Cubic Feet / Second	448.831	Gallon / Min
	Cubic Feet / Sec	0.1247	Gallon / Sec
	Liter / Min	0.0005885	Cubic Feet / Sec
	Barrel / Day	0.02917	Gallon / Min
	Barrel / Hour	0.700	Gallon / Min

PRESSURE	Multiply	By	To Obtain
	Cm. of Mercury	0.01316	Atmospheres
	Cm. of Mercury	135.95	Kgs. / Sq. Meter
	Cm. of Mercury	0.1934	Lbs. / Sq. Inch
	Inches of Mercury	0.03342	Atmospheres
	Inches of Mercury	0.03453	Kgs. / Sq. Cm.
	Inches of Mercury	0.4912	Lbs. / Sq. Inch
	Kgs. / Sq. Cm.	0.9678	Atmospheres
	Kgs. / Sq. Cm.	28.96	Inches of Mercury
	Kgs. / Sq. Cm.	2048.0	Lbs. / Sq. Foot
	Kgs. / Sq. Cm.	14.223	Lbs. / Sq. Inch
	Lbs. / Sq. Inch	0.06805	Atmospheres
	Lbs. / Sq. Inch	2.036	Inches of Mercury
	Lbs. / Sq. Inch	0.07031	Kgs. / Sq. Cm.
	Lbs. / Sq. Inch	0.0004882	Kgs. / Sq. Cm.
	Lbs. / Sq. Inch	6.8948	Kilopascal
	Atmospheres	1.01325	BAR
	Atmospheres	29.92	Inches of Mercury
	Atmospheres	1.0332	Kgs. / Sq. Cm.
	Kilopascal	100.0	BAR
	Kilopascal	0.14504	Lbs. / Sq. Inch
	Joule	0.73756	Ft.-Lbs. (Force)
	BAR	14.504	Lbs. / Sq. Inch

GOR or PI	Multiply	By	To Obtain
	M ³ / Day / (Kg/cm ²)	0.44217	Barrels / Day / PSI
	Barrels / Day / PSI	2.261574	M ³ / Day / (Kg/cm ²)
	Meter ² / Meter ²	5.6145	Cubic Feet / Barrel
HEAD	Cubic Feet / Barrel	0.17811	Barrels / Day / PSI

HEAD	Multiply	By	To Obtain
	Feet of Water	0.02945	Atmospheres
	Feet of Water	0.8811	Inches of Mercury
HEAD	Feet of Water	0.03042	Kgs. / Sr. Cm.

	Feet of Water	0.4328	Lbs. / Sq. In.
	Inch of Water	0.002454	Atmospheres
	Pound / Foot	1.4882	Kgs. / Meter
	Kilogram / Meter	0.6720	Lbs. /Ft.

VOLUME	Multiply	By	To Obtain
	Gallon (US)	0.02381	Barrel (Oil)
	Gallon (US)	0.003785	Cubic Meter
	Gallon (US)	0.00495	Cubic Yard
	Gallon (US)	0.83267	Gallon (Imperial)
	Gallon Water (US)	0.338	Pound of Water
	Gallon (Imperial)	1.20095	Gallon (US)
	Cubic Feet	0.1781	Barrels
	Cubic Feet	0.02832	Cubic Meter
	Cubic Feet	7.48052	Gallons
	Cubic Feet	28.32	Liter
	Cubic Meter	6.289	Barrels
	Ton (Metric)	7.454	Barrels (36°API)
	Cubic Inch	16.387	Cubic Centimeter
	Cubic Inch	0.01639	Liter
	Cubic Centimeter	0.00003531	Cubic Feet
	Cubic Centimeter	0.002113	Pints (Liquid)
	Liter	0.03531	Cubic Feet
	Liter	61.02	Cubic Inch
	Liter	0.2642	Gallon (US)
	Liter	1.057	Quarts (Liquid)
	Barrel	5.6146	Cubic Feet
	Barrel	0.15898	Cubic Meter
	Barrel (Oil)	42.0	Gallons (US)
	Barrels (36°API)	0.1342	Metric Ton
	Cubic Meter	35.315	Cubic Feet
	Cubic Meter	1.308	Cubic Yard
	Cubic Meter	264.17	Gallons
	Cubic Yard	0.7646	Cubic Meter
	Cubic Yard	201.97	Gallons
	Quart (Liquid)	0.946	Liter

LENGTH	Multiply	By	To Obtain
	Feet	30.48	Centimeters
	Feet	0.3048	Meters
	Inches	2.540	Centimeters
	Kilometers	3281.0	Feet
	Kilometers	0.6214	Miles
	Kilometers	1094.00	Yards
	Meters	3.281	Feet
	Meters	39.37	Inches
	Meters	1.094	Yards
	Centimeters	0.3937	Inches
	Miles	1.609	Kilometers



AREA	Multiply	By	To Obtain
	Sq. Inches	6.4516	Sq. Centimeters
	Sq. Meters	10.764	St. Feet
	Sq. Feet	0.0929	Sq. Meters

WEIGHT and DENSITY	Multiply	By	To Obtain
	Ounce	16.0	Drams
	Ounce	0.0625	Pounds (Avoir.)
	Dram	27.34375	Grains
	Grams / Liter	58.417	Grain / Gallon (US)
	Pound / Cubic Feet	0.01602	Grams / Cubic Cm.
	Pound / Cubic Feet	16.02	Kgs. / Cubic Meter
	Pound / Cubic Inch	27680.00	Kgs. / Cubic Meter
	Pound	0.45359	Kilograms
	Pound (US)	16.0	Ounces
	Pound (Troy)	12.0	Ounces
	Pound (Troy)	0.0003674	Tons (Long)
	Pound (Troy)	0.0003732	Tons (Metric)
	Pound (Troy)	0.0004114	Tons (Short)
	Ton (Metric)	2205.0	Pounds (Avoir.)
	Ton (Short)	2000.0	Pounds (Avoir.)
	Ton (Short)	907.18486	Kilograms
	Ton (Long)	2240.0	Pounds (Avoir.)
	Part / Million	8.328	Lbs. / Million Gal.
	Grain (Troy)	0.0022857	Ounces (Troy)
	Grain / US Gallon	17.118	Parts / Million
	Gram / Centimeter	0.0056	Pounds / Inch

VELOCITY and ACCELERATION	Multiply	By	To Obtain
	Rev. / Min. (RPM)	0.1047	Radian / Sec
	Foot / Min	0.5080	Centimeter / Sec.
	Foot / Min.	0.01829	Kilometer / Hr.
	Foot / Min.	0.3048	Meter / Min.
	Centimeter / Second	1.969	Foot / Min.
	Centimeter / Second	0.6	Meter / Min.
	Meter / Min.	3.281	Foot / Min
	Meter / Min.	0.05468	Foot / Sec.
	Meter / Sec.	3.281	Foot / Sec.
	Mile / Hr.	1.609	Kilometer / Hr.
	Kilometer / Hr.	16.67	Meter / Min.
	Kilometer / Hr.	0.6214	Mile / Hr.
	Foot / Sec ²	30.48	Centimeter / Sec ²

POWER and FORCE	Multiply	By	To Obtain
	Horsepower	33000.00	Ft-Lbs. (Force) / Min.
	Horsepower	550.00	Ft-Lbs. (Force) / Sec.
	Horsepower	1.0139	Horsepower (Metric)
	Horsepower	745.7	Watt
	Watt	44.254	Ft-Lbs. (Force) / Min.
	Watt	0.0013410	Horsepower
	Watt	1.0	Joule / Sec.

	Newton	0.10197	Kilogram (Force)
	Newton	0.2248	Pound (Force)
	Watt-Hour	2655.00	Ft-Lbs. (Force)
	Watt-Hour	3600.00	Joule
	Watt-Hour	367.10	Kilogram (Force) – M.

TEMPERATURE	Multiply	By	To Obtain
	Centigrade	+273	Kelvin
	Fahrenheit	+460	Rankine
	Centigrade	+32 x 1.8	Fahrenheit
	Fahrenheit	-32 x 1/1.8	Centigrade



Table 20 Useful Formulas

Velocity (ft/sec)	$\frac{\text{GPM}}{2.46 \times D^2}$	$\frac{\text{LPM}}{4.76 \times D^2}$
	$\frac{0.321 \times \text{GPM}}{\text{Area}}$	$\frac{0.165 \times \text{LPM}}{\text{Area}}$
GPM	$\frac{\text{Area} \times \text{Velocity}}{0.321}$	$\frac{\text{Area} \times \text{Velocity}}{0.165}$
	$2.448 \times V \times D^2$	$4.76 \times V \times D^2$
Pipe Diameter (Inches)	$[\text{GPM} / (2.45 \times V)]^{0.5}$	$[\text{LPM} / (4.76 \times V)]^{0.5}$
GPD	$3530 \times V \times D^2$	$6850 \times V \times D^2$

Definitions

A:	Area in Square Inches	A:	Area in Centimeters
D:	ID of Pipe in Inches	D:	ID of Pipe in Centimeters
GPM:	Gallons per Minute	LPM:	Liters per Minute
GPD:	Gallons per Day	LPD:	Liters per Day
V:	Velocity in Feet / Second	V:	Velocity in Meters / Second
BEP:	= Best Efficient Point	I:	Amperes
BPD:	= Barrels per Day	Kp:	Kilopascal
M ³ /Day: =	Cubic Meters per Day	KW:	Kilowatts
Eff: =	Efficiency (Decimal)	M:	Meters
GPM: =	Gallons per Minute	PF:	Power Factor (Decimal)
H:	Head	RPM:	Revolutions per Minute
HP:	Horsepower	Sp.Gr.:	Specific Gravity

TO FIND	IMPERIAL UNITS	METRIC UNITS
Brake HP	$\frac{\text{GPM} \times \text{TDH (feet)} \times \text{Sp. Gr.}}{3960 \times \text{Pump Eff.}}$	$\frac{\text{m}^3/\text{Day} \times \text{TDH (meter)} \times \text{Sp. Gr.}}{6570 \times \text{Pump Eff.}}$
	$\frac{\text{BPD} \times \text{TDH (feet)} \times \text{Sp. Gr.}}{135788 \times \text{Pump Eff.}}$	$\frac{\text{m}^3/\text{Day} \times \text{TDH (meter)} \times \text{Sp. Gr.}}{64300 \times \text{Pump Eff.}}$
	$\frac{\text{GPM} \times \text{TDH (psi)} \times \text{Sp. Gr.}}{1715 \times \text{Pump Eff.}}$	$\frac{\text{m}^3/\text{Day} \times \text{TDH (Bar)} \times \text{Sp. Gr.}}{643 \times \text{Pump Eff.}}$
	$\frac{\text{BPD} \times \text{TDH (psi)} \times \text{Sp. Gr.}}{58807 \times \text{Pump Eff.}}$	
Pump Efficiency	$\frac{\text{GPM} \times \text{Head (feet)} \times \text{Sp. Gr.}}{3960 \times \text{Brake HP}}$	$\frac{\text{m}^3/\text{Day} \times \text{Head (meter)} \times \text{Sp. Gr.}}{6570 \times \text{Brake Horsepower}}$

TO FIND	IMPERIAL UNITS	METRIC UNITS
GPM	$\frac{\text{Pounds Fluid per Hour}}{500 \times \text{Sp. Gr.}}$	$\frac{\text{Kilograms Fluid per Hour}}{227 \times \text{Sp. Gr.}}$
BPD	$\frac{\text{Pounds Fluid per Hour}}{14.6 \times \text{Sp. Gr.}}$	
m ³ /Day	$\frac{\text{Pounds Fluid per Hour}}{91.8 \times \text{Sp. Gr.}}$	$\frac{\text{Kilograms Fluid per Hour}}{41.7 \times \text{Sp. Gr.}}$
HP Input (3-Phase)	$\frac{I \times V \times 1.73 \times \text{PF}}{746}$	
	KW x 1.34	
KVA (3-Phase)	$\frac{V \times I \times 1.73}{1000}$	
Torque/ Ft. Pounds	$\frac{\text{BHP} \times 5250}{\text{RPM}}$	



Table 21 SubPump – Pump Performance with Gas Graph – Example VFD application

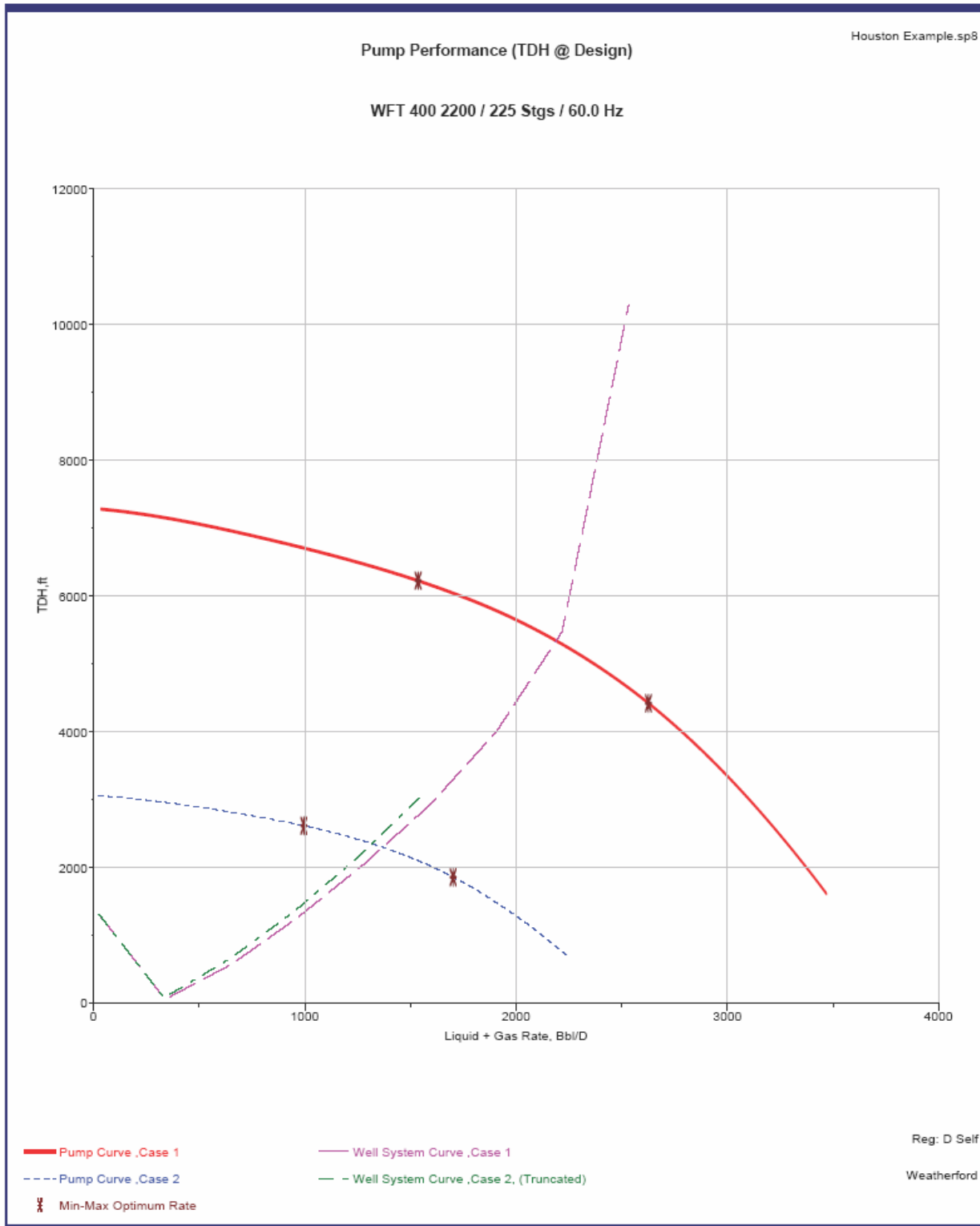




Table 22 SubPump Total Volume through Pump Graph – Example VFD application

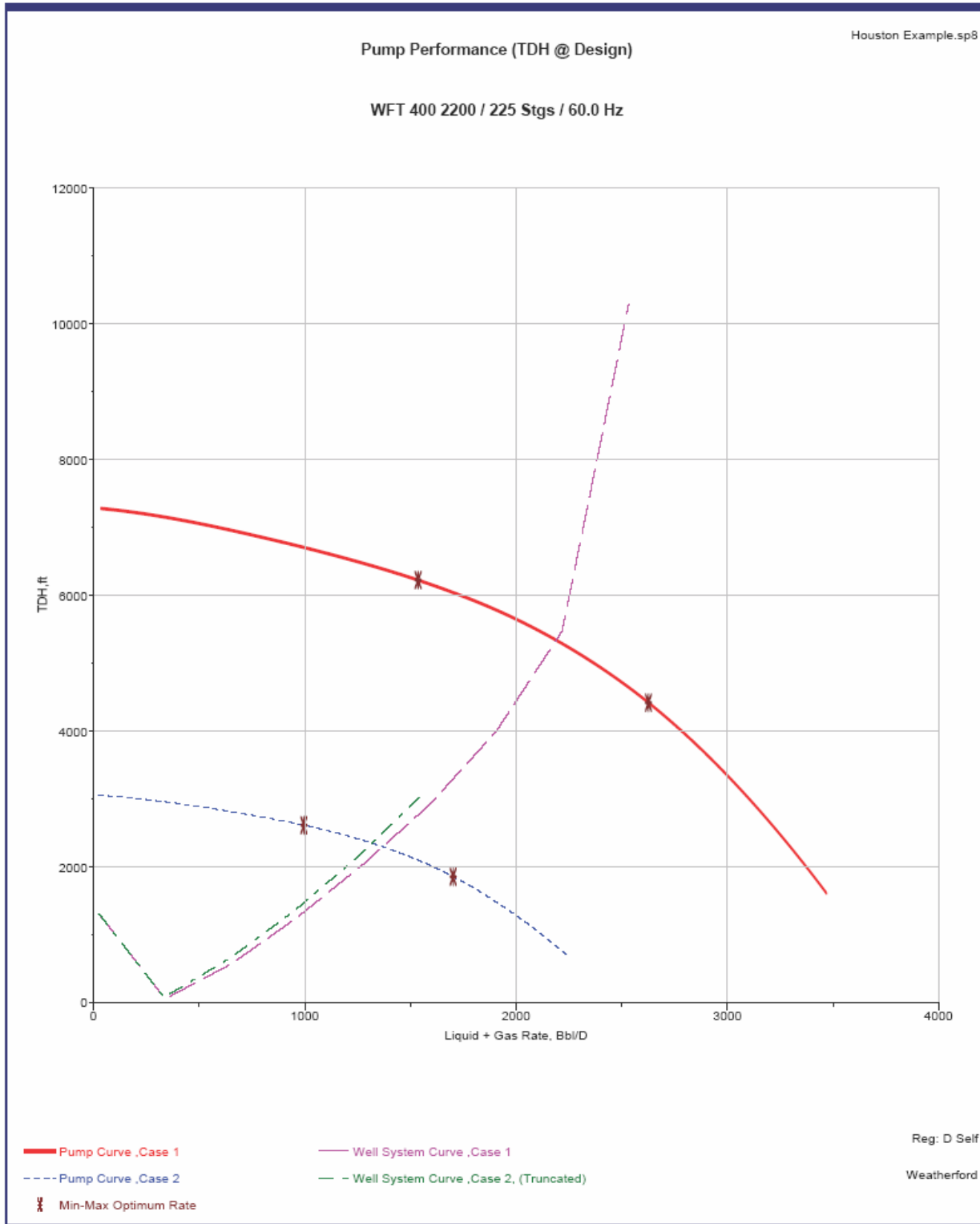




Table 23 SubPump Pump TDH Graph – Example VFD application

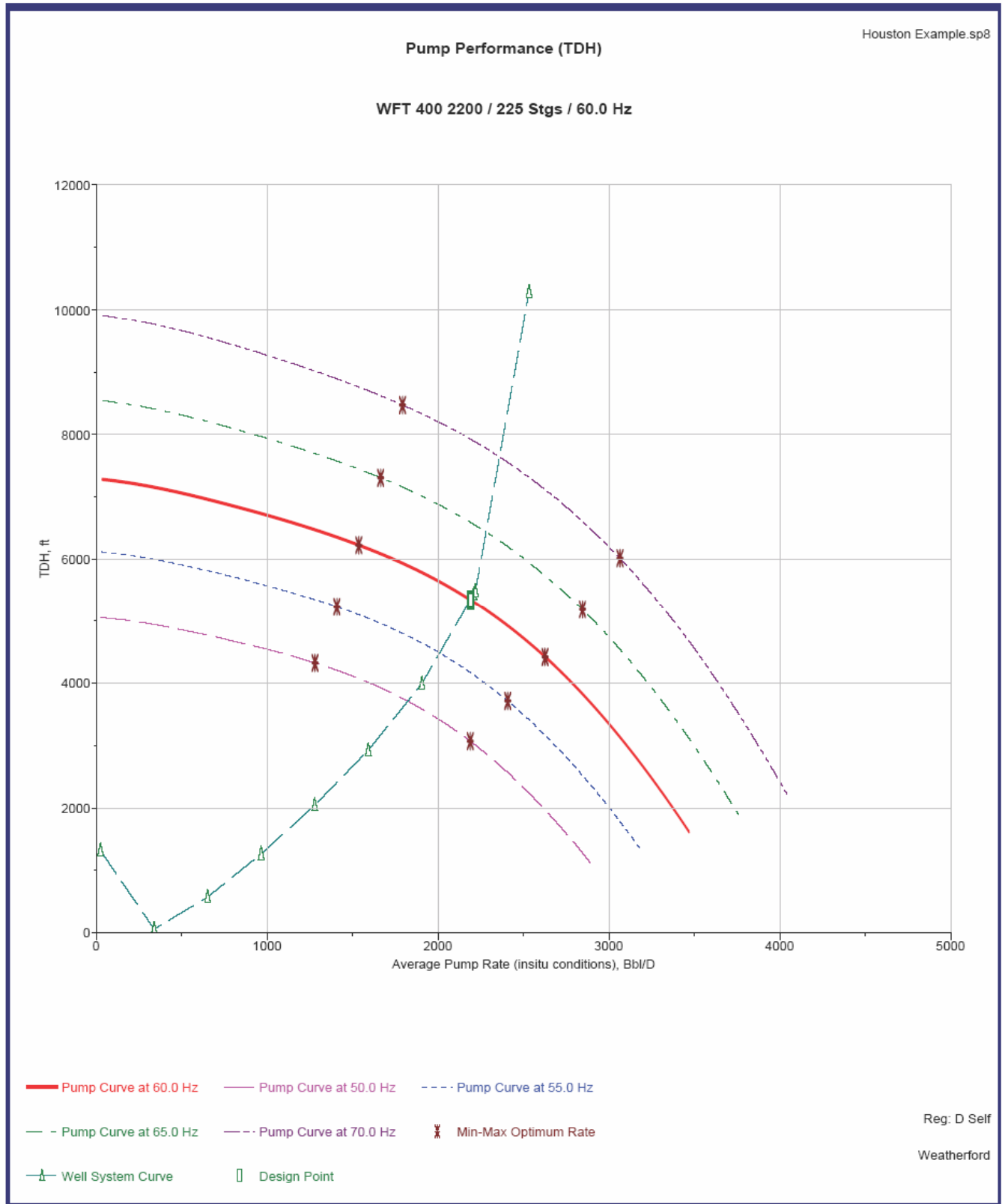




Table 24 SubPump Summary Run – Example VFD Application 2000 & 1200 BPD

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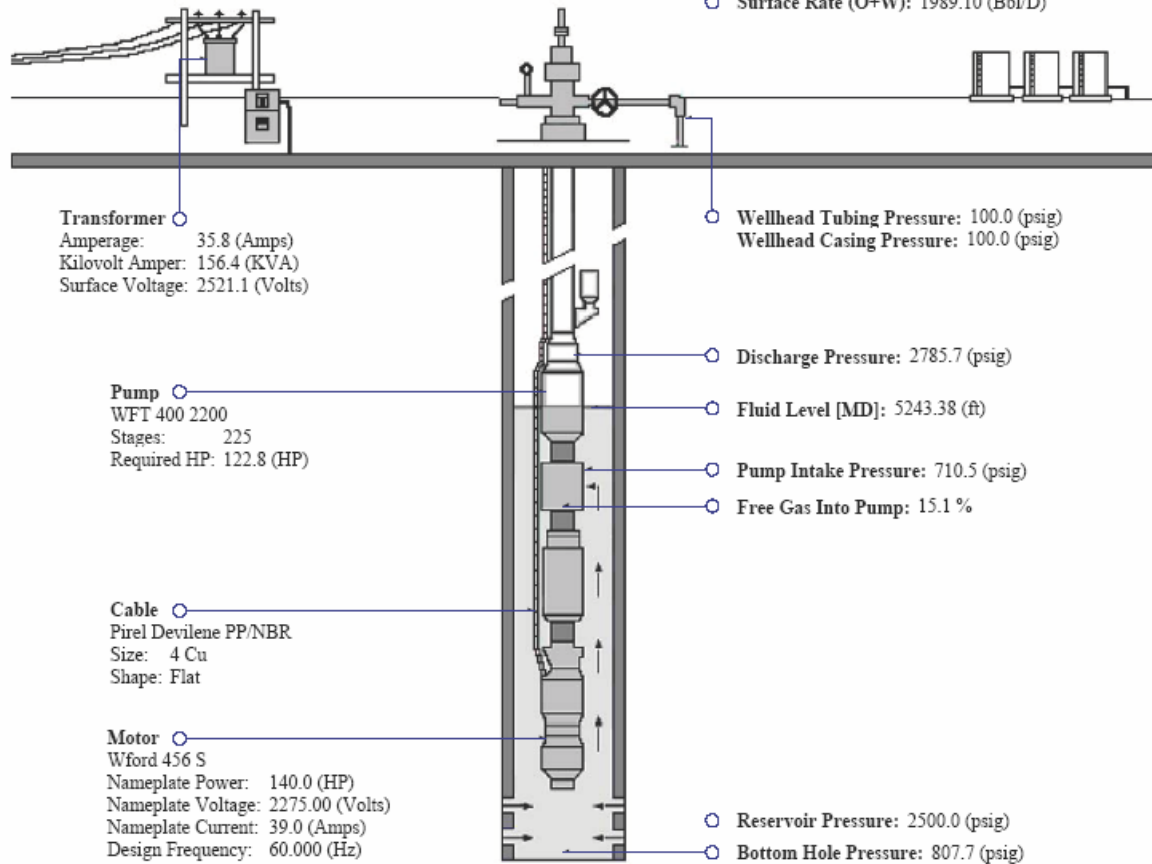
SubPUMP Summary Report

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FINAL WELL OPERATION DIAGRAM (Case 1)

Note: "Surface" rates are calculated at standard conditions.

○ Surface Rate (O+W): 1989.10 (Bbl/D)





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SubPUMP Summary Report

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GENERAL DESCRIPTION

Company Name:	Weatherford	
Well Name:	5-25-06	
Field Name:	Houston	
Reservoir Name:	Example	
Analyst:		
Date:	May 26, 2006	
Case:	(1)	(2)
Comments:	2035 Liquid BPD	1200 Liquid BPD

WELLBORE AND FLOWLINE

Casing OD, in:	5.500 (7500.00 ft)	5.500 (7500.00 ft)
Tubing OD, in:	2.875 (7000.00 ft)	2.875 (7000.00 ft)
Pump Depth MD/TVD, ft:	7000.00/7000.00	7000.00/7000.00
Top of Formation MD/TVD, ft:	7250.00/7250.00	7250.00/7250.00
Downhole Temp, °F:	180.0	180.0
Oil Rate, Bbl/D:	700.00	420.00
Oil Gravity, °API:	30.0 (0.7 cP)	30.0 (0.7 cP)
Water Rate, Bbl/D:	1300.00	780.00
Water Sp. Grav., fw=1:	1.050	1.050
Gas Rate, Mscf/D:	319.13	191.48
Gas Sp. Grav., air=1:	0.700	0.700
Liquid (O+W) @ Surf., Bbl/D:	2000.00	1200.00
Tubing Pressure, psig:	100.0	100.0
Casing Pressure, psig:	100.0	100.0
Tubing Outflow Correlation:	Hagedorn & Brown (1963)	Hagedorn & Brown (1963)
Flowline Correlation:	None Calculated	None Calculated
Correlation Tuning Factor (wellbore) :	1.000	1.000



OPERATION AND PERFORMANCE

Case:	(1)	(2)
Comments:	2035 Liquid BPD	1200 Liquid BPD
Frequency, Hz:	60.000	38.300
Oper. Motor Load @ Design Hz, HP:	124.5	32.3
Oper. Motor Load @NP, HP:	122.8	51.6
Operating Speed, RPM:	3482.7	2250.0
Operating Current, Amps:	35.8	22.6
Operating Voltage, Volts:	2275.0	1452.2
Operating Power Factor :	0.792	0.520
Adjusted for Motor Slip :	Yes	Yes
Pump Efficiency, %:	67.1	66.6
Motor Efficiency, %:	80.8	66.1
Operating Thrust Load, lb:	N/A	N/A
Maximum Thrust Load, lb:	N/A	N/A
Surf. Final Liq. Rate(O+W), Bbl/D:	1989.10	1210.55
Total Stages :	225	225
Avg. Pump Final Fluid, Bbl/D:	2197.66	1337.47
Avg. Pump Total Rate, Bbl/D:	2208.94	1309.84
Free Gas by Volume @ Pump, %:	31.0	8.3
Free Gas by Volume into Pump, %:	15.1	2.0
Total Dynamic Head (TDH), ft:	5383.32	2352.62
Pump Intake Pressure, psig:	710.5	1545.2
Pump Operating Power, HP:	119.3	32.0
Fluid Level [MD], ft:	5243.38	2733.43



Table 25 SubPump Detail Run – Example VFD Application 2000 BPD

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SubPUMP Detailed Report: CASE 1

Page 1

GENERAL DESCRIPTION

Company Name: Weatherford
Well Name: 5-25-06
Field Name: Houston
Reservoir Name: Example
Analyst:
Calculation Type: Rigorous Design
Location:

Comments: 2035 Liquid BPD

Date: May 26, 2006

WELLBORE DATA

	<u>No.</u>	<u>OD</u> <u>in</u>	<u>Wt</u> <u>lb/ft</u>	<u>ID</u> <u>in</u>	<u>Rough</u> <u>in</u>	<u>Bottom MD</u> <u>ft</u>	<u>Top MD</u> <u>ft</u>
<u>Tubing</u>	1	2.875	6.50	2.441	0.0006500	7000.00	
<u>Casing</u>	1	5.500	17.00	4.892	0.0006500	7500.00	0.00

Pump Depth, ft: 7000.00
Top of Perfs. or Datum Pt. (MD), ft: 7250.00
Reservoir Temperature, °F: 180.0
Wellhead Temperature, °F: 120.0
Fluid Temperature Before Motor, °F: 177.9
Fluid Temperature After Motor, °F: 177.9

Tubing Outflow Correlation: Hagedorn & Brown (1963)
Correlation Tuning Factor: 1.000



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SubPUMP Detailed Report: CASE 1

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FLOWLINE DATA

Flowline Correlation
Choke Correlation

None Calculated
No choke calculations done



PVT CORRELATIONS

Description

Method

Oil Viscosity, (Dead)	Beggs & Robinson
Oil Viscosity, (Saturated)	Beggs & Robinson
Oil Viscosity, (UnSaturated)	Vasquez & Beggs
Gas Viscosity	Lee
Water Viscosity	Matthews & Russell
Oil Formation Volume Factor	Vasquez & Beggs
Water Formation Volume Factor	HP41C
Oil Density	Katz
Z-Factor	Dranchuk, Purvis
Oil Isothermal Compressibility	Vasquez & Beggs
Water Isothermal Compressibility	Meehan
Solution Gas/Oil Ratio	Lasater
Bubble Point Pressure	Lasater
Gas Density	Beggs
Water Density	Beggs
Oil Surface Tension Correlation	Baker and Swerdloff
Water Surface Tension Correlation	Hough



FLUID DATA

Standard Temperature, °F:	60.0	Prod. Gas/Oil Ratio, scf/bbl:	455.90
Standard Pressure, psi:	14.7	Prod. Gas/Liq. Ratio, scf/bbl:	159.57
Oil Gravity, °API:	30.0	Bubble Point Pressure, psia:	2500.0
Gas Specific Gravity:	0.700	Solution GOR, scf/bbl:	438.27
Water Specific Gravity:	1.050	Average Fluid Viscosity, cP:	0.7
Water Salinity, ppm:	67197	Fluid Grad. @ Pump Intake, psi/ft:	0.327
Water Cut, %:	65.0		



INFLOW DATA

IPR Calculation Method: Vogel

Fluid Rate at Test BHP, Bbl/D:	900.00
Productivity Index, blpd/psi:	1.669
Bubble Point Rate, Bbl/D:	24.54
Max. Oil Flow Rate, Bbl/D:	2343.23
Max. Total Flow Rate, Bbl/D:	2343.23

	<u>Static</u>	<u>Test</u>
Casing Pressure, psig:	0.0	0.0
Fluid Level, ft:	0.00	0.00
Bottom Hole Pressure (BHP), psig:	2500.0	1900.0

PRESSURE/RATE DATA

Design Criteria - Solved For: Pump Intake Conditions

Total Fluid Rate, Bbl/D:	2000.00
Pump Depth, ft:	7000.00
Fluid Over Pump, ft:	1753.47
Pumping Fluid Level, ft:	5246.53
Pump Intake Pressure, psig:	694.8
Total Dynamic Head, ft:	5439.71
Tubing Pressure, psig:	100.0
Casing Pressure, psig:	100.0
Bottom Hole Pressure, psig:	791.6
Gas through Pump:	Gas Into Solution

Fluid level Above Pump: Clean Oil Gradient



GAS SEPARATOR PERFORMANCE

Packer Installed?	No
Free Gas Avail. at Pump, %:	31.0
Natural Gas Separation efficiency, %:	0.0
Gas Separator Installed?	Yes
Gas Separator Efficiency, %:	59.2
Free Gas into Pump, %:	15.5

WELL SYSTEM CURVE DETAIL

<u>Point #</u>	<u>(+) Tubing Head ft</u>	<u>(-) PIP Head ft</u>	<u>(=) TDH ft</u>	<u>Surface Rate O+W Bbl/D</u>	<u>Avg. Pump Rate O+W+G Bbl/D</u>	<u>Pumping Fluid Level ft</u>
1	7099.82	5787.98	1311.84	22.91	25.32	201.67
2	5402.57	5347.06	55.51	306.46	338.59	719.51
3	5451.29	4881.97	569.31	590.00	651.87	1277.36
4	5653.31	4395.15	1258.16	873.55	965.14	1884.11
5	5911.15	3869.05	2042.11	1157.09	1278.42	2555.04
6	6212.34	3290.72	2921.62	1440.64	1591.69	3312.95
7	6628.57	2636.46	3992.11	1724.18	1904.97	4196.48
8	7324.68	1837.65	5487.04	2007.73	2218.25	5280.35
PumpOff	10735.14	453.10	10282.04	2291.27	2531.52	6992.57
Design	7302.23	1862.52	5439.71	2000.00	2209.71	5246.53

Fluid level Above Pump: Clean Oil Gradient

THEORETICAL PUMP PERFORMANCE

	<u>Intake</u>	<u>Discharge</u>	<u>Surface</u>
Oil Rate, Bbl/D:	790.06	813.98	696.18
In-situ Gas Rate Through Pump, Bbl/D:	388.80	0.00	N/A
In-situ Gas Rate from Casing, Bbl/D:	564.21	N/A	N/A
Total Gas Rate, Mscf/D:	N/A	N/A	317.39
Free Gas Percent, %:	15.05	0.00	N/A
Water Rate, Bbl/D:	1331.38	1325.35	1292.91
Total Liquid Rate, Bbl/D:	2121.44	2139.33	1989.10
Pressure, psig:	710.5	2785.7	100.0 (wellhead)
Specific Gravity Liquid, wtr=1:	0.94	0.94	N/A
Specific Gravity Mixture, wtr=1:	0.81	0.94	N/A
Liquid Density, lb/cf:	58.874	58.733	N/A
Mixture Density, lb/cf:	50.360	58.733	N/A
Mixture Viscosity, cStokes:	N/A	N/A	N/A
Mixture Viscosity, SSU:	N/A	N/A	N/A
Solution GOR, scf/bbl:	126.24	253.21	N/A
Solution GWR, scf/bbl:	4.00	6.41	N/A
Liquid FVF, res/surf:	1.07	1.08	N/A
Mixture FVF, res/surf:	1.26	1.08	N/A
Gas Deviation Factor:	0.928	0.858	N/A



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SubPUMP Detailed Report: CASE 1

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PUMP DATA 1

Manufacturer: WFT
Series: 400
Model: 2200

Minimum Recommended Rate, Bbl/D: 1542.33**
Maximum Recommended Rate, Bbl/D: 2636.88**
Rate at Peak Efficiency, Bbl/D: 2245.00**
Power at Peak Efficiency, HP: 132.4**
Frequency, Hz: 60.0

** = Corrected for Frequency & Viscosity

Number of Stages: 225
Stages with Free Gas: 80

Viscosity Correction Factors

User Entered

Rate:	1.000	No
Head:	1.000	No
Power:	1.000	No
Efficiency:	1.000	
Viscosity: Fluid		

Design

Catalog

Total Dynamic Head (TDH), ft:	5439.71	5435.79
Surface Rate (O+W), Bbl/D:	2000.00	1999.31
Avg. Rate per Stage (O+G+W), Bbl/D:	N/A	2208.94
Pump Intake Pressure, psig:	694.8	695.8
Operating Power, HP:	N/A	124.5
Pump Efficiency, %:	N/A	65.9

Housing Data

NO HOUSINGS SELECTED



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SubPUMP Detailed Report: CASE 1

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MOTOR DATA

Manufacturer: Wford
Series: 456
Type: S - Double

Nameplate Power, HP: 140.0
Nameplate Voltage, Volts: 2275.00
Nameplate Current, Amps: 39.0

Nameplate Frequency, Hz: 60

Adjust for Motor Slip: Yes
Design Frequency, Hz: 60.0
Operating Motor Load, HP: 124.5
(@ Design Frequency)
Operating Motor Load, %: 87.71
Operating Speed, RPM: 3483

Operating Current, Amps: 35.8
Operating Voltage, Volts: 2275.00
Operating Power Factor, frac: 0.792
Operating Efficiency, %: 80.83
Fluid Velocity, ft/sec: 7.553

	<u>Catalog</u>	<u>Actual</u>
Total Stages	225	225
Slip Stages	0	0
Total Dynamic Head (TDH), ft	5435.79	5383.32
Surface Rate (O+W), Bbl/D	1999.31	1989.10
Avg. Rate per Stage (O+G+W), Bbl/D	2208.94	2197.66
Pump Intake Pressure, psig	695.8	710.5
Operating Power, HP	124.5	119.3
Pump Efficiency, %	65.9	67.1
Operating Speed, RPM	3500	3482

SEAL DATA

NO SEAL SELECTED



CABLE DATA

Manufacturer: Pirel
Type: Devilene PP/NBR
Size: 4 Cu
Shape: Flat
Conductor Type: Solid
Max. Cond. Temp., °F: 205.0
Cable Length, ft: 7100.00

Solve for: Surface Voltage
Electric Cost, \$/kWh: 0.10

Note: *Indicates the cable is out of range!

	<u>Design Freq.</u>
Frequency, Hz	60
Max. Allowable Amps, Amps	38.1
Amperage, Amps	35.8
Kilovolt Amper, KVA	156.4
Kilowatts, kW	123.9
Kilowatts, \$/mo	9044
Surface Voltage, Volts	2521.1
Voltage Drop @ 68.0 °F, Volts	198.5
Voltage Drop @ 180.0 °F BHT, Volts	246.1
Kilowatt Loss, kW	12.1
Cost of Voltage Loss, \$/mo	882
Required Motor Voltage, Volts	2275.0
Downhole Voltage at Motor, Volts	2275.0
In-rush Motor Voltage Drop, Volts	984.3
Motor Startup Voltage, Volts	1536.8
Startup/Operating Ratio, Ratio	0.7

Note: "Surface" rates are calculated at standard conditions.



Table 26 SubPump Detail Run – Example VFD application 1200 BPD

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SubPUMP Detailed Report: CASE 2

Page 1

GENERAL DESCRIPTION

Company Name: Weatherford
Well Name: 5-25-06
Field Name: Houston
Reservoir Name: Example
Analyst:
Calculation Type: Rigorous Design
Location:

Comments: 1200 Liquid BPD

Date: May 26, 2006

WELLBORE DATA

	<u>No.</u>	<u>OD</u> <u>in</u>	<u>Wt</u> <u>lb/ft</u>	<u>ID</u> <u>in</u>	<u>Rough</u> <u>in</u>	<u>Bottom MD</u> <u>ft</u>	<u>Top MD</u> <u>ft</u>
<u>Tubing</u>	1	2.875	6.50	2.441	0.0006500	7000.00	
<u>Casing</u>	1	5.500	17.00	4.892	0.0006500	7500.00	0.00

Pump Depth, ft: 7000.00
Top of Perfs. or Datum Pt. (MD), ft: 7250.00
Reservoir Temperature, °F: 180.0
Wellhead Temperature, °F: 120.0
Fluid Temperature Before Motor, °F: 177.9
Fluid Temperature After Motor, °F: 177.9

Tubing Outflow Correlation: Hagedorn & Brown (1963)
Correlation Tuning Factor: 1.000



FLOWLINE DATA

Flowline Correlation
Choke Correlation

None Calculated
No choke calculations done



PVT CORRELATIONS

<u>Description</u>	<u>Method</u>
Oil Viscosity, (Dead)	Beggs & Robinson
Oil Viscosity, (Saturated)	Beggs & Robinson
Oil Viscosity, (UnSaturated)	Vasquez & Beggs
Gas Viscosity	Lee
Water Viscosity	Matthews & Russell
Oil Formation Volume Factor	Vasquez & Beggs
Water Formation Volume Factor	HP41C
Oil Density	Katz
Z-Factor	Dranchuk, Purvis
Oil Isothermal Compressibility	Vasquez & Beggs
Water Isothermal Compressibility	Meehan
Solution Gas/Oil Ratio	Lasater
Bubble Point Pressure	Lasater
Gas Density	Beggs
Water Density	Beggs
Oil Surface Tension Correlation	Baker and Swerdloff
Water Surface Tension Correlation	Hough



FLUID DATA

Standard Temperature, °F:	60.0	Prod. Gas/Oil Ratio, scf/bbl:	455.90
Standard Pressure, psi:	14.7	Prod. Gas/Liq. Ratio, scf/bbl:	159.57
Oil Gravity, °API:	30.0	Bubble Point Pressure, psia:	2500.0
Gas Specific Gravity:	0.700	Solution GOR, scf/bbl:	438.27
Water Specific Gravity:	1.050	Average Fluid Viscosity, cP:	0.7
Water Salinity, ppm:	67197	Fluid Grad. @ Pump Intake, psi/ft:	0.405
Water Cut, %:	65.0		



INFLOW DATA

IPR Calculation Method: Vogel

Fluid Rate at Test BHP, Bbl/D:	900.00
Productivity Index, blpd/psi:	1.669
Bubble Point Rate, Bbl/D:	24.54
Max. Oil Flow Rate, Bbl/D:	2343.23
Max. Total Flow Rate, Bbl/D:	2343.23

	<u>Static</u>	<u>Test</u>
Casing Pressure, psig:	0.0	0.0
Fluid Level, ft:	0.00	0.00
Bottom Hole Pressure (BHP), psig:	2500.0	1900.0

PRESSURE/RATE DATA

Design Criteria - Solved For: Pump Intake Conditions

Total Fluid Rate, Bbl/D:	1200.00
Pump Depth, ft:	7000.00
Fluid Over Pump, ft:	4336.99
Pumping Fluid Level, ft:	2663.01
Pump Intake Pressure, psig:	1554.1
Total Dynamic Head, ft:	2317.84
Tubing Pressure, psig:	100.0
Casing Pressure, psig:	100.0
Bottom Hole Pressure, psig:	1659.6
Gas through Pump:	Gas Into Solution

Fluid level Above Pump: Clean Oil Gradient

GAS SEPARATOR PERFORMANCE

Packer Installed?	No
Free Gas Avail. at Pump, %:	8.3
Natural Gas Separation efficiency, %:	0.0
Gas Separator Installed?	Yes
Gas Separator Efficiency, %:	78.2
Free Gas into Pump, %:	1.9

WELL SYSTEM CURVE DETAIL

Point #	(+) Tubing Head ft	(-) PIP Head ft	(=) TDH ft	Surface Rate O+W Bbl/D	Avg. Pump Rate O+W+G Bbl/D	Pumping Fluid Level ft
1	7099.77	5786.98	1312.79	22.91	24.99	201.67
2	5427.87	5347.51	80.36	306.46	334.28	719.51
3	5527.77	4878.07	649.71	590.00	643.56	1277.36
4	5750.40	4391.17	1359.23	873.55	952.85	1884.11
5	6042.92	3865.40	2177.52	1157.09	1262.13	2555.04
6	6394.46	3285.82	3108.64	1440.64	1571.42	3312.95
7	6839.89	2631.32	4208.57	1724.18	1880.70	4196.48
8	7610.93	1831.43	5779.50	2007.73	2189.99	5280.35
PumpOff	11211.04	450.36	10760.67	2291.27	2499.27	6992.57
Design	6099.68	3781.84	2317.84	1200.00	1308.94	2663.01

Fluid level Above Pump: Clean Oil Gradient

THEORETICAL PUMP PERFORMANCE

	Intake	Discharge	Surface
Oil Rate, Bbl/D:	506.76	507.80	423.69
In-situ Gas Rate Through Pump, Bbl/D:	25.79	0.00	N/A
In-situ Gas Rate from Casing, Bbl/D:	92.58	N/A	N/A
Total Gas Rate, Mscf/D:	N/A	N/A	193.16
Free Gas Percent, %:	1.96	0.00	N/A
Water Rate, Bbl/D:	809.42	807.46	786.85
Total Liquid Rate, Bbl/D:	1316.18	1315.26	1210.55
Pressure, psig:	1545.2	2492.2	100.0 (wellhead)
Specific Gravity Liquid, wtr=1:	0.93	0.93	N/A
Specific Gravity Mixture, wtr=1:	0.92	0.93	N/A
Liquid Density, lb/cf:	58.179	58.303	N/A
Mixture Density, lb/cf:	57.143	58.303	N/A
Mixture Viscosity, cStokes:	N/A	N/A	N/A
Mixture Viscosity, SSU:	N/A	N/A	N/A
Solution GOR, scf/bbl:	283.84	318.61	N/A
Solution GWR, scf/bbl:	6.85	7.35	N/A
Liquid FVF, res/surf:	1.09	1.09	N/A
Mixture FVF, res/surf:	1.11	1.09	N/A
Gas Deviation Factor:	0.869	0.851	N/A



PUMP DATA 1

Manufacturer: WFT
Series: 400
Model: 2200

Minimum Recommended Rate, Bbl/D: 996.41**
Maximum Recommended Rate, Bbl/D: 1703.54**
Rate at Peak Efficiency, Bbl/D: 1433.06**
Power at Peak Efficiency, HP: 34.4**
Frequency, Hz: 38.3

** = Corrected for Frequency & Viscosity

Number of Stages: 225
Stages with Free Gas: 39

Viscosity Correction Factors

User Entered

Rate: 1.000 No
Head: 1.000 No
Power: 1.000 No
Efficiency: 1.000
Viscosity: Fluid

Design

Catalog

Total Dynamic Head (TDH), ft: 2317.84 2320.16
Surface Rate (O+W), Bbl/D: 1200.00 1200.83
Avg. Rate per Stage (O+G+W), Bbl/D: N/A 1309.84
Pump Intake Pressure, psig: 1554.1 1553.4
Operating Power, HP: N/A 32.3
Pump Efficiency, %: N/A 66.5

Housing Data

NO HOUSINGS SELECTED



MOTOR DATA

Manufacturer:	Wford		
Series:	456		
Type:	S - Double		
Nameplate Power, HP:	140.0	Nameplate Frequency, Hz:	60
Nameplate Voltage, Volts:	2275.00		
Nameplate Current, Amps:	39.0		
Adjust for Motor Slip:	Yes	Operating Current, Amps:	22.6
Design Frequency, Hz:	38.3	Operating Voltage, Volts:	1452.21
Operating Motor Load, HP:	32.3	Operating Power Factor, frac:	0.520
(@ Design Frequency)			
Operating Motor Load, %:	36.85	Operating Efficiency, %:	66.07
Operating Speed, RPM:	2250	Fluid Velocity, ft/sec:	4.597

	<u>Catalog</u>	<u>Actual</u>
Total Stages	225	225
Slip Stages	0	0
Total Dynamic Head (TDH), ft	2320.16	2352.62
Surface Rate (O+W), Bbl/D	1200.83	1210.55
Avg. Rate per Stage (O+G+W), Bbl/D	1309.84	1320.44
Pump Intake Pressure, psig	1553.4	1545.2
Operating Power, HP	32.3	32.0
Pump Efficiency, %	66.5	66.6
Operating Speed, RPM	2234	2249

SEAL DATA

NO SEAL SELECTED



CABLE DATA

Manufacturer: Pirel
Type: Devilene PP/NBR
Size: 4 Cu
Shape: Flat
Conductor Type: Solid
Max. Cond. Temp., °F: 205.0
Cable Length, ft: 7100.00

Solve for: Surface Voltage
Electric Cost, \$/kWh: 0.10

Note: *Indicates the cable is out of range!

	<u>Design Freq.</u>
Frequency, Hz	38
Max. Allowable Amps, Amps	38.1
Amperage, Amps	22.6
Kilovolt Amper, KVA	62.9
Kilowatts, kW	32.7
Kilowatts, \$/mo	2387
Surface Voltage, Volts	1607.4
Voltage Drop @ 68.0 °F, Volts	125.2
Voltage Drop @ 180.0 °F BHT, Volts	155.2
Kilowatt Loss, kW	3.2
Cost of Voltage Loss, \$/mo	230
Required Motor Voltage, Volts	1452.2
Downhole Voltage at Motor, Volts	1452.2
In-rush Motor Voltage Drop, Volts	620.7
Motor Startup Voltage, Volts	986.7
Startup/Operating Ratio, Ratio	0.7

Note: "Surface" rates are calculated at standard conditions.